Coarse-Graining and Long-Range Response in Generated Systems

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Abstract: We study when a reduced description preserves the strength and singularities of the response to an applied source. For systems defined by state updates and observations, eliminating variables changes the feedback, source coupling, readout and direct response together. For jointly C^1 updates and observations and bounded surjective linear coarse maps between Banach spaces, fibre consistency yields differentiable descent of the update and observation, and hence preservation of their joint source response on the common invertible domain. For approximate descent, we derive an exact response-error identity that separates defects in the update, source, readout and direct term without requiring an inverse coarse map. Resolvent and contour estimates propagate reference-branch, source and readout errors to visible poles and their residues at displaced critical points. In a reciprocal Hilbert-space realisation with a self-adjoint endpoint operator, an isolated finite-dimensional critical subspace, a bounded complementary inverse and a positive crossing matrix, the visible residue is positive semidefinite and finite rank; source overlap determines which critical directions remain visible. Arithmetic and reflecting-heat examples distinguish static response preservation from memoryless dynamical closure. In the two-mode example, at fixed geometric memory decay $0 < d < 1$, retaining m terms gives pole displacement $O(d^m)$ and residue error $O(md^m)$. In a vacuum quantum-field-theory realisation, when the exact unfiltered-covariance, projection, source-pairing, continuation, subtraction and endpoint matching hypotheses hold, the response residue is the zero-mass atom of the corresponding positive spectral measure.

Keywords: Coarse-graining; Linear response; Operator approximation; Schur complements; Memory kernels; Analytic Fredholm theory

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1 Introduction

A macroscopic description replaces many microscopic variables by a few collective quantities. Agreement on an equilibrium value, however, leaves open how those quantities respond to an applied disturbance. Unresolved variables can carry the disturbance back to the retained ones, change the coupling to an external source, and contribute to the observed output. Near a collective critical mode, a small change in these effects can move a response pole or substantially change its strength. The physical question is which microscopic distinctions can be discarded while preserving this collective response.

This question is more restrictive than reproducing an equilibrium value. Two microscopic states may look identical now and still differ in the response that a later source

intervention reveals. The coarse state must therefore preserve future distinguishability under the allowed operations, not only the information visible at one instant. The response is the quantity that makes this requirement quantitative.

We formulate the problem through a sourced state equation $X = F(X, h; z)$ and an observation $O(X, h; z)$. Differentiating both with respect to h follows the disturbance from its entry into the system to its observed effect. Differentiable descent of both maps preserves this derivative; approximate coarse-graining introduces errors whose amplification depends on the collective modes. We establish conditions for exact preservation and estimates for the response, pole positions and residues when the coarse description is approximate.

For given generating operations, source interventions and observation maps, two states are equivalent when every allowed future experiment gives the same output. This defines the coarsest observation-sufficient quotient. Its universal property concerns sets; the topology needed for differentiable response is treated separately. For a bounded surjective linear map between Banach spaces, constancy of the projected update and observation on each fibre gives a unique differentiable coarse realisation. Thus the information to be retained is fixed by the operations and observations, while regularity determines whether the resulting description supports linear response.

The future-output equivalence extends the state distinction used by Moore for finite sequential machines [1]. In linear finite-dimensional systems, Ho–Kalman realisation reconstructs a minimal state space from input–output data [2]. We compare specified state equations and their coarse descriptions, including nonlinear updates on Banach spaces. The quotient identifies which observations descend; the response-error identity quantifies the failure of differentiable descent for a given approximation.

Differentiation gives the response $D+R(I-B)^{-1}S$, where B is the linearised update, S the source map, R the readout and D the direct term. Theorem 2.2 proves its preservation under joint descent of the update and observation. Source-dependent coarse coordinates contribute compensating derivatives to S and D. The implicit state equation gives the same response independently of a numerical iteration used to solve it.

The four factors have a direct physical reading. S carries an applied disturbance into the internal state, $(I-B)^{-1}$ sums its repeated transmission through the coupled variables, R converts the resulting state change into an output, and D records the part of the output that changes directly at fixed state. The construction is organised around this source–feedback–readout path.

For approximate descent, the response difference separates into four defects, one for each map. The comparison map may be noninvertible, so the result covers the removal of state coordinates. The estimates retain the displacement of the reference branch and extend to contour integrals of the response. Classical Schur–Feshbach reduction [3] supplies the complementary description by exact block elimination. We keep its source, readout and direct term together with the effective kernel, since all four enter the observable residue. In evolution equations the eliminated coordinates instead appear through an initial-state term and memory of states and sources, as in projection-operator descriptions of memory [4]. The block inverse and memory mechanism are classical. The contribution developed here is a response comparison within one source–observation system: irreversible reduction,

reference-branch displacement, and errors in the source and readout are carried through to visible poles and their residues. The four-defect identity of Theorem 4.1 and the displaced-root estimates of Proposition 4.2 make this comparison quantitative.

Hydrodynamics gives a useful physical comparison. Near local equilibrium, conservation laws identify slow sectors that a reduced description must retain, while constitutive data determine the response within those sectors. When the remaining variables are eliminated, their delayed influence reappears as constitutive memory unless a separate scale-separation argument makes it local.

For linear dynamical systems, projection-based model reduction already compares input–output transfer functions, with $\mathcal{H}_2$ and $\mathcal{H}_\infty$ error measures for asymptotically stable systems (Ref. [5], Sections 2.2–2.4). Our comparison is local near an isolated critical response, where it tracks the root and residue rather than a global transfer-function norm. The full and coarse maps may be derivatives of nonlinear state equations at different reference states. Their comparison requires bounds on that state displacement, the four map defects, and the slope of the critical zero; it does not construct an optimal reduced basis.

Once the response has been constructed, analytic Fredholm theory describes its isolated singularities (Ref. [6], Chapter VII). A simple crossing of a kernel eigenvalue through one gives a resolvent pole; source and readout overlaps determine whether it is observable. In a reciprocal Hilbert-space realisation, a positive crossing matrix on the isolated critical subspace gives a positive finite-rank response residue. For approximations, norm and Riesz-projection estimates (Ref. [7], Chapter 5) control the displacement of critical roots and the change of their residues. Resolvent bounds retain the sensitivity of nonnormal kernels, for which spectral distances alone can be misleading [8].

An arithmetic merge from Ref. [9] and a two-mode reflecting-heat system make the distinction between static and dynamical reduction concrete. We prove the finite identities used in both examples. Static susceptibility survives Schur elimination even where a memoryless update fails to descend; the reflecting-heat example also permits explicit calculation of the memory error and visible critical residue. Corollary 5.1 gives pole displacement of order d^m and residue error of order md^m when m memory terms are retained at fixed $0 < d < 1$. The extra factor m comes from differentiating the memory remainder, so kernel accuracy and residue accuracy impose different requirements even in this two-mode system.

Gaussian averaging gives a continuous family of operations for these constructions. We give a self-contained characterisation within the class of weakly continuous, centred, orthogonally invariant convolution semigroups with finite nonzero variance and square-root self-similarity; this is a special case of the theory of stable laws [10]. Heat smoothing and an information-losing quotient have different roles. We also distinguish averaging in external and relative coordinates, since their momentum multipliers act on different parts of a response. The reflecting-window example connects this averaging law to calibrated correlation endpoints, a chosen boundary condition, and an explicit heat spectrum.

The word “unique” enters in three logically distinct places. The observation-sufficient quotient is unique because all future allowed operations and observations determine which states remain distinguishable. The Gaussian semigroup is unique within the stated composition and self-similarity class. The spectral atom is unique because the same centred

Euclidean correlator has a unique Laurent coefficient at its isolated pole, and the matched Källén–Lehmann representation identifies that coefficient with the massless atom of a positive measure. These are, respectively, uniqueness of a representation, of an averaging law within a class, and of a spectral identification.

Quantum field theory provides a further realisation of the response construction. Vacuum positivity and covariance then give particle content to the response: after channel projection and spectral matching, the pole residue becomes a positive measure atom. Renormalised operator mixing and physical symmetry quotients specify the channels; Ward identities and functional flows then constrain and compute their responses. For a centred, connected vacuum covariance of local physical insertions or of a directly reduced helicity representation, matched physical projections, continuation and subtraction conventions identify a Fredholm residue with a Källén–Lehmann spectral atom. This spectral identification is additional to the positive-residue theorem for reciprocal response. The full covariant stress-tensor case also illustrates the force of the representation hypotheses: its null-shell transverse-traceless atom is excluded by the local null-shell argument. Scattering interpretations use the soft-spin-two and Lorentz-covariance consistency conditions [11, 12]. The free two-helicity example evaluates both sides of this identification on a flat background. Interpreting the resulting tensor response as gravity requires the matter couplings and constraints of an effective action.

The following companion works realise selected parts of this framework in specific models. The tensor-response application in Ref. [13] treats matter coupling and the resulting long-range exchange. The compact internal construction of Ref. [14] specifies gauge and matter channels, whose effective couplings and mass scales are evaluated in the companion application Ref. [15] using *Coarse-Graining Genesis Framework V5* [16]. That companion work also describes a coupled matter–amplitude evolution with a phase-clock readout and finite-interval propagation bounds (Appendix G.6, Proposition G.7 and Lemma G.8 of that reference). This provides a model-specific realisation of algebraic elimination alongside the dynamical memory examples studied here. The general response results are proved within the present paper; the companion constructions provide model-specific applications. For time-dependent generated spectra, Refs. [17, 18] treat the interaction bounds, controlled limits and residual estimates that propagate a finite approximation along a specified evolution. The viscous-flow realization in Ref. [19] develops state-dependent moving spectral windows, tail budgets and residual control for incompressible evolution, while the cosmic-structure application in Ref. [20] uses finite-window evolution and explicit source transport. These evolution problems require propagation estimates beyond the local response comparison proved here.

Sections 2–3 establish descent and the response criteria; Section 4 treats errors and memory. The examples are in Section 5. The appendices contain the spectral, operator and analytic details invoked in these results.

2 Generated operations, sufficient coarse states, and response

The choice of a coarse state expresses which microscopic distinctions matter for the specified physical questions. Agreement of present observations is sufficient only if subsequent allowed operations never reveal a difference. This requirement first determines a set quotient; its regular realisation makes it possible to differentiate the response to a source. In a physical realisation, observations are the specified physical outputs of the system and need not involve a measuring apparatus.

2.1 Operations and the observation-sufficient quotient

Let X be a common invariant state domain and let $F_a : \mathsf{X} \to \mathsf{X}$ be the allowed deterministic operations. The label a includes each allowed source intervention, not just the unforced update. A finite word w specifies an ordered composition F_w; the empty word acts as the identity. Let $\mathscr{O}$ be the declared observation family. Its members may have different output spaces, but equality of their outputs is fixed before taking a quotient. These data come from a specified protocol. For finite sequential machines, distinguishing states by their future outputs is the viewpoint of Moore's experiments [1]. Here the state and output sets may be infinite, and the regularity needed for response is imposed below.

In a generated physical realisation, the operations, allowed sources and observations belong to one fixed construction. The arithmetic operations of Ref. [9] and the spectral products and covariant transport of Ref. [14] give companion realisations. Their continuous extensions use the specified common domains and convergence norms. For time evolution, Ref. [17] gives model-specific interaction hypotheses and uniform bounds under which finite constructions and their controlled limits define the same observation. These constructions provide applications of the quotient below; its proof uses only the fixed family of operations and observations declared here.

For example, observe only x in a state (x, u) whose next observed coordinate is $x' = ax + bu$, with $b \neq 0$. States with the same x and different u agree at present but give different outputs after one update. The equivalence below tests whether such a distinction can appear after any allowed sequence of operations.

Theorem 2.1 (Coarsest observation-sufficient quotient)**.** *Define*

$$x \sim y \quad \Longleftrightarrow \quad O(F_w x) = O(F_w y) \quad \text{for every } O \in \mathscr{O} \text{ and } w. \tag{2.1}$$

Then $\sim$ is an equivalence relation, and every allowed operation and observation descends to $q : \mathsf{X} \to \mathsf{X}/\!\sim$. If another map C admits exact descended operations and observations on its image, there is a unique surjection $\pi : C(\mathsf{X}) \to \mathsf{X}/\!\sim$ with $q = \pi C$. It intertwines the descended operations and observations. Thus the quotient is unique up to its canonical bijection among coarsest exact observation-sufficient descriptions.

Proof. Equality of all specified outputs is reflexive, symmetric, and transitive. If $x \sim y$, append F_a at the input of any word to obtain $F_a x \sim F_a y$. The empty word gives $O(x) = O(y)$, so both operations and observations descend. If $Cx = Cy$, induction along a word using the descended C-operations gives $CF_w x = CF_w y$; the descended observations then

give $x \sim y$. Therefore $\pi(Cx) = [x]$ is well defined and surjective. This formula proves uniqueness and both intertwining claims. □

The quotient is relative to the chosen operations and observations. Enlarging either family can distinguish previously equivalent states and therefore refine the quotient. The theorem concerns deterministic operations on a common invariant domain; its set-theoretic conclusion leaves the dimension and regularity of the quotient to be determined from the particular system.

A coarse state therefore encodes future-response sufficiency. Two states are identified only when no allowed sequence can bring their difference back to an observed output. The quotient is thus the smallest state description that remains sufficient for the declared source interventions, rather than a compression chosen solely from the present configuration.

2.2 Regular descent and its limits

For a fixed coarse map C, exact descent on its actual image has a simple test. No linear structure is needed for the equivalence

$$CF = \bar{F}C \text{ for a unique } \bar{F} : C(\mathsf{X}) \to C(\mathsf{X}) \quad \Longleftrightarrow \quad Cx = Cy \implies CF(x) = CF(y). \tag{2.2}$$

Indeed, the only possible definition is $\bar{F}(Cx) = CF(x)$, and the right-hand condition is exactly its well-definedness. Uniqueness is on $C(\mathsf{X})$, not on a larger ambient space, and does not select C or F themselves.

This test answers a question about values: does the coarse output depend only on the coarse input? Response requires more, since it differentiates that dependence. The required regularity statement is standard but worth recording because it fixes the Banach-space hypotheses used below.

Regular Banach descent. Let $\mathcal{X}, \mathcal{Y}, \mathcal{J}, \mathcal{Z}$ be Banach spaces, $C : \mathcal{X} \to \mathcal{Y}$ a bounded linear surjection, $U \subset \mathcal{X}$, $V \subset \mathcal{J}$ open, and $F : U \times V \to \mathcal{X}$ continuously Fréchet differentiable. Assume fibre consistency for every $h \in V$. Then the unique map $\bar{F} : C(U) \times V \to \mathcal{Y}$ is C^1, $C(U)$ is open, and

$$\bar{F}(Cx, h) = CF(x, h), \qquad D_y\bar{F}(Cx, h)C = CD_xF(x, h), \qquad D_h\bar{F}(Cx, h) = CD_hF(x, h). \tag{2.3}$$

A C^1 observation $O : U \times V \to \mathcal{Z}$ descends with the same regularity if it is constant on each C-fibre at every h.

The quotient-norm proof is given in Appendix C. Bounded surjectivity makes $\operatorname{Ker} C$ closed and identifies $\mathcal{X}/\operatorname{Ker} C$ with $\mathcal{Y}$ by a bounded isomorphism. The quotient argument applies without a bounded linear section or a complemented kernel. Repeated iteration additionally requires the relevant invariant domains. A full-system fixed point projects to a coarse fixed point; the converse need not hold: $CF(x) = Cx$ permits $F(x) - x \neq 0$ in $\operatorname{Ker} C$.

A Gaussian heat map $\mathcal{H}_\tau$ on $L^2(\mathbb{R}^d)$ has the strictly positive Fourier multiplier $e^{-\tau|q|^2}$. It is injective, but its multiplier tends to zero at large $|q|$. Unit vectors supported at

successively higher frequencies therefore have images tending to zero; the inverse on the range is unbounded and that range is nonclosed for $\tau > 0$ in the inherited L^2 norm. It is not a surjection onto that L^2 space. Giving its range the norm $\|\mathcal{H}_\tau f\|_{\mathrm{tr}} = \|f\|_2$ makes it a Banach isomorphic copy of the original space, rather than an information-losing quotient. A genuine projection or observation quotient must still satisfy Eq. (2.2). Heat smoothing alone cannot repair nonlinear failure of that condition: on $[0, \pi]$, let $C = P_0$ be the constant-mode projection and $F(u) = H_t(u^2)$ on the finite cosine core. Then $C(a\cos nx) = C0 = 0$, whereas $CF(a\cos nx) = a^2/2 \neq CF(0)$ for $a \neq 0$, $n \geq 1$. These are function amplitudes, not probabilities.

2.3 Linear response of the sourced state equation

An applied source can change both the state and the observation at fixed state. These two contributions explain the direct term in the response. Likewise, eliminating a state can change how the source enters and how the remaining state is observed. Differentiating the update and observation together keeps all these contributions in the same calculation.

On a regular state space, specify an actual update $F(X, h; z)$ and observation $O(X, h; z)$ with finite-dimensional source and output spaces. The parameter z is an analytic response coordinate; η denotes an independent control parameter. Neither is intrinsically a momentum or a clock. The symbols B, S, R, D below are the four derivatives of this rule at a declared fixed-point branch $X_*(z)$ and $h = 0$:

$$B = D_X F, \qquad S = D_h F, \qquad R = D_X O, \qquad D = D_h O. \tag{2.4}$$

Here D is the direct source response, not a state-space operator. We write $\mathscr{C}$ for the measured response and reserve C in the general descent results for the coarse map. The QFT realisation later writes the measured matrix as $C(Q^2)$ and its kernel as $\mathcal{B}$.

Same-rule response. Suppose F, O are C^1 near the reference solution and $I - B$ is boundedly invertible. The local fixed-point branch is differentiable in h and

$$\mathscr{C} = D_h[O(X(h), h)]_{h=0} = D + R(I - B)^{-1}S. \tag{2.5}$$

More generally, if a C^1 residual equation $\mathcal{E}(X, h; z) = 0$ defines the branch and $L = D_X\mathcal{E}$ is boundedly invertible, then

$$\mathscr{C} = O_h - O_X L^{-1}\mathcal{E}_h. \tag{2.6}$$

The Banach implicit-function theorem gives the local branch. Differentiating its defining equation gives $(I - B)X_h = S$, or $LX_h = -\mathcal{E}_h$. Differentiating the observation gives $\mathscr{C} = D + RX_h$, proving both formulae.

The linearised equation $\delta X = B\delta X + S\delta h$ describes the source-induced change and its internal feedback. When $\|B\| < 1$, successive substitution converges in operator norm and gives

$$\mathscr{C} = D + \sum_{n=0}^{\infty} RB^n S = D + RS + RBS + RB^2S + \cdots . \tag{2.7}$$

Here S injects the disturbance, each B transmits it through the coupled state, and R reads out the result. The direct term D records the source dependence of the observation at fixed state. In a stationary problem these are repeated feedback contributions; they represent successive evolution steps only when the update has that physical meaning. Away from the norm-convergent regime, Eq. (2.5) still defines the response wherever the inverse exists. Eliminating variables must preserve the source and readout together with the internal feedback to preserve this sum.

For the residual convention put $S = -\mathcal{E}_h$. The displayed formula uses the residual derivatives and makes no choice of numerical iteration. Convergence of a solver and stability of the actual update are thus separate questions. The common origin of the four derivatives guarantees consistency, not reciprocity or positivity. Those properties will be verified in the examples or supplied by an independently established physical representation.

Theorem 2.2 (Preservation of the complete visible response)**.** *Suppose $F, O, \bar{F}, \bar{O}$ are jointly C^1 in state and source, and $CF(X,h) = \bar{F}(CX,h)$ and $O(X,h) = \bar{O}(CX,h)$ hold in compatible open neighbourhoods, with fixed bounded linear C and corresponding reference branches. If both $I - B$ and $I - \bar{B}$ are invertible, then*

$$CB = \bar{B}C, \quad \bar{S} = CS, \quad R = \bar{R}C, \quad D = \bar{D}, \qquad \mathscr{C} = \bar{\mathscr{C}}. \tag{2.8}$$

If both responses have meromorphic extensions to the same connected domain from a common nonempty open invertible set, their visible poles and output Laurent coefficients agree there.

Proof. Differentiate the two neighbourhood identities. Multiplying $(I - \bar{B})C = C(I - B)$ by the inverses gives $(I - \bar{B})^{-1}C = C(I - B)^{-1}$. Substitution in Eq. (2.5) gives equality; the identity theorem gives the meromorphic assertion. □

This theorem preserves the source-to-observation response, not every internal eigenvalue. The observation hypothesis is essential. For $B(z) = \mathrm{diag}(0, 1 - z)$, $S = (0,1)^{\mathsf{T}}$, $R = (0,1)$, and $C = (1,0)$, the update descends to $\bar{B} = 0$, but the full response is $1/z$. It cannot be recovered from the retained coordinate because the readout does not descend.

State- and source-dependent coarse maps. Let $F, O, C = C(X,h)$ and the descended maps $\bar{F}, \bar{O}$ be jointly C^1 in state and source on compatible open neighbourhoods of the reference states and source, with the following compositions defined. Suppose $C(F(X,h),h) = \bar{F}(C(X,h),h)$ and $O(X,h) = \bar{O}(C(X,h),h)$ hold near an actual sourced fixed point. Set $T = D_X C$ and $C_h = D_h C$ there. Then

$$\begin{aligned} TB &= \bar{B}T, \quad \bar{S} = TS + (I - \bar{B})C_h, \\ R &= \bar{R}T, \quad D = \bar{D} + \bar{R}C_h. \end{aligned} \tag{2.9}$$

On the common invertible domain the full responses still agree. At a fixed point the derivatives of C on both sides are evaluated at the same full state. The state derivative gives the first identity; the source derivative gives $TS + C_h = \bar{B}C_h + \bar{S}$. The observation

chain rule gives the second line. Inserting all four identities into Eq. (2.5) cancels the C_h terms.

These formulae concern fixed-source stationary response. Time-varying windows or sources require the actual adjacent-time coarse maps and their history. Discarding C_h would already change the static response.

This also explains the distinction between static elimination and dynamical reduction. In a stationary calculation an unresolved variable can adjust to the current source and appear as an effective stiffness. During an actual update it carries its own initial data and relaxation law; after it is removed, the retained variables remember the past through the induced kernel.

2.4 Gaussian operations and their representations

Gaussian averaging is one of the allowed operations when its spatial domain has been specified. It is distinct from the sufficient quotient. It changes the resolution with which a response is represented: rapidly varying components receive less weight. Interaction-induced collective feedback instead enters through the sourced state equation and its derivative. Keeping these roles explicit separates attenuation of an existing response from a change in the dynamics that produces it. The following Euclidean results distinguish averaging the external coordinates of a correlator from averaging its internal relative coordinates. For $\tau > 0$, define the normalised heat kernel and its action by

$$g_\tau(x) = \frac{1}{(4\pi\tau)^{d/2}} e^{-|x|^2/(4\tau)}, \qquad (\mathcal{H}_\tau f)(x) = (g_\tau * f)(x). \tag{2.10}$$

We use $\widehat{f}(q) = \int e^{\mathrm{i}q\cdot x} f(x)\, d^d x$, so that $\widehat{\partial_\mu f}(q) = -\mathrm{i}q_\mu \widehat{f}(q)$. In this convention,

$$\widehat{\mathcal{H}_\tau f}(q) = e^{-\tau|q|^2} \widehat{f}(q), \qquad \mathcal{H}_{\tau_1}\mathcal{H}_{\tau_2} = \mathcal{H}_{\tau_1+\tau_2}. \tag{2.11}$$

For a translationally invariant two-point function, smearing each insertion with $\mathcal{H}_{\tau_Q/2}$ gives the external representation

$$\mathcal{G}^{\mathrm{ext}}_{\tau_Q} C(Q^2) := e^{-\tau_Q Q^2} C(Q^2). \tag{2.12}$$

If a Gaussian is instead written as $\exp[-|x|^2/(2\sigma^2)]$, its Fourier multiplier is $\exp[-\sigma^2|q|^2/2]$, so that $\tau = \sigma^2/2$. Equivalently, the heat-kernel parameter used here is one half of the one-coordinate variance. This convention is stated once because one-insertion, two-insertion, and relative-coordinate averages otherwise differ by factors of two.

Composition across scales describes successive independent averaging steps: their displacements add, so their laws convolve. Centring and orthogonal invariance express the absence of drift and a preferred direction; finite nonzero variance sets a resolution length. We require the full displacement law to retain its shape when that length grows as the square root of the scale parameter. Within this composition class, the following proposition determines the Gaussian semigroup.

Proposition 2.3 (Conditional uniqueness of the Gaussian semigroup)**.** *Let $\{\mu_\tau\}_{\tau\geq 0}$ be probability measures on $\mathbb{R}^d$ satisfying: weak continuity with $\mu_0 = \delta_0$; the convolution law*

$\mu_{\tau+s} = \mu_\tau * \mu_s$; *centring; orthogonal invariance; a finite, nonzero second moment; and exact square-root self-similarity* $(D_{\sqrt{\tau}})_\# \mu_1 = \mu_\tau$. *Then there is a constant* $D > 0$ *such that*

$$\widehat{\mu}_\tau(q) = e^{-D\tau|q|^2}. \tag{2.13}$$

Thus the semigroup is Gaussian, up to the choice of the scale unit D.

The self-contained proof in Appendix C uses the characteristic semigroup and its quadratic homogeneity. For the stable-law background, see Chapter 3 of Ref. [10]; a related Gaussian derivation from different axioms appears in Section 2 of Ref. [9]. The conclusion belongs to this probabilistic Euclidean class. An FRG regulator is a separate scale-dependent modification of the inverse propagator.

There is a complementary entropy characterisation at fixed covariance. If a density f has the same mean and positive covariance as the Gaussian g, and the entropy integrals exist, then $\int f \log(f/g) \geq 0$ and $\int f \log g = \int g \log g$ imply $-\int f \log f \leq -\int g \log g$, with equality only for $f = g$ almost everywhere. This fixed-moment variational statement does not replace the composition and exact self-similarity assumptions of Proposition 2.3.

Theorem 2.4 (Tensor lift and momentum-routing form)**.** *Let* $F \in \mathcal{S}'((\mathbb{R}^d)^n)$ *and apply* $\mathcal{H}_{a_i\tau}$ *to its* i*th coordinate, where* $a_i > 0$. *In Fourier space the tensor-product action is*

$$\widehat{F}_\tau(p_1, \ldots, p_n) = \exp\left[-\tau \sum_{i=1}^{n} a_i |p_i|^2\right] \widehat{F}(p_1, \ldots, p_n). \tag{2.14}$$

Suppose, in addition, that translation invariance has been used to factor out the single overall momentum-conservation distribution. Let $\mathsf{E} = (\mathbb{R}^d)^n$, *define* $\Pi : \mathsf{E} \to \mathbb{R}^d$ *by* $\Pi(p_1, \ldots, p_n) = \sum_i p_i$, *and set* $r = d(n-1)$. *Choose linear maps*

$$A : \mathbb{R}^d \to \mathsf{E}, \qquad L : \mathbb{R}^r \to \mathsf{E}, \qquad \Pi A = I_d, \qquad \operatorname{Im} L = \operatorname{Ker} \Pi, \tag{2.15}$$

with L *injective. Then* $T(P, \boldsymbol{\xi}) = AP + L\boldsymbol{\xi}$ *is a linear coordinate isomorphism from* $\mathbb{R}^d \times \mathbb{R}^r$ *onto* E, *and* $L\boldsymbol{\xi}$ *parametrises the entire momentum-conserving subspace. Let* $\widehat{F}_{\text{full}}$ *denote the coordinate representation of* $\widehat{F}$ *under* T, *defined for test functions* Φ *by*

$$\langle \widehat{F}_{\text{full}}, \Phi \rangle := |\det T|^{-1} \langle \widehat{F}, \Phi \circ T^{-1} \rangle. \tag{2.16}$$

In these adapted variables assume

$$\widehat{F}_{\text{full}}(P, \boldsymbol{\xi}) = (2\pi)^d \delta^{(d)}(P) \widehat{F}_{\text{red}}(\boldsymbol{\xi}), \qquad \widehat{F}_{\text{red}} \in \mathcal{S}'(\mathbb{R}^r), \tag{2.17}$$

in the distributional sense that

$$\langle \widehat{F}_{\text{full}}, \Phi \rangle = (2\pi)^d \langle \widehat{F}_{\text{red}}, \Phi(0, \cdot) \rangle \quad \text{for every } \Phi \in \mathcal{S}(\mathbb{R}^d \times \mathbb{R}^r), \tag{2.18}$$

where $\delta^{(d)}$ *is normalised by* $\int \delta^{(d)}(P) f(P)\, dP = f(0)$. *This convention includes the Jacobian associated with the declared adapted coordinates; the equality then fixes the normalisation of* $\widehat{F}_{\text{red}}$. *The induced action on the reduced distribution is then*

$$\widehat{F}_{\text{red},\tau}(\boldsymbol{\xi}) = \exp[-\tau \boldsymbol{\xi}^{\mathsf{T}} M_L \boldsymbol{\xi}]\, \widehat{F}_{\text{red}}(\boldsymbol{\xi}), \qquad M_L = L^{\mathsf{T}} \operatorname{diag}(a_1 I_d, \ldots, a_n I_d) L \succeq 0. \tag{2.19}$$

If V *is a routing subspace on which* $M_L \succeq cI$ *for some* $c > 0$, *then the multiplier is at most* $e^{-\tau c q_0^2}$ *on* $\{\boldsymbol{\xi} \in V : \|\boldsymbol{\xi}\| \geq q_0\}$. *Directions in* $\operatorname{Ker} M_L$ *are not suppressed by this statement.*

The reduced-distribution hypothesis is the standard translation-invariant n-point convention: the overall momentum delta is removed before independent routing variables are introduced. Equation (2.19) does not assert the restriction of an arbitrary distribution to a linear subspace. Its test-function proof, including the normalisation in Eq. (2.18), is given in Appendix C.

For two equally weighted coordinates, the routing can be displayed without matrix notation. With

$$P_{\mathrm{c}} := p_1 + p_2, \qquad q_{\mathrm{r}} := p_1 - p_2, \qquad p_{1,2} = \frac{1}{2}(P_{\mathrm{c}} \pm q_{\mathrm{r}}), \tag{2.20}$$

one has the exact identity

$$|p_1|^2 + |p_2|^2 = \frac{1}{2}\big(|P_{\mathrm{c}}|^2 + |q_{\mathrm{r}}|^2\big). \tag{2.21}$$

Thus an equal-weight tensor Gaussian remains diagonal in centre and relative momentum, with multiplier $\exp[-a\tau(|P_{\mathrm{c}}|^2 + |q_{\mathrm{r}}|^2)/2]$. For unequal weights, the same substitution also produces the mixed term $(a_1 - a_2)P_{\mathrm{c}} \cdot q_{\mathrm{r}}/2$; it cannot be discarded unless the weights or the kinematics remove it.

For a two-point function with $p_1 = Q$, $p_2 = -Q$, and $a_1 = a_2 = 1/2$, Eq. (2.14) reduces to Eq. (2.12). For an internal transfer, Eq. (2.19) generally contains a different quadratic form and may contain mixed momentum terms. We denote that separately specified action by

$$\mathcal{G}^{\mathrm{rel}}_{\tau_q} A(\boldsymbol{\xi}) := e^{-\tau_q \boldsymbol{\xi}^{\mathsf{T}} M_L \boldsymbol{\xi}} A(\boldsymbol{\xi}). \tag{2.22}$$

Equation (2.22) for an internal loop momentum follows from the declared coordinate action and routing. These data must be supplied separately from any external smearing of the composite insertion.

2.5 Gaussian momentum-transfer criterion

Let $A(q)$ denote a projected contribution as a function or distribution of a specified set of transferred momenta. Once the coordinate action and the routing have supplied a positive-semidefinite matrix M, set

$$A_\tau(q) := \mathcal{G}^{\mathrm{rel}}_\tau A(q) = e^{-\tau q^{\mathsf{T}} M q} A(q). \tag{2.23}$$

This definition quantifies the Gaussian action on a specified contribution. The corresponding graph and the scale dependence of the unfiltered amplitude remain part of the effective dynamics.

Theorem 2.5 (Fixed-transfer suppression)**.** *For $\delta > 0$, let $\Omega_\delta = \{q : q^{\mathsf{T}} M q \geq \delta^2\}$. The following statements hold.*

(i) If $A \in L^\infty(\Omega_\delta)$, then

$$\|A_\tau\|_{L^\infty(\Omega_\delta)} \leq e^{-\tau\delta^2} \|A\|_{L^\infty(\Omega_\delta)} \xrightarrow[\tau\to\infty]{} 0. \tag{2.24}$$

(ii) If A is a tempered distribution and $\chi \in C_c^\infty(\mathbb{R}^d)$ has support in Ω_δ, then

$$\langle A_\tau, \chi \rangle = \langle A, e^{-\tau q^\mathsf{T} M q} \chi \rangle \xrightarrow[\tau\to\infty]{} 0. \tag{2.25}$$

(iii) Suppose a scale-dependent contribution has the form $A_\tau^{\mathrm{full}} = e^{-\tau q^\mathsf{T} M q} \widetilde{A}_\tau$ and

$$\|\widetilde{A}_\tau\|_{L^\infty(\Omega_\delta)} \le C e^{a\tau} \quad \textit{with} \quad a < \delta^2. \tag{2.26}$$

Then $A_\tau^{\mathrm{full}} \to 0$ in $L^\infty(\Omega_\delta)$.

The proof is given in Appendix C. The L^∞ bounds use essential suprema. For bounded continuous representatives they also give ordinary uniform bounds on the indicated region. Part (ii) is the appropriate formulation for amplitudes containing threshold distributions or other non-function singularities. Part (iii) records the additional hypothesis required when the amplitude itself runs with the averaging scale. Without such a bound, growth of $\widetilde{A}_\tau$ can offset the Gaussian multiplier. The convergence is not uniform in a shrinking neighbourhood of the origin. Indeed, for $q = u/\sqrt{\tau}$,

$$e^{-\tau q^\mathsf{T} M q} = e^{-u^\mathsf{T} M u}, \tag{2.27}$$

so the soft scaling region, together with $\operatorname{Ker} M$, remains in the problem. Its integrated weight also depends on the volume of that region. For example, if $M \succ 0$ and A is bounded and continuous at the origin, the substitution $q = u/\sqrt{\tau}$ gives

$$\begin{aligned} I_\tau &:= \int_{\mathbb{R}^d} e^{-\tau q^\mathsf{T} M q} A(q)\, d^d q \\ &= \tau^{-d/2} \int_{\mathbb{R}^d} e^{-u^\mathsf{T} M u} A(u/\sqrt{\tau})\, d^d u, \\ \lim_{\tau\to\infty} \tau^{d/2} I_\tau &= \frac{\pi^{d/2}}{\sqrt{\det M}} A(0). \end{aligned} \tag{2.28}$$

The Gaussian supplies an integrable majorant, so the last step follows by dominated convergence. A regular soft amplitude thus carries a phase-volume factor $\tau^{-d/2}$ even though the multiplier equals one at zero. Singular amplitudes and undamped directions in $\operatorname{Ker} M$ retain their own scaling and integration measures. Integrating the damped variables into an effective action can also leave local operators, derivative corrections, and running couplings in the retained variables.

The multiplier has a simple physical interpretation: it is a dephasing filter in the transferred momentum. Contributions with a resolved nonzero transfer acquire rapidly varying phases under spatial averaging, whereas back-to-back internal momenta with zero net transfer do not dephase. This is a kinematic retention statement. The size and topology of the contribution that survives in the soft region are still determined by the response dynamics.

Momentum routing specifies which combinations of momenta are damped; graph topology specifies how interactions connect the internal lines. Bubbles, crossed ladders, and vertex corrections can all contribute in the soft region. Their relative weights follow from the interaction vertices and propagators of the response equation.

Preservation of an isolated pole. Let a projected Euclidean form factor be meromorphic near $Q^2 = 0$ and let $w(Q^2)$ be analytic there with $w(0) = 1$ and no zero in a neighbourhood of the origin. Then multiplication by w preserves both the order and residue of a possible simple pole at $Q^2 = 0$ and cannot create such a pole from a regular form factor. In particular, $w(Q^2) = e^{-\tau_Q Q^2}$ preserves

$$Z_0 = \lim_{Q^2 \to 0} Q^2 C(Q^2). \tag{2.29}$$

This statement concerns $\mathcal{G}^{\mathrm{ext}}_{\tau_Q}$ acting on a fixed correlator. Interactions integrated along an RG flow can change the correlator through the effective action; that dynamical effect is addressed by the kernel criterion below. The positive spectral measure continues to be defined from the original Wightman correlator. In particular, multiplication turns a contact polynomial into an analytic, non-polynomial short-range term. The physical spectral decomposition and its contact subtraction are therefore defined before the diagnostic filter is applied.

2.6 An additive-source realisation

The response operator should be identified from the linearisation of the self-consistent dynamics rather than from the topology of a selected set of graphs. An additive-source specialisation of Eq. (2.5) uses a Banach response space $\mathcal{X}_{\mathfrak{C}}$ and maps $\mathsf{S}_X, \mathsf{R}_X$ of the types in Eq. (2.4). Let $\mathfrak{F}_{\tau_q}(\,\cdot\,; z, \eta) : \mathcal{X}_{\mathfrak{C}} \to \mathcal{X}_{\mathfrak{C}}$ and consider, near a reference solution, the sourced equation

$$X = \mathfrak{F}_{\tau_q}(X; z, \eta) + \mathsf{S}_{X,\tau_Q}(z) J. \tag{2.30}$$

Here $J \in \mathbb{C}^m$, while $\mathsf{S}_{X,\tau_Q} : \mathbb{C}^m \to \mathcal{X}_{\mathfrak{C}}$ and $\mathsf{R}_{X,\tau_Q} : \mathcal{X}_{\mathfrak{C}} \to \mathbb{C}^m$ may include external Gaussian factors. The relative scale τ_q, if present, belongs to the self-consistent map itself. The analytic term C_{reg} is understood in the same external convention as the source and readout.

For a local solution branch $X(J; z, \eta)$, its measured output is a map into the external operator coordinates,

$$\begin{aligned} \mathcal{Y}(J; z, \eta) &= \mathcal{Y}_{\mathrm{reg}}(J; z, \eta) \\ &\quad + \mathsf{R}_{X,\tau_Q}(z) X(J; z, \eta), \\ C_{\tau_Q,\tau_q} &:= D_J \mathcal{Y}\big|_{J=0}, \qquad\qquad C_{\mathrm{reg}} := D_J \mathcal{Y}_{\mathrm{reg}}\big|_{J=0}. \end{aligned} \tag{2.31}$$

Thus the correlator or susceptibility is the Jacobian from the declared physical source coordinates to the declared readout coordinates. Both maps are needed: a critical response direction contributes only when it is excited by the source and detected by the readout. When the sources couple to physical insertions and the output is their normalised expectation value, differentiation gives the connected two-point matrix, with the specified source-pairing convention and local contact terms. This source derivative fixes the physical meaning of every factor: the source perturbs the state equation, the state adjusts, and the output records that adjustment together with its direct source dependence.

Proposition 2.6 (Response factorisation)**.** *Suppose $\mathfrak{F}_{\tau_q}$ is continuously Fréchet differentiable in X on a neighbourhood of a solution X_* of Eq. (2.30) with $J = 0$, and suppose $\mathcal{Y}_{\mathrm{reg}}$ is Fréchet differentiable in J at the origin. Set*

$$\mathcal{B}_{\tau_q}(z;\eta) := D_X \mathfrak{F}_{\tau_q}(X_*; z, \eta). \tag{2.32}$$

If $I - \mathcal{B}_{\tau_q}$ is invertible, the Banach-space implicit-function theorem gives a differentiable local solution branch, and its measured response has the form

$$C_{\tau_Q,\tau_q}(z;\eta) = C_{\mathrm{reg}}(z;\eta) + \mathsf{R}_{X,\tau_Q}(z)[I - \mathcal{B}_{\tau_q}(z;\eta)]^{-1}\mathsf{S}_{X,\tau_Q}(z). \tag{2.33}$$

At a critical point the same identity holds as a meromorphic continuation whenever the Fredholm hypotheses below are satisfied.

Thus an external Gaussian changes the source and readout maps, but not the spectrum of the fixed response operator in Eq. (2.32). For a source–source matrix with the same scalar filter on both insertions, one may write on the Euclidean axis

$$\begin{aligned} \mathsf{S}_{X,\tau_Q}(z) &= \mathsf{S}_{X,0}(z) W_{\tau_Q}(z), \\ \mathsf{R}_{X,\tau_Q}(z) &= W_{\tau_Q}(z) \mathsf{R}_{X,0}(z), \\ C_{\mathrm{reg},\tau_Q}(z) &= W_{\tau_Q}(z) C_{\mathrm{reg},0}(z) W_{\tau_Q}(z), \\ W_{\tau_Q}(z) &= e^{-\tau_Q z/2} I_m. \end{aligned} \tag{2.34}$$

Since $W_{\tau_Q}(0) = I_m$, this factorisation also makes the preservation of an existing residue manifest. Internal averaging affects that spectrum through a specified τ_q-dependence of $\mathfrak{F}_{\tau_q}$. This dependence must be derived from the declared Gaussian action.

3 Channels, elimination, and visible critical structure

The sourced state equation of Section 2 supplies a response space and the four maps B, S, R, D. A channel is a specified source-to-observation problem on this space. A bounded projector P_X selects a retained subspace and $Q_X = I_X - P_X$ its complement. The retained sector is closed only when the update does not send it into the complement. Here $\mathcal{B} = B$, $\mathsf{S}_X = S$, $\mathsf{R}_X = R$, and $C_{\mathrm{reg}} = D$ retain the notation also used in the QFT realisation below.

3.1 Reference-state symmetry and exact elimination

The relevant symmetry is the symmetry of the reference state as well as of the update. A transformation that moves the reference state relates two different linear-response problems; it need not split either one into invariant channels.

Equivariant linearisation at the reference state. Let a group act by bounded linear isomorphisms U_g on the state space, and suppose $F(U_g X) = U_g F(X)$. At X_*,

$$DF(U_g X_*) U_g = U_g DF(X_*). \tag{3.1}$$

Consequently $DF(X_*)$ commutes with U_g for the stabilizer $G_{X_*} = \{g : U_g X_* = X_*\}$. The same fixed-background restriction applies to source and observation intertwiners when their equivariance holds. Differentiate the equivariance identity with respect to X. For $g \in G_{X_*}$ the two Jacobians have the same base point. Differentiating the sourced equivariance and observation identities gives their intertwining relations. A broken symmetry instead relates different backgrounds and need not block-diagonalise one of their Jacobians.

Symmetry provides invariant sectors only after these fixed-background conditions and the relevant bounded projectors have been established. It does not supply the nonzero source and readout overlaps required below. Exact finite systems and finite approximations to infinite systems both admit block elimination, but only the latter need cutoff convergence to identify an infinite target.

Proposition 3.1 (Response-closed channel)**.** *The spectrum of the restriction $\mathcal{B}|_{\operatorname{Ran} P_X}$ is represented by $P_X \mathcal{B} P_X$ only if*

$$Q_X \mathcal{B} P_X = 0. \tag{3.2}$$

If Eq. (3.2) fails and $I_X - Q_X \mathcal{B} Q_X$ is invertible in the domain under consideration, elimination of the complementary sector gives the exact effective response operator

$$\mathcal{B}_{\text{eff}} = P_X \mathcal{B} P_X + P_X \mathcal{B} Q_X (I_X - Q_X \mathcal{B} Q_X)^{-1} Q_X \mathcal{B} P_X. \tag{3.3}$$

Writing $D_Q := I_X - Q_X \mathcal{B} Q_X$, the associated source and readout maps are

$$\mathsf{S}_{X,\text{eff}} = P_X \mathsf{S}_X + P_X \mathcal{B} Q_X D_Q^{-1} Q_X \mathsf{S}_X, \tag{3.4}$$

$$\mathsf{R}_{X,\text{eff}} = \mathsf{R}_X P_X + \mathsf{R}_X Q_X D_Q^{-1} Q_X \mathcal{B} P_X. \tag{3.5}$$

If the unreduced response contains the analytic term C_{reg} of Eq. (2.33), its reduced value is

$$C_{\text{reg,eff}} = C_{\text{reg}} + \mathsf{R}_X Q_X D_Q^{-1} Q_X \mathsf{S}_X. \tag{3.6}$$

The full response is therefore $C_{\text{reg,eff}} + \mathsf{R}_{X,\text{eff}} (I_X - \mathcal{B}_{\text{eff}})^{-1} \mathsf{S}_{X,\text{eff}}$. Consequently, the spectrum of $P_X \mathcal{B} P_X$ alone has no channel interpretation when the omitted sector remains coupled. If $\mathcal{B}(z)$ is a compact analytic family and $D_Q(z)^{-1}$ exists throughout a neighbourhood, then $\mathcal{B}_{\text{eff}}(z)$ is a compact analytic family on $\operatorname{Ran} P_X$. On a connected neighbourhood where $I_X - \mathcal{B}_{\text{eff}}$ is invertible at one point, analytic Fredholm theory gives a meromorphic inverse. With analytic source, readout and background maps, and no continuum threshold in that neighbourhood, the effective response is meromorphic as well.

Proof. Write $I - \mathcal{B}$ in $P \oplus Q$ block form and solve the Q equation before the P equation. This gives Eqs. (3.3)– (3.6). Compactness of $\mathcal{B}_{\text{eff}}$ follows because bounded left and right multiplication preserve compactness, while analyticity follows from analyticity of the inverse map on the open set of bounded invertible operators. □

This is the Feshbach–Schur block elimination adapted to response maps between the stated spaces [3]. A channel that fails Eq. (3.2) can be enlarged to include the coupled states, or described by the energy-dependent operator Eq. (3.3). The second term accounts

for excursions into the eliminated sector and their return to the retained one; the source, readout and direct terms account for the other ways that sector contributes to the observation. Thus a retained channel can acquire collective feedback through other channels even when its restricted update is noncritical. Whether the feedback reaches criticality is determined by the full effective operator, including these excursions.

If the source lies in the retained sector and the readout annihilates its complement, then $Q_X \mathsf{S}_X = 0$ and $\mathsf{R}_X Q_X = 0$. The general block formulae then reduce to

$$\mathsf{S}_{X,\mathrm{eff}} = \mathsf{S}_X, \qquad \mathsf{R}_{X,\mathrm{eff}} = \mathsf{R}_X, \qquad C_{\mathrm{reg,eff}} = C_{\mathrm{reg}}. \tag{3.7}$$

These simplifications require both compatibility identities. A general source and observation can couple to Q_X, in which case all four unsimplified terms must be retained.

For differentiation with respect to a parameter a, the inverse identity is $\partial_a D_Q^{-1} = D_Q^{-1} Q_X (\partial_a \mathcal{B}) Q_X D_Q^{-1}$ for fixed projectors. The product rule then contributes derivatives of both off-diagonal blocks, sources, readouts, and direct term. Derivatives along a reference branch include its state displacement. A moving projector additionally contributes its own derivatives. Appendix C records the fixed-projector formula. Frequency dependence generated by this elimination can represent memory, not a memoryless one-step update.

An eliminated dynamical variable with independent initial data generally leaves both memory and an initial-state contribution. The situation is different for a local auxiliary field: if its algebraic equation has an invertible derivative with respect to that variable, the implicit-function theorem determines it locally from the retained fields, their derivatives and sources, without an independent initial history. Substitution into the common quadratic action gives the corresponding Schur correction. The auxiliary collective radius in the companion application Ref. [15], Appendix G.6, Proposition G.7 and Lemma G.8, provides a model-specific realisation along a coupled FLRW solution. In that realisation the regular radius constraint permits elimination; positivity of physical propagation additionally uses elimination of the gravitational constraints and analysis of the full principal part. The response Schur identity alone supplies no such stability conclusion, and the evolving background is not a stationary fixed point of the response equation considered here.

3.2 Isolated Fredholm response

Near an isolated critical mode, an internally transmitted disturbance nearly reproduces itself. An eigenvalue approaching one measures this feedback in the linearised equation. For it to appear in an observation, the source must excite the corresponding direction and the readout must detect it. The next result separates feedback criticality from visibility; the reciprocal and spectral realisations then determine the sign and physical interpretation of the residue.

The operator $I - \mathcal{B}$ is the restoring operator of the coupled channel, and its inverse is the corresponding susceptibility. When an eigenvalue of $\mathcal{B}$ reaches one, one collective direction reproduces its disturbance after a complete feedback cycle, so the restoring coefficient along that direction vanishes. The condition $\partial_z \lambda \neq 0$ specifies how this softening is crossed in

the declared response coordinate; the source and readout overlaps then decide whether the softened direction is visible.

Theorem 3.2 (Kernel criterion for an isolated algebraic pole)**.** *Fix $\eta = \eta_c$ and let U be a connected neighbourhood of $z = 0$. Suppose that $\mathcal{B}(z; \eta_c)$ is a compact analytic operator family on U, that $I - \mathcal{B}(z; \eta_c)$ is invertible at least at one point of U, and that U contains no continuum threshold or other non-meromorphic singularity. Suppose also that the response-space source $\mathsf{S}_X(z)$, readout $\mathsf{R}_X(z)$, and matrix background $C_{\mathrm{reg}}(z)$ in Eq. (2.33) are analytic on U. Suppress the fixed parameter η_c and consider that matrix response. Then:*

(i) if $1 \notin \operatorname{spec} \mathcal{B}(0)$, the response has no pole at $z = 0$ generated by this kernel;

(ii) if $\mathcal{B}(z)$ has an isolated algebraically simple eigenvalue $\lambda(z)$ with

$$\lambda(0) = 1, \qquad \partial_z \lambda(0) \neq 0, \tag{3.8}$$

then the resolvent has a simple pole. The measured response contains that pole precisely when both $\mathsf{R}_X(0)|r\rangle \neq 0$ and $\langle \ell | \mathsf{S}_X(0) \neq 0$;

(iii) with $\langle \ell | r \rangle = 1$, its algebraic residue matrix is

$$Z_{\mathrm{alg}} = -\frac{\mathsf{R}_X(0)|r\rangle\langle \ell|\mathsf{S}_X(0)}{\partial_z \lambda(0)}. \tag{3.9}$$

This formula is an algebraic residue. Its sign is not fixed by these hypotheses, and its interpretation as a Lorentzian mass atom requires matching to a positive physical spectral representation.

The proof in Appendix C is an application of the analytic Fredholm theorem and the Riesz projection onto the critical eigenspace. On the simple branch, choose local analytic eigenvectors with $\langle \ell(z) | r(z) \rangle = 1$. The singular contribution is

$$\frac{\mathsf{R}_X(z)|r(z)\rangle\langle \ell(z)|\mathsf{S}_X(z)}{1 - \lambda(z)}.$$

The two numerator factors measure excitation and detection of the collective direction. Spectral isolation permits the response to be resolved into this collective mode and an analytic complementary part. The bounded complementary inverse controls the response of the remaining directions while the critical mode becomes soft. Expanding $1 - \lambda(z) = -\lambda'(0)z + O(z^2)$ gives the residue in Eq. (3.9). Its magnitude therefore depends both on the coupling to the mode and on how rapidly its inverse response vanishes. This is a local statement in z; its interpretation as a frequency or momentum singularity uses the specified physical realisation. Condition (3.8) is transversality in the declared complex response coordinate z. For a real eigenvalue branch along a real control parameter, $\partial_\eta \lambda(0; \eta_c) \neq 0$ certifies a first-order transverse crossing of one. It is not necessary for every crossing: $\lambda(z; \eta) = 1 - z + \eta^3$ crosses at $\eta = 0$ despite a vanishing first control derivative. Higher-order crossings require their own analysis and do not change the simple-pole criterion in z.

When $\eta = k$ is an RG scale, a crossing at intermediate k is a flow diagnostic; a physical-state claim requires the pole conditions for the exact endpoint response. If a continuum threshold starts at $z = 0$, the meromorphic-neighbourhood hypothesis of Theorem 3.2 fails in general. The channel is then labelled *threshold-sensitive* and requires a limiting-absorption or explicit spectral analysis.

3.3 Reciprocal response and a finite-rank positive residue

The Fredholm result accommodates nonnormal kernels and independent source and readout maps. A reciprocal Hilbert-space realisation permits an additional positive statement before any Lorentzian representation. The singular part is determined on the critical subspace: the invertible complement contributes only analytic terms, while the derivative of the response operator fixes the sign of the residue. Source overlap then determines which critical directions are visible.

Theorem 3.3 (Positive residue for reciprocal response)**.** *Let $L(z)$ be analytic in operator norm near zero on a Hilbert space $\mathcal{H}$. Suppose $L(0)$ is self-adjoint, its kernel is nonzero and finite dimensional with orthogonal projector P, and $QL(0)Q|_{Q\mathcal{H}}$ has a bounded inverse, where $Q = I - P$. If*

$$G_0 = PL'(0)P|_{P\mathcal{H}} \succ 0, \tag{3.10}$$

then, on a sufficiently small punctured neighbourhood,

$$L(z)^{-1} = \frac{PG_0^{-1}P}{z} + A(z), \qquad A \text{ analytic at } 0. \tag{3.11}$$

Let $S(z) : \mathbb{C}^m \to \mathcal{H}$, $R(z) : \mathcal{H} \to \mathbb{C}^m$ and $D(z)$ be analytic, with $R(0) = S(0)^*$ in the declared source–output pairing. The response $D + RL^{-1}S$ has residue

$$Z_{\text{native}} = S(0)^* PG_0^{-1} PS(0) \succeq 0, \qquad \operatorname{rank} Z_{\text{native}} = \operatorname{rank}(PS(0)). \tag{3.12}$$

A pole is visible precisely when $PS(0) \neq 0$.

Proof. Self-adjointness makes both off-diagonal blocks of $L(0)$ zero. The inverse of the Q block persists analytically. The Schur complement on $P\mathcal{H}$ is $zG_0 + O(z^2)$, since the product of the off-diagonal blocks is $O(z^2)$. Its inverse is $z^{-1}G_0^{-1}$ plus an analytic remainder. Block inversion proves Eq. (3.11); the off-diagonal factors cancel the possible z^{-1} singularity outside the P block. Taking the residue gives Eq. (3.12). Its factorisation as $(G_0^{-1/2}PS(0))^*(G_0^{-1/2}PS(0))$ proves positivity and rank. □

For a source direction j, the residue obeys $j^* Z_{\text{native}} j = \|G_0^{-1/2} PS(0) j\|^2$. It is positive exactly when that source excites the critical subspace. Reciprocity makes readout the adjoint of excitation in the same pairing; the two mode overlaps therefore combine into a squared norm. The positive crossing matrix supplies the weight with which that norm enters the pole. Dynamical stability is determined separately by the actual evolution; the sign of an interaction energy also depends on how physical sources couple to this response.

This theorem requires neither a compact B nor a Lorentzian vacuum, but it does require the stated Hilbert pairing, reciprocity, isolated critical subspace, and actual positive

crossing matrix. For $L = I - \mathcal{B}$ the crossing is $-P\mathcal{B}'(0)P$, not a derivative in an unrelated control parameter. A one-dimensional kernel alone is insufficient: $\mathrm{diag}(0, 1, 1/2, \ldots)$ has no bounded complementary inverse and adding zI produces singularities accumulating at zero. Unbounded operators require a separately verified common-domain or closed-form analytic realisation; this bounded theorem does not assert one. No relation $R = S^*$ is imposed on a genuinely nonreciprocal update.

For a finite-dimensional real affine response $L(z) = H + zK$ under the hypotheses of the theorem, with $H = H^{\mathsf{T}}$ and reciprocal source pairing, a basis of the critical space gives a direct residue formula. For rational matrices this basis can be obtained by exact elimination. If N is a full kernel basis of H and $N^{\mathsf{T}}KN \succ 0$, the residue is

$$Z = S^{\mathsf{T}}N(N^{\mathsf{T}}KN)^{-1}N^{\mathsf{T}}S. \tag{3.13}$$

To see this without orthogonalising N, complete it to an invertible basis and apply the same Schur argument by congruence. This expression is invariant under $N \mapsto NT$ for invertible T and has rank $\mathrm{rank}(N^{\mathsf{T}}S)$. The software of Ref. [16] provides supplementary rational finite-matrix evaluations of this formula. A small positive eigenvalue instead gives a regular response at zero, since $\lim_{z\downarrow 0} z/(\epsilon+z) = 0$ for every $\epsilon > 0$. The residue here refers to the specified analytic parameter z. An interpretation as a spatial propagation pole or a vacuum spectral atom additionally requires a representation of that parameter and of the same source-contracted response. Iteration stability is determined by the update spectrum, independently of the sign of this residue.

3.4 Quantum-field-theory realisation

For the local-operator realisation, work with a local, unitary, Poincaré-invariant quantum field theory admitting a Euclidean continuation for the physical correlation functions considered below. Definition A.1 also specifies the directly reduced helicity realisation, whose physical Hilbert space and source covariance enter the same response factorisation. The invariant-mass spectral assertions use a Poincaré-invariant positive-energy vacuum, unique in the chosen physical representation, as specified in Appendix A. Whenever a response is identified with a physical two-point function, its insertions are centred, $\Theta_i = O_i - \langle O_i \rangle I$, and its correlation matrix is connected. Subtracting these one-point products precedes, and is distinct from, the subtraction of local contact terms. Gauge theories are understood in a renormalised BRST formulation without an uncancelled gauge anomaly. Let $\mathcal{V}_{\mathrm{phys}}$ be the renormalised operator space after passage to the physical quotient described in Appendix B. The quotient is algebraic. There is no canonical Hilbert norm on a space of local composite operators, so every infinite-dimensional boundedness statement below is made relative to a specified channel norm for which $\mathcal{V}_{\mathrm{phys}}$ has been completed to a Banach space and all stated maps extend continuously; we use the same symbol for that completion. Thus "bounded" always refers to this declared channel norm, not to an implicit canonical norm on local operators. A bounded projector $\mathcal{P}_{\mathfrak{C}}$ selects a Lorentz and internal-symmetry sector, with the closed Banach subspace $\mathcal{H}_{\mathfrak{C}} = \mathrm{Ran}\,\mathcal{P}_{\mathfrak{C}}$. For a finite operator block the choice of norm is immaterial because all norms are equivalent. A practical calculation may later

replace this exact space by a finite block; that replacement is part of the approximation, not part of the definition of the physical channel.

The operator space specifies which insertions are probed; the response space carries the variables used to solve their sourced equations. The realisation maps connect these descriptions, while the regulator and truncation data record how a concrete calculation is performed.

Definition 3.4 (Physical channel)**.** *A physical operator channel is the tuple*

$$\mathfrak{C}_{\mathrm{phys}} = \big(\mathsf{T}, |\Omega\rangle, \mathcal{V}_{\mathrm{phys}}, \mathcal{P}_{\mathfrak{C}}, \mathcal{S}, \mathcal{E}, \mathsf{S}_{\mathcal{O}}, \mathsf{R}_{\mathcal{O}}\big), \tag{3.14}$$

where T *is the theory,* $|\Omega\rangle$ *the state or vacuum,* $\mathcal{S}$ *the exact symmetry and anomaly data, and* $\mathcal{E}$ *the external kinematics. For a general state the response variables must respect its unbroken symmetries; the vacuum identification* $z = Q^2$ *and the invariant-mass atom test do not extend to an ensemble without a separate spectral argument. The bounded maps*

$$\mathsf{S}_{\mathcal{O}}(z) : \mathbb{C}^m \longrightarrow \mathcal{H}_{\mathfrak{C}}, \qquad \mathsf{R}_{\mathcal{O}}(z) : \mathcal{H}_{\mathfrak{C}} \longrightarrow \mathbb{C}^m \tag{3.15}$$

specify the physical operator insertions and their output coordinates. To use these insertions in a response equation, one must also specify how they are realised on the response space. Let $\mathcal{X}_{\mathfrak{C}}$ *be a Banach response space, let* $P_X \in \mathcal{L}(\mathcal{X}_{\mathfrak{C}})$ *be a bounded projector, and let*

$$\iota_{\mathfrak{C}}(z) : \mathcal{H}_{\mathfrak{C}} \longrightarrow \mathcal{X}_{\mathfrak{C}}, \qquad \pi_{\mathfrak{C}}(z) : \mathcal{X}_{\mathfrak{C}} \longrightarrow \mathcal{H}_{\mathfrak{C}} \tag{3.16}$$

be bounded source-realisation and readout-realisation maps. They need not be mutual inverses. Compatibility with the selected response block means

$$P_X \iota_{\mathfrak{C}} = \iota_{\mathfrak{C}}, \qquad \pi_{\mathfrak{C}} P_X = \pi_{\mathfrak{C}}. \tag{3.17}$$

The maps that enter a response equation are then

$$\mathsf{S}_X := \iota_{\mathfrak{C}} \mathsf{S}_{\mathcal{O}} : \mathbb{C}^m \longrightarrow \mathcal{X}_{\mathfrak{C}}, \qquad \mathsf{R}_X := \mathsf{R}_{\mathcal{O}} \pi_{\mathfrak{C}} : \mathcal{X}_{\mathfrak{C}} \longrightarrow \mathbb{C}^m. \tag{3.18}$$

Their explicit construction is part of the physical model, not a choice of notation. In particular, a critical response eigenspace can fail to couple to either map.

The operator part of the tuple is an exact channel *when* $\operatorname{Ran} \mathcal{P}_{\mathfrak{C}}$ *is stable under the renormalisation map and the unbroken symmetries recorded in* $\mathcal{S}$*. A finite-dimensional projected space* $P_N \mathcal{V}_{\mathrm{phys}}$ *is a* computational block*. It is exact only when closure under renormalisation, symmetry, and response has been proved; otherwise conclusions drawn from it carry the truncation label. The classification applies to the full block and its correlation matrix, not to a preferred basis vector. Here boundedness is taken in the Banach completion and channel norm declared as part of the response data below. The algebraic BRST/EOM quotient precedes this completion; choosing a completion does not alter which representatives define physical cohomology classes.*

For later reference, the response and approximation data are recorded as

$$\mathfrak{I} = \big(\mathcal{X}_{\mathfrak{C}}, P_X, \iota_{\mathfrak{C}}, \pi_{\mathfrak{C}}, \mathcal{H}_\tau, \mathcal{G}^{\mathrm{ext}}_{\tau_Q}, \mathcal{G}^{\mathrm{rel}}_{\tau_q}, \mathfrak{F}_{\tau_q}, \mathcal{B}(z;\eta), \mathcal{R}_{\mathrm{ren}}, \mathcal{T}_N\big), \tag{3.19}$$

where $\mathcal{H}_\tau$ is the one-coordinate heat semigroup, $\mathcal{G}^{\mathrm{ext}}$ and $\mathcal{G}^{\mathrm{rel}}$ are the two induced representations of Section 2, $\mathfrak{F}_{\tau_q}$ is a self-consistent response map and $\mathcal{B}$ its linearisation, $\mathcal{R}_{\mathrm{ren}}$ is the renormalisation and regulator prescription, and $\mathcal{T}_N$ is a truncation when an infinite target is approximated, with $N = \infty$ denoting that exact hierarchy. A finite system defined by an exact operation protocol has no such limiting requirement.

The following variables will not be interchanged. The invariant $s \geq 0$ labels Lorentzian spectral mass. The variable $Q^2 \geq 0$ is the squared external Euclidean momentum of a projected source–source correlator. The scale k labels a Wilsonian or functional RG trajectory. Within a diagram or kernel, $q \in \mathbb{R}^d$ denotes the Euclidean momentum transferred between a fast subgraph and the external channel. In this QFT realisation, z denotes the complexified external invariant: our convention is $z = Q^2$ on the Euclidean axis and $z = -p^2 - \mathrm{i}0$ at the Lorentzian physical boundary. Finally, η denotes a control parameter such as a coupling, background, or RG scale. A derivative with respect to z is therefore not a flow derivative with respect to η or k.

3.5 Operator blocks and the physical quotient

Begin with the renormalised local operators of fixed Lorentz representation, internal quantum numbers, and ghost number. Their exact completion can be infinite dimensional. A finite set selected by a canonical-dimension bound or a numerical basis is denoted $\mathcal{V}_N$ and is treated as a truncation unless a separate closure argument applies. Renormalisation takes the matrix form

$$[\mathcal{O}_i]_R(\mu) = Z_{ij}(\mu, \mathcal{R})\mathcal{O}_j^{\mathrm{bare}}. \tag{3.20}$$

In a gauge theory this set may contain gauge-invariant representatives, BRST-exact operators, equation-of-motion operators, and total derivatives. The triangular renormalisation system retains these directions before passage to physical matrix elements [21].

Let $\overline{\mathcal{V}} = \mathcal{V}/\mathcal{I}_{\mathrm{EOM}}$ be the on-shell local operator space and assume that the renormalised BRST differential s_{B} descends to it. For physical separated-point matrix elements, the relevant ghost-number-zero cohomology is

$$\mathcal{V}_{\mathrm{phys}} = H^0(\mathsf{s}_{\mathrm{B}}, \overline{\mathcal{V}}) = \frac{\mathrm{Ker}\Big(\mathsf{s}_{\mathrm{B}} : \overline{\mathcal{V}}^0 \longrightarrow \overline{\mathcal{V}}^1\Big)}{\mathrm{Ran}\Big(\mathsf{s}_{\mathrm{B}} : \overline{\mathcal{V}}^{-1} \longrightarrow \overline{\mathcal{V}}^0\Big)}, \tag{3.21}$$

with contact terms retained separately. The cohomological treatment of local operator renormalisation is developed in Section 8.6 of Ref. [22]. The on-shell gluonic matrix elements at zero momentum transfer studied in Ref. [23] illustrate the infrared care needed when taking exceptional limits. Here $\mathcal{I}_{\mathrm{EOM}}$ is represented quantum mechanically by the renormalised field-redefinition insertions $\mathcal{E}_F^R$ of Eq. (B.3), including the measure divergence and operator-mixing counterterms. Total derivatives require separate treatment. They are redundant for a constant-source integrated insertion, or at zero insertion momentum, only when the boundary term vanishes and the indicated limit is well defined. For a nonconstant source, integration by parts retains a source-gradient insertion; at nonzero insertion momentum the descendants must remain in the mixing block. Equation (3.21) is the local

cohomological construction reviewed in Appendix B; it is not used to discard anomalous, descendant, or contact contributions.

For non-gauge theories the BRST quotient is absent, but operator mixing and EOM redundancies remain. For the scalar improvement $T_{\mu\nu} \mapsto T_{\mu\nu} + (\partial_\mu\partial_\nu - \eta_{\mu\nu}\Box)X$, with X a renormalised Lorentz scalar, the separated-point TT projection is unchanged, whereas the spin-zero response and the poles visible in an individual representative can change. If the improvement descendant is retained in a closed operator block, a finite nonsingular constant basis change preserves the pole structure of the full correlation matrix and the rank of each isolated residue: the full matrix and its residue transform by $C \mapsto MCM^\dagger$ with M constant and nonsingular.

This definition accommodates mixing between scalar operators such as mass terms, interaction densities, and total derivatives, and between tensor operators with the same physical spin. It also prevents unlike objects from being entered into a common table. A running coupling is a coordinate on theory space; a local composite operator is an insertion. They are related through source derivatives and beta functions but are not the same classification object.

Renormalisation closure does not by itself imply closure under the dynamical response. The operator projector $\mathcal{P}_{\mathfrak{C}}$ acts on $\mathcal{V}_{\text{phys}}$, whereas the compatible response projector P_X from Eq. (3.17) acts on $\mathcal{X}_{\mathfrak{C}}$. Put $Q_X = I_X - P_X$; the two projectors are related by the realisation maps but are not identified.

3.6 Symmetry-allowance criterion

The original QED Ward identity and its Ward–Takahashi generalisation relate the fermion propagator and current vertex [24, 25]. More generally, a symmetry identity relates current insertions to variations of the other fields. For a current J_A^μ and renormalised insertions Φ_r, the position-space identity, allowing for an anomaly, has the schematic form

$$\begin{aligned}\partial_\mu^x \langle T J_A^\mu(x)\Phi_1(x_1)\cdots\Phi_n(x_n)\rangle = & -\mathrm{i}\sum_{r=1}^{n}\delta^{(d)}(x-x_r)\langle T\Phi_1\cdots(\delta_A\Phi_r)\cdots\Phi_n\rangle \\ & + \langle T\mathcal{A}_A(x)\Phi_1\cdots\Phi_n\rangle, \end{aligned} \tag{3.22}$$

where $\mathcal{A}_A$ is an anomaly insertion and the displayed delta functions are contact terms. Translation Ward identities for $T_{\mu\nu}$ have the same logical structure, with derivatives acting on the other insertions. After Fourier transformation, these identities fix divergences and selected normalisations; they do not determine the unconstrained transverse form factors.

For a gauge theory, physical local insertions are represented by BRST cohomology classes rather than by the condition of BRST closure alone. The corresponding constraints on gauge-fixed Green functions are the Slavnov–Taylor identities [26, 27]. Writing the BRST differential as s_{B}, ghost-number-zero closure $\mathsf{s}_{\mathrm{B}}\mathcal{O} = 0$ is supplemented by the identification $\mathcal{O} \sim \mathcal{O} + \mathsf{s}_{\mathrm{B}}\Psi$ after passage to the equation-of-motion quotient. A total derivative dY is also quotiented for a constant-source integrated functional with vanishing boundary term, or at zero insertion momentum when that limit is well defined. A nonconstant source leaves a source-gradient insertion after integration by parts, and descendants are

retained at nonzero insertion momentum. A colour component $F^a_{\mu\nu}$ transforms covariantly and is not by itself a gauge-invariant scalar observable, whereas $\operatorname{tr} F_{\mu\nu}F^{\mu\nu}$ defines a scalar gauge-invariant class.

Scope of a symmetry identity. Fix a renormalised physical operator block, its state, and a projector. Assume the relevant Ward or Slavnov–Taylor identity is anomaly free in that block. After the contact and anomaly terms have been kept explicitly, the identity has the following consequences:

(i) set symmetry-forbidden projected components to zero up to specified contact terms;

(ii) determine longitudinal components in terms of lower-point functions;

(iii) fix a zero-transfer form-factor normalisation when stable external states carry a nonzero conserved charge, LSZ reduction applies, and the diagonal matrix element has a continuous zero-transfer limit;

(iv) leave the remaining physical transverse components symmetry allowed.

None of these conclusions, without additional state and dynamical information, implies a nonzero vacuum spectral atom, a nonzero shell contribution, or an isolated pole. After Fourier transformation, projection, and, for part (iii), LSZ amputation, Eq. (3.22) together with the stated representation and regularity assumptions gives (i)–(iii). Part (iv) is the complement of those constraints in the physical operator block.

This leaves form factors not fixed by the specified symmetry to the dynamical calculation. Symmetry thus organises which disturbances can couple to which outputs and fixes selected normalisations, while the state and interactions determine the remaining spectral weight. Comparing channels requires both parts of this information. Composite scalar and fermion-bilinear channels may contain bound-state poles even though no conserved-current normalisation fixes their residues.

The Bethe–Salpeter construction describes bound states through a two-particle kernel [28]. In a sourced channel this kernel enters the inhomogeneous response equation

$$\Gamma_{\mathfrak{C}}(z) = \Gamma^{(0)}_{\mathfrak{C}}(z) + \mathcal{B}_{\mathfrak{C}}(z;\eta)\Gamma_{\mathfrak{C}}(z), \qquad \mathcal{B}_{\mathfrak{C}}(z;\eta) = \mathcal{K}_{\mathrm{2PI},\mathfrak{C}}(z;\eta)G_{0,\mathfrak{C}}(z;\eta). \tag{3.23}$$

The dimensionless eigenvalue problem is the product of the irreducible kernel $\mathcal{K}_{\mathrm{2PI}}$ with the two-particle propagation operator G_0. For an eigenvalue-based solution of the homogeneous bound-state equation, see Ref. [29]. A concrete application must show that its 2PI, Bethe–Salpeter, Dyson–Schwinger, or FRG construction represents the Jacobian in Eq. (2.32), exactly or with a stated defect.

A vacuum spectral representation gives the residue a particle interpretation. Resolving the centred insertions into physical intermediate states writes their Wightman matrix as a positive measure of excitation amplitudes. This is the intermediate-state construction of the Källén–Lehmann representation [30, 31]. Compression onto positive-norm polarisations preserves that positivity. Euclidean continuation then expresses the same source response as a mass spectral integral, with local subtractions recorded separately. Matching these

insertions and conventions to the response equation allows its Laurent coefficient to be read as the weight of the zero-mass atom. The next theorem makes this identification precise.

Theorem 3.5 (Fredholm–KL bridge)**.** *Let the hypotheses of Theorem 3.2(ii)–(iii) hold and set $z = Q^2$ on the Euclidean axis. Assume in addition that:*

(i) at the physical endpoint and with $\tau_Q = 0$, the matrix in Eq. (2.33) is exactly the projected, unfiltered, connected Euclidean covariance of a physical spectral realisation in Definition A.1, with the vacuum and spectral assumptions of Appendix A;

(ii) the tensor or spin part of $\mathcal{P}_{\mathfrak{C}}$ uses the physical-polarisation compression in Assumption A.2, including its separately defined zero-mass helicity endpoint, and the external basis, vacuum, centring convention, source pairing, continuation convention, and subtraction polynomial $P(Q^2)$ are the same in the response and spectral constructions;

(iii) after subtraction of $P(Q^2)$, the direct term and the complementary Fredholm resolvent contribution are analytic at $Q^2 = 0$;

(iv) the continuum part of the physical matrix-valued spectral measure is locally finite, has no atom at zero, and satisfies the ultraviolet integrability hypotheses of Proposition A.4.

Then the algebraic residue is the physical zero-mass atom matrix:

$$Z_{\mathrm{alg}} = \lim_{Q^2 \downarrow 0} Q^2 [C(Q^2) - P(Q^2)] = Z_0 \succeq 0. \tag{3.24}$$

Consequently, a nonzero observable Fredholm pole satisfying these matching hypotheses is a physical zero-mass spectral atom. Conversely, if an exact calculation that purports to satisfy (i)–(iv) produces a non-Hermitian or indefinite Z_{alg}, at least one response-realisation, continuation, subtraction, or positivity premise has failed.

Proof. The Fredholm decomposition gives $C(Q^2) = Z_{\mathrm{alg}}/Q^2 + A(Q^2)$ with A analytic at the origin under assumption (iii). Hence $\lim_{Q^2 \downarrow 0} Q^2[C(Q^2) - P(Q^2)] = Z_{\mathrm{alg}}$. The same connected matrix is, by assumptions (i)–(ii), the correlator entering the subtracted Källén–Lehmann representation. Assumption (iv) permits Proposition A.4 to identify the same limit with Z_0. Positivity of the Wightman measure gives $Z_0 \succeq 0$. □

The bridge applies to the isolated meromorphic endpoint specified in its hypotheses. At the level of the spectral measure, a zero-mass atom may also coexist with continuum weight accumulating at $s = 0$. Proposition A.4 still extracts the atomic coefficient under its local-finiteness and integrability hypotheses, while the threshold density determines an additional non-meromorphic contribution and its own long-distance tail. Such an endpoint lies outside the analytic-remainder hypothesis of the bridge and is classified as *threshold-sensitive*. Accordingly, *isolated Fredholm pole* denotes the meromorphic response structure, whereas *zero-mass atom* denotes the atomic part of the physical spectral measure.

Spatial response of the pole. The spatial meaning can be made explicit in a translation-invariant realisation with three spatial dimensions. Suppose a fixed physical source contraction j has static pole contribution $Z_j/|\boldsymbol{q}|^2$, where $Z_j = j^* Z_0 j > 0$ is independent of $\boldsymbol{q}$ and $Q^2 = |\boldsymbol{q}|^2$ at zero frequency. For the external Gaussian representation already defined, put $r = |\boldsymbol{x}| > 0$ and $\tau_Q > 0$. The pole alone gives

$$\begin{aligned} K^{\mathrm{pole}}_{\tau_Q}(r) &= Z_j \int \frac{\mathrm{d}^3 q}{(2\pi)^3} \frac{e^{-\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{x}} e^{-\tau_Q|\boldsymbol{q}|^2}}{|\boldsymbol{q}|^2} = Z_j \int_{\tau_Q}^{\infty} \frac{e^{-r^2/(4u)}}{(4\pi u)^{3/2}}\,\mathrm{d}u \\ &= \frac{Z_j}{4\pi r}\,\mathrm{erf}\!\left(\frac{r}{2\sqrt{\tau_Q}}\right). \end{aligned} \tag{3.25}$$

Here $\mathrm{erf}(v) = (2/\sqrt{\pi}) \int_0^v e^{-t^2}\,\mathrm{d}t$. The integral representation follows by writing

$$\frac{e^{-\tau_Q|\boldsymbol{q}|^2}}{|\boldsymbol{q}|^2} = \int_{\tau_Q}^{\infty} e^{-u|\boldsymbol{q}|^2}\,\mathrm{d}u.$$

The Gaussian Fourier integral and the substitution $r/(2\sqrt{u})$ give the remaining equalities. At distances large compared with $\sqrt{\tau_Q}$, this contribution approaches $Z_j/(4\pi r)$. External smoothing resolves its short-distance behaviour while preserving its long-distance coefficient. For a tensor source with a momentum-dependent numerator, that numerator remains inside the transform. The complete spatial response also includes the transformed regular and continuous-spectrum terms; dominance of the pole requires bounds on those terms. Equation (3.25) is a response kernel; a force interpretation requires the corresponding matter coupling and interaction energy.

A realised helicity response. The bridge can be evaluated explicitly for the two physical helicities of a free gauge tensor on Minkowski space. Let ϕ_λ, $\lambda = \pm 2$, denote unit-normalised transverse, traceless amplitudes with reduced action

$$S_2 = \frac{Z}{2} \sum_{\lambda=\pm 2} \int [(\partial_t \phi_\lambda)^2 - |\nabla \phi_\lambda|^2]\,\mathrm{d}^4 x, \qquad Z > 0.$$

The one-particle space is $L^2(H_0^+, \mathrm{d}\Omega_0) \otimes \mathbb{C}^2$, with the unitary massless helicity representations $+2$ and -2. Its symmetric Fock space has the Poincaré-invariant free vacuum and centred insertions $\Theta_\lambda = g\phi_\lambda$ paired with physical helicity test sections. Their positive-frequency measure and continued covariance are

$$\begin{aligned} \widehat{W}_{\lambda\lambda'}(p) &= \frac{2\pi g^2}{Z} \theta(p^0)\delta(p^2)\delta_{\lambda\lambda'}, \\ C_{\lambda\lambda'}(Q^2) &= \frac{g^2}{ZQ^2}\delta_{\lambda\lambda'}. \end{aligned} \tag{3.26}$$

The Fourier convention absorbs the displayed 2π in the mass-shell measure. At fixed $p = |\boldsymbol{p}| > 0$, the reflected time covariance is $g^2 e^{-p(t_i+t_j)}/(2Zp)$ times the helicity identity, a positive Gram matrix. This establishes the directly reduced realisation of Definition A.1(ii), including its continuation and null-shell source pairing. The subtraction polynomial, direct

term and complementary contribution all vanish. Thus the two helicity channels separately satisfy the simple-eigenvalue and realisation hypotheses of Theorem 3.5; their direct sum gives the displayed two-dimensional residue. Equivalently, Theorem 3.3 has $L(z) = ZzI_2$, $P = I_2$ and $S = gI_2$, giving the same matrix $Z_0 = (g^2/Z)I_2$. The two constructions use the same vacuum, source normalisation and helicity endpoint. This response is the physical quotient of the quadratic macroscopic tensor action. Its covariant Einstein–matter completion [32] supplies a common matter source and constraint propagation, while the displayed translation-invariant spectral measure describes the flat-background linear response. The coupling of this tensor response to stationary matter and the separate potential and force estimates form a companion application in Ref. [13]. The response and spectral-identification theorems above are proved independently.

Finite truncations. An exact finite operation protocol can satisfy the native response and positive-residue theorems without an infinite-cutoff limit. Its interpretation as a QFT spectral atom still needs a corresponding physical representation. By contrast, a truncation approximating an infinite target must establish that limiting identification. A finite response truncation that satisfies the algebraic pole and overlap tests produces a candidate residue $Z_{\mathrm{alg},N}$. Even if this matrix is Hermitian and positive semidefinite, it represents a physical spectral atom only after the realisation maps, kernels, z-derivatives, source and readout maps, subtraction convention, and projected correlators converge so that the critical root remains isolated and simple, its slope stays bounded away from zero, and the hypotheses of Theorem 3.5 hold in the limit. Proposition C.1 gives the corresponding convergence result. This distinction is necessary because positivity of a finite matrix does not identify the limiting correlator or its spectral measure.

Ladder diagrams provide a possible approximation to $\mathcal{K}_{\mathrm{2PI}}$; a complete response may also contain crossed ladders, vertex corrections, ghost contributions, and higher-particle sectors. Likewise, the sign of a closed fermion loop does not by itself determine the sign of a projected response eigenvalue; vertices, group factors, propagators, and the convention used for the kernel enter the result.

3.7 Classification and scope of the conclusions

The classification returns four complementary entries:

$$\mathfrak{R}(\mathfrak{C}_{\mathrm{phys}}; \mathfrak{I}) = \big(R_{\mathrm{kin}}, R_{\mathrm{sym}}, R_{\mathrm{dyn}}, R_{\mathrm{scope}}\big). \tag{3.27}$$

Here R_{kin} records suppression away from the routed zero set or soft scaling; R_{sym} is forbidden, constrained, allowed, or anomalous; R_{dyn} records the strongest dynamical conclusion established; and R_{scope} records whether the mechanism is described by the response kernel, supplied by a topological construction, or assigned to a different response hierarchy. Keeping the entries separate allows a channel to be Gaussian-suppressed away from the routed zero set, symmetry allowed in its soft region, and dynamically undecided at the same time.

The four entries should be read as four successive questions: where does averaging leave support, what does symmetry permit, does the full response kernel become critical, and which physical realisation of that criticality has actually been established? The order

is logical rather than temporal: symmetry narrows the dynamical problem, but the system does not pass through a symmetry stage before entering a spectral one.

Within the isolated simple-eigenvalue setting, the conclusions can be established in successive steps. Spectral criticality means $1 \in \operatorname{spec} \mathcal{B}(0)$. A simple Fredholm zero additionally requires $\partial_z \lambda(0) \neq 0$. An observable pole additionally requires both response-space source–readout overlaps to be nonzero. A native positive residue follows when Theorem 3.3 applies; this is an additional certification, not a necessary condition on every nonreciprocal response. A Lorentzian physical atom additionally requires the exact unfiltered-correlator identification and spectral hypotheses of Theorem 3.5; Hermiticity and positive semidefiniteness then follow from the Wightman measure rather than being imposed as a substitute for that bridge. The labels *dynamically open*, *threshold-sensitive*, and *representation-outside* distinguish unresolved dynamics, threshold behaviour, and responses outside the stated representation. A negative result is labelled *exact-noncritical*, *certified-truncation-noncritical*, or *truncation-only*; an exact exclusion requires the exact operator of the declared system, or a certified approximation to it. A finite exact system is not a truncation merely because its dimension is finite.

Compatibility with a physical realisation. For a QFT channel, the operator quotient and projector $\mathcal{P}_{\mathfrak{C}}$ first determine the source and readout $\mathsf{S}_{\mathcal{O}}, \mathsf{R}_{\mathcal{O}}$. Their realisation on the Banach response space uses P_X and $\iota_{\mathfrak{C}}, \pi_{\mathfrak{C}}$, subject to Eqs. (3.17)– (3.18). The kernel is the Jacobian of the same sourced state equation. An approximate coarse equation also requires comparison at actual reference solutions. If the coarse solution differs from the image of the full solution, its displacement must be bounded; state derivatives of the kernel, source and readout then convert that bound into response errors. Residues must be compared at the two actual critical roots, including the variation of their slopes.

If $\operatorname{Ran} P_X$ is not invariant, the four Schur–Feshbach terms of Proposition 3.1 retain the omitted feedback. On the resulting Fredholm domain, the simple zero and the two overlaps determine the observable pole. For a finite approximation to an infinite target, operator-norm, derivative and Riesz-projection convergence transfer this conclusion to the limiting response.

Its spectral interpretation concerns the same unfiltered connected endpoint correlator. Common vacuum assumptions, centring, physical projection, continuation, source pairing and subtraction polynomial are precisely the matching hypotheses of Theorem 3.5. They identify the residue with a physical atom. Theorem 3.3 instead establishes positivity directly for the reciprocal Hilbert-space response, without asserting such a QFT realisation.

Two exclusions complement these positive conclusions. External Gaussian filtering preserves an existing zero-mass residue and cannot create a pole from a regular form factor, as shown around Eq. (2.29). If an exact symmetry sets the complete separated-point block to zero, uniqueness of its spectral decomposition gives $Z_0 = 0$: a contact polynomial has no nonlocal spectral weight. Neither statement evaluates an unevaluated, symmetry-allowed response kernel.

3.8 Four-axis classification

For a channel block $\mathfrak{C}_{\mathrm{phys}}$ and a computational realisation $\mathfrak{I}$, the result $\mathfrak{R}(\mathfrak{C}_{\mathrm{phys}};\mathfrak{I})$ of Eq. (3.27) is read as follows.

Kinematic entry. *Fixed-transfer suppressed* means that Theorem 2.5 applies on support separated from the zero set of the routed quadratic form. *Soft retained* means that $q = O(\tau^{-1/2})$ or a direction in $\operatorname{Ker} M$ remains and must be analysed. Both labels normally occur for the same channel in their respective momentum regions.

Symmetry entry. *Forbidden* refers to a projected component set to zero by an exact selection rule. *Constrained* refers to a component or normalisation determined by a Ward or Slavnov–Taylor identity. *Allowed* means only that no stated identity removes the physical block. *Anomalous/broken* records that an apparent classical conservation law is modified by an anomaly, explicit breaking, or the state.

Dynamical entry. *Spectral-critical* means only that a specified response operator has eigenvalue one. *Simple-zero*, *observable-pole*, and *physical-atom* add, in order, z-transversality, the two nonzero source–readout overlaps, and the physical spectral test. *Noncritical* is used without qualification only for an exact or certified limiting operator; otherwise the label is *truncation-only*. *Threshold-sensitive* records a collision with a continuum threshold. *Dynamically open* is the default when symmetry has left the channel open but the relevant kernel has not been evaluated.

Scope entry. The response-kernel label identifies the mechanism treated here. A topological label records a zero mode fixed by a separate index or cohomological construction. An outside label is used when no response realisation covered by the stated theorems has been established. In particular, the native positive-residue theorem also covers isolated finite-dimensional critical subspaces without requiring a compact response kernel or a two-body representation.

These entries answer different questions. In particular, neither "allowed" nor "constrained" entails spectral criticality, and fixed-transfer suppression is compatible with a nontrivial soft response.

3.9 Representative channels

Table 1 applies the four-axis language to a representative set of channels. The dynamical column states the calculation required; no criticality claim is made where the relevant kernel has not been evaluated.

Several distinctions in the table are essential. The translation Ward identity alone does not fix the transverse-traceless stress-tensor form factor. With the stronger complete-tensor positivity, exact Lorentz covariance, conservation, and invariant-vacuum hypotheses of Proposition A.3, however, the zero-mass TT atom is excluded. The spin-two spectrum at $s > 0$ still requires a dynamical calculation. This distinction concerns the QFT representation; it does not exclude a native tensor response or a discrete transfer residue that

Table 1. Representative renormalised channels. Under the hypotheses of Theorem 2.5, "suppressed; soft retained" means suppression on support separated from the zero set of the routed Gaussian quadratic form, with the soft or degenerate region of Eq. (2.27) still requiring analysis.

Channel	R_{sym}	R_{kin}	R_{dyn}	R_{scope}
$T^{\mathrm{TT}}_{\mu\nu}$	conserved; full tensor covariance and positivity	suppressed; soft retained	$s > 0$: spin-two kernel; $s = 0$: TT atom excluded	QFT response kernel; $d = 4$ vacuum
conserved J_μ	charge normalisation conditional; transverse weight unfixed	suppressed; soft retained	vector kernel calculation required	response kernel
$[\bar\psi\psi]_R$ block	no current normalisation	suppressed; soft retained	scalar bound-state kernel calculation required	response kernel
longitudinal nonsinglet J^5_μ block	exact anomaly-free chiral identity; state dependent	suppressed; soft retained	Goldstone test with nonzero symmetry-breaking expectation; kernel otherwise	Ward identity / response
longitudinal J^5_μ block with anomaly or explicit breaking	anomaly or explicit breaking	suppressed; soft retained	pseudoscalar kernel calculation required	response kernel
$[\mathrm{tr}\, F^2]_R$ block	physical BRST cohomology class	suppressed; soft retained	scalar glueball kernel calculation required	response kernel
$[H^\dagger H]_R$, $[(H^\dagger H)^2]_R$	gauge invariant; no charge normalisation	suppressed; soft retained	mixed scalar kernel calculation required	response kernel
internal gauge zero modes	topology and background dependent	not applicable	index or harmonic analysis required	topological; outside

Charge normalisation assumes stable external states, LSZ reduction, and a continuous diagonal zero-transfer limit. The stress-tensor row assumes positivity and exact covariance of the complete tensor before TT compression; its zero-mass endpoint is fixed by Proposition A.3, not by a massive-projector limit. The axial rows refer to longitudinal current–scalar or pseudoscalar mixing blocks, not the transverse axial response. A required kernel calculation is not a pole certificate: identification of a physical zero-mass atom uses the centred, connected vacuum correlator and all matching, positivity, and threshold hypotheses of Theorem 3.5.

has not been identified with that complete Wightman tensor measure. The same separation applies to a conserved vector current: an exactly normalised charge can coexist with a gapped current–current spectral function.

The scalar bilinear is not assigned a zero source. Depending on the theory and state, its kernel can contain a bound-state pole. The axial rows concern longitudinal mixing blocks; transverse axial response requires a separate projection and kernel calculation. An anomaly-free nonsinglet current in a chirally symmetric Lagrangian obeys a different identity from a singlet anomalous current or a current with explicit mass breaking. In the exact case a Goldstone pole additionally requires a nonzero symmetry-breaking vacuum

expectation in the current Ward identity; the operator name alone does not decide the result.

The notation $\mathrm{tr}\, F^2$ denotes the scalar 0^{++} gauge-invariant composite channel. It must not be identified with a spin-one force-carrier channel. Gauge-field zero modes on a compact internal space concern the kernel of a background differential operator and constitute a separate topological question. The compact geometry and gauge and matter representations in the companion construction Ref. [14] provide an internal-space realisation in which these channels can be specified. The absence of a nontrivial scalar index neither forbids a glueball pole nor removes the constant kernel of a scalar Laplacian.

Finally, $\|DH\|^2$ and four-fermion monomials are local composite operators. Whether they are independent representatives depends on integration by parts, field redefinitions, and EOM relations in the chosen operator basis. They cannot be excluded merely because one is a kinetic insertion and the other is a four-point interaction.

3.10 Symmetry permission and dynamical existence

For a symmetry-allowed block, the principal output is a well-posed dynamical question. If a symmetry forbids a projected block, no kernel calculation is needed for that block. If the block is allowed, the classification identifies the response operator, quantum-number sector, and soft kinematics in which a pole could occur. The existence question is thereby reduced to a specified spectral problem.

The distinction can be phrased directly in spectral language. For a Hermitian operator block, symmetry controls linear relations among the matrix elements of the positive spectral measure $d\rho_{ij}(s)$. It does not ordinarily fix whether ρ_{ij} has an atom at $s = 0$. Such an atom can be forced when additional state information closes a Ward identity, as in the Goldstone theorem. A critical response kernel yields an observable pole under the hypotheses and source–readout visibility conditions of Theorem 3.2; under the additional matching hypotheses of Theorem 3.5, its residue is the physical zero-mass atom matrix. Otherwise its weight is a dynamical observable.

This observation eliminates a false dichotomy between "protected" and "unprotected" operators. Protection can refer to current conservation, gauge-parameter independence, a nonrenormalisation statement, or membership in physical BRST cohomology. These notions are related but not equivalent. The classification therefore records the actual identity and its consequence instead of assigning a universal binary attribute.

3.11 External basis covariance, response coordinates, and schemes

Two different transformations must be kept separate. First let $O = (O_1, \ldots, O_m)^T$ be the external physical operator basis and let

$$O' = MO, \qquad M \in GL(m, \mathbb{C}), \tag{3.28}$$

be a finite change of renormalised basis. The Wightman spectral matrix, Euclidean correlator, and any isolated residue matrix transform by congruence,

$$d\rho' = M\, d\rho\, M^\dagger, \qquad C' = MCM^\dagger, \qquad Z' = MZM^\dagger. \tag{3.29}$$

This is an external change of source and readout coordinates. The same physical source has coordinates $j' = (M^\dagger)^{-1} j$, hence $j'^\dagger Z' j' = j^\dagger Z j$. The pole strength measured by that source is unchanged even when its individual matrix components change.

Second, let $T : \mathcal{X}_{\mathfrak{C}} \to \mathcal{X}'_{\mathfrak{C}}$ be a bounded isomorphism that changes only the internal coordinates of the response state. Then

$$\begin{aligned} \mathcal{B}_T &= T\mathcal{B}T^{-1}, \quad & P_{X,T} &= TP_XT^{-1}, \\ \mathsf{S}_{X,T} &= T\mathsf{S}_X, & \mathsf{R}_{X,T} &= \mathsf{R}_XT^{-1}. \end{aligned} \tag{3.30}$$

It leaves the output matrix unchanged. If an external basis change M and an internal coordinate change T are performed together, the complete transformation law is

$$\begin{aligned} \mathcal{B}' &= T\mathcal{B}T^{-1}, \\ P'_X &= TP_XT^{-1}, \\ \mathsf{S}'_X &= T\mathsf{S}_XM^\dagger, \\ \mathsf{R}'_X &= M\mathsf{R}_XT^{-1}, \\ C'_{\mathrm{reg}} &= MC_{\mathrm{reg}}M^\dagger. \end{aligned} \tag{3.31}$$

Therefore

$$C' = C'_{\mathrm{reg}} + \mathsf{R}'_X(I - \mathcal{B}')^{-1}\mathsf{S}'_X = MCM^\dagger, \tag{3.32}$$

as required by the Wightman construction. The adjoint on the source leg follows the convention $C_{ij} = \langle O_i O_j^\dagger \rangle$; in a real bilinear convention it is replaced consistently by the corresponding transpose.

Proposition 3.6 (External and internal covariance)**.** *Under the finite nonsingular external transformation M and bounded internal isomorphism T, the following exact channel data are invariant: the location and order of poles of the full correlator matrix; the rank and positive-semidefinite character of an isolated residue matrix; the forbidden, constrained, and allowed symmetry subspaces up to the isomorphism induced by M (and hence their dimensions); the spectrum of the response operator; and the vanishing or nonvanishing of the source and observable overlaps at an isolated critical eigenvalue. Individual matrix entries and residue components need not be invariant.*

The proof is given in Appendix B. For a scale-dependent finite transformation $M(\mu)$, the anomalous-dimension matrix obeys

$$\gamma' = M\gamma M^{-1} - \left(\mu \frac{dM}{d\mu}\right) M^{-1}. \tag{3.33}$$

The extra term is why a running operator basis cannot be treated as a constant similarity transformation.

Exact physical pole positions are independent of a consistent renormalisation scheme. Intermediate Wilson coefficients, contact terms, operator normalisations, and individual entries of a mixing matrix are not. At finite truncation, even the inferred kernel eigenvalues and crossing scales generally depend on the regulator, basis size, and projection

prescription. Claims of scheme independence must therefore be demonstrated by the exact transformation law or replaced by a convergence and regulator-variation study.

For Higgs-sector applications, the construction identifies the scalar mixing block and the kernel governing its correlations. Ultraviolet boundary conditions for the mass and quartic coupling remain independent renormalisation or phenomenological inputs.

4 Coarse-response errors, residual states, and evolution

Exact observation-sufficient descent preserves the full response. When a chosen coarse description is only approximate, the first object to compare is that response, even if the state dimensions differ. The comparison below needs no inverse coarse map. Its four defects distinguish the effects of unresolved states, displaced reference branches and approximate source couplings.

4.1 A response error identity without an inverse coarse map

The two state spaces may have different dimensions, so their resolvents cannot in general be subtracted directly. The map C places the update defect in a common space, while the common source and output spaces allow the observable responses to be compared. The identity below propagates each of the four defects to that output.

Theorem 4.1 (Irreversible coarse-response error)**.** *Let B, S, R, D and $\bar{B}, \bar{S}, \bar{R}, \bar{D}$ be bounded response data on Banach state spaces $\mathcal{X}, \mathcal{Y}$, with the same source and output spaces. Let $C : \mathcal{X} \to \mathcal{Y}$ be any bounded linear comparison map. Set $G = (I - B)^{-1}$, $\bar{G} = (I - \bar{B})^{-1}$ where both inverses exist, and define*

$$\begin{aligned} E_B &= \bar{B}C - CB, \quad & E_S &= \bar{S} - CS, \\ E_R &= \bar{R}C - R, \quad & E_D &= \bar{D} - D. \end{aligned} \tag{4.1}$$

Then the complete responses obey the exact identity

$$\bar{\mathscr{C}} - \mathscr{C} = E_D + \bar{R}\bar{G}E_S + \bar{R}\bar{G}E_BGS + E_RGS. \tag{4.2}$$

In particular, writing $e_i = \|E_i\|$ in the specified operator norms,

$$\|\bar{\mathscr{C}} - \mathscr{C}\| \le e_D + \|\bar{R}\|\|\bar{G}\|e_S + \|\bar{R}\|\|\bar{G}\|e_B\|G\|\|S\| + e_R\|G\|\|S\|. \tag{4.3}$$

Neither injectivity nor surjectivity of C is required.

Proof. From $(I-\bar{B})C = C(I-B) - E_B$ obtain $\bar{G}C - CG = \bar{G}E_BG$. Inserting $\bar{S} = CS + E_S$ into $\bar{D} + \bar{R}\bar{G}\bar{S} - (D + RGS)$ and using $\bar{R}C = R + E_R$ gives Eq. (4.2). Submultiplicativity gives the norm bound. □

The update defect is multiplied by both resolvents, whereas the source and readout defects each carry one. Thus near criticality the response error depends on where a defect enters, as well as on its norm. For example, in a scalar reciprocal response with $S = R = 1$,

$D = 0$ and $B = 1 - \epsilon$, $0 < \epsilon < 1$, the susceptibility is ϵ^{-1}. Changing B to $B + \delta b$ changes it by

$$\frac{1}{\epsilon - \delta b} - \frac{1}{\epsilon} = \frac{\delta b}{\epsilon(\epsilon - \delta b)}.$$

For $0 < |\delta b| \ll \epsilon$, the relative error is approximately $|\delta b|/\epsilon$. Thus the distance from criticality sets the required accuracy of the reduced feedback. Comparing the displaced roots and their residues describes this sensitivity even when no uniform pointwise comparison across the poles is possible. Directional amplification also matters for nonnormal updates. With the Euclidean norm on $\mathbb{R}^2$ and $g \in \mathbb{R}$, take

$$B = \begin{pmatrix} 0 & g \\ 0 & 0 \end{pmatrix}, \qquad S = e_2, \quad R = e_1^{\mathsf{T}}, \quad D = 0.$$

Here $B^2 = 0$ and both eigenvalues vanish, but $R(I - B)^{-1}S = R(I + B)S = g$. The source excites the second coordinate, which the update transfers into the observed first coordinate with gain g. This finite response can be large at a fixed distance from the kernel spectrum; the resolvent norms in the error estimate account for such amplification as well as for proximity to a pole. This role of the resolvent is central to the pseudospectral analysis of nonnormal operators [8]. For the source-dependent coarse map in Eq. (2.9), apply this theorem with $C = T$ and the corrected data

$$\bar{S}_{\mathrm{corr}} = \bar{S} - (I - \bar{B})C_h, \qquad \bar{D}_{\mathrm{corr}} = \bar{D} + \bar{R}C_h. \tag{4.4}$$

They give exactly the same coarse response since the two corrections cancel in $\bar{D}_{\mathrm{corr}} + \bar{R}\bar{G}\bar{S}_{\mathrm{corr}}$.

All comparisons are at the actual reference states. For a fixed linear C, let $\delta = \|\bar{X}_* - CX_*\|$ be independently bounded and let $e_i^{(0)}$ compare the data at X_* and the transported point CX_*. If the coarse data have state-Lipschitz constants L_i on the intervening neighbourhood, the resulting bounds are

$$\begin{aligned} e_B &\le e_B^{(0)} + L_B\|C\|\delta, \quad e_S \le e_S^{(0)} + L_S\delta, \\ e_R &\le e_R^{(0)} + L_R\|C\|\delta, \quad e_D \le e_D^{(0)} + L_D\delta. \end{aligned} \tag{4.5}$$

For a moving map its T, C_h errors must also enter the corrected defects. Near criticality neither a small stationary residual nor the norm of C alone bounds δ. Resolvent growth in Eq. (4.3) records the associated sensitivity rather than removing it.

Contour control of the total output residue. Suppose both responses extend meromorphically to a neighbourhood of a simple closed positively oriented rectifiable contour Γ and its interior, with finitely many poles inside and none on Γ. If Eq. (4.3) is bounded by b_Γ on Γ, then

$$\left\| \sum_{\Gamma} \operatorname{Res} \bar{\mathscr{C}} - \sum_{\Gamma} \operatorname{Res} \mathscr{C} \right\| \le \frac{\operatorname{length}(\Gamma)}{2\pi} b_\Gamma. \tag{4.6}$$

Indeed, integration of Eq. (4.2), followed by the matrix-valued residue theorem and the contour norm bound, gives Eq. (4.6).

This controls a sum of output residues, not a one-to-one correspondence of full and coarse kernel roots. A branch cut inside the contour would contribute its jump integral and invalidate a pole-only interpretation. Individual roots additionally require an isolated spectral branch, a simple zero with slope bounded away from zero, and control of its Riesz projection. These conditions distinguish a root comparison from a bound on the sum of residues inside a contour.

4.2 Residual states and exact memory

When fibre consistency fails, the next coarse state depends on the unresolved coordinate. Its elimination produces memory, as in the projection-operator formulation of Mori [4]. For the bounded discrete linear system considered here, the terms follow directly from the block equations:

$$\begin{aligned} x_{n+1} &= A_{11}x_n + A_{12}u_n + S_1h_n, \\ u_{n+1} &= A_{21}x_n + A_{22}u_n + S_2h_n. \end{aligned} \tag{4.7}$$

Induction gives

$$u_n = A_{22}^n u_0 + \sum_{j=0}^{n-1} A_{22}^{n-1-j}(A_{21}x_j + S_2h_j). \tag{4.8}$$

Substitution into the first equation produces the initial-residual term $A_{12}A_{22}^n u_0$ and two convolution memories, one of the retained state and one of the source. An observation $R_1x_n + R_2u_n + Dh_n$ retains the same hidden history. Static Schur elimination is the corresponding stationary or transform-domain identity, not a proof that these time-domain terms vanish.

The same distinction appears in projection-operator descriptions of transport: eliminating microscopic coordinates leaves a generalized force with memory, and a local constitutive coefficient requires a further Markovian or scale-separation approximation [4]. Here the static Schur identity closes the stationary response, while the actual update retains the delayed influence of the eliminated coordinates.

Each factor $A_{12}A_{22}^{n-1-j}A_{21}$ describes a disturbance entering the hidden sector, evolving there, and returning to the retained one. The analogous factor with S_2 records a source applied directly to that sector. Memory therefore preserves the delayed physical influence of eliminated variables, while the initial term preserves the influence of their preparation. A closed description can retain these effects through additional state coordinates or through the history terms.

If $\|A_{22}^k\| \le M\rho^k$ with $0 < \rho < 1$, discarding lags $k \ge m$ incurs, on a history with bounds X_H, H_H, at most

$$\frac{\|A_{12}\|M\rho^m}{1-\rho}(\|A_{21}\|X_H + \|S_2\|H_H) \tag{4.9}$$

in one retained update; the initial term remains separately bounded by $\|A_{12}\|M\rho^n\|u_0\|$. Replace A_{12} by R_2 for the readout tail. Summing a geometric series proves these bounds. They are local update or readout errors, not automatically uniform trajectory errors. If an independently verified augmented evolution bound gives $e_{n+1} \le qe_n + \epsilon_n$, then induction

gives $e_n \le q^n e_0 + \sum_{j<n} q^{n-1-j}\epsilon_j$. In particular, $0 \le q < 1$ and bounded forcing errors give a uniform trajectory bound. For nonlinear dynamics the corresponding propagation bound must be established from the nonlinear evolution.

Companion applications illustrate how the additional propagation input may be supplied in particular models. For generated semilinear spectra, common stronger bounds and a weaker difference norm give estimates on the interval where the Duhamel map is contractive [17]; the finite approximation then retains both the initial discrepancy and the discarded nonlinear source. A specified Neumann evolution is treated in Ref. [18], and the cosmological application [20] uses its declared matter trajectory, transport law, and continuous-equation residual. The local response and memory statements above have self-contained proofs; these additional bounds implement the propagation step in the stated models.

The arithmetic and two-mode examples below distinguish a faithful environment retaining all full-system states from an environment sufficient for one observation protocol. Minimal dimension is proved there within a specified finite-dimensional linear class. The set quotient fixes observational equivalence; the distinct exponential rates and the Hankel rank determine the minimal linear realisation in these examples.

4.3 Actual updates and stationary solvers

Four transformations should be distinguished: internal coordinate similarity, external source-basis congruence, equation preconditioning, and actual coarse compression. Only the last can forget state directions. For a residual $\mathcal{E}(X,h) = 0$, let M be a differentiable bounded isomorphism from the state space to the residual space, with bounded inverse, and introduce the solver $F_{\mathrm{solve}} = X - M^{-1}\mathcal{E}$. At a solution, derivatives of M multiply $\mathcal{E} = 0$, so

$$I - B_{\mathrm{solve}} = M^{-1}L, \qquad S_{\mathrm{solve}} = -M^{-1}\mathcal{E}_h, \qquad (I - B_{\mathrm{solve}})^{-1}S_{\mathrm{solve}} = -L^{-1}\mathcal{E}_h. \tag{4.10}$$

This proves invariance of the complete static response after the source is transformed consistently. It does not preserve the solver spectrum or its convergence rate. For $\mathcal{E}(X,h) = X - h$, $M = I$ gives derivative zero, while $M = I/3$ gives derivative -2 and a divergent iteration; both stationary responses are one. Numerical convergence therefore establishes no physical stability unless the iteration has separately been identified with the actual update. An update index, a Gaussian scale, an RG coordinate, and physical time likewise need distinct definitions.

4.4 Invertible sector comparisons and displaced branches

When the comparison map has a bounded inverse on a specified subspace, one can also transport the response operator and its spectral data between the two sectors. This additional hypothesis is useful for coordinate changes and convergent approximations. General irreversible maps remain covered by Theorem 4.1. Let $\mathcal{C}_{\tau_q} : \mathcal{X}_{\mathfrak{C}} \to \mathcal{X}_{\mathfrak{C},\tau_q}$ be the stated coarse-graining map and define

$$E_{\tau_q}(X;z) := \mathcal{C}_{\tau_q}\mathfrak{F}(X;z) - \mathfrak{F}_{\tau_q}(\mathcal{C}_{\tau_q}X;z). \tag{4.11}$$

If $X_*(z) = \mathfrak{F}(X_*(z); z)$ and $\bar{X}_*(z) := \mathcal{C}_{\tau_q} X_*(z)$, then

$$E_{\tau_q}(X_*(z); z) = \bar{X}_*(z) - \mathfrak{F}_{\tau_q}(\bar{X}_*(z); z). \tag{4.12}$$

Thus $E_{\tau_q}(X_*; z)$ is the residual of the transported point in the coarse fixed-point equation. Its derivative compares Jacobians at that same transported point; it does not, by itself, compare with a Jacobian evaluated at a different coarse solution. If $E_{\tau_q} = 0$ on a neighbourhood of the reference solution, differentiation gives the exact intertwining relation

$$\mathcal{C}_{\tau_q} D_X \mathfrak{F}(X_*; z) = D_X \mathfrak{F}_{\tau_q}(\bar{X}_*; z) \mathcal{C}_{\tau_q}. \tag{4.13}$$

For an approximate closure, both the base-point residual and its derivative must therefore be accounted for.

The following stability result uses exact matching of the reference branches. With this matching, the derivative defect compares the two response Jacobians at actual solutions, without a separate state-displacement term.

The estimates proceed from the derivative defect to the critical spectral subspace, then to the location of a simple root and the residue at that root. Source and readout errors enter when passing from the internal mode to the measured response. Resolvent factors record how these errors can be amplified near criticality.

Proposition 4.2 (Stability under a closure defect)**.** *Let $X_*(z) = \mathfrak{F}(X_*(z); z)$ be a reference branch on U, set $\bar{X}_*(z) := \mathcal{C}_{\tau_q} X_*(z)$, and assume the pointwise reference-branch matching condition*

$$E_{\tau_q}(X_*(z); z) = 0, \qquad z \in U. \tag{4.14}$$

Then $\bar{X}_*(z)$ is an actual coarse fixed point. Define the response Jacobians on the two matched branches by

$$\mathcal{B}(z) := D_X \mathfrak{F}(X_*(z); z), \qquad \mathcal{B}_{\tau_q}(z) := D_X \mathfrak{F}_{\tau_q}(\bar{X}_*(z); z). \tag{4.15}$$

Let $Y \subset \mathcal{X}_{\mathfrak{C}}$ and $\widetilde{Y} \subset \mathcal{X}_{\mathfrak{C},\tau_q}$ be comparison spaces invariant under $\mathcal{B}(z)$ and $\mathcal{B}_{\tau_q}(z)$, respectively, and suppose $\mathcal{C}_{\tau_q} : Y \to \widetilde{Y}$ is a bounded isomorphism. Assume $\operatorname{Ran} \mathsf{S}_X(z) \subset Y$ *and $\mathsf{R}_X(z)|_Y : Y \to \mathbb{C}^m$ is bounded. Suppose all response, source, and readout families below are analytic in z. In the remainder of the proposition, $\mathcal{B}(z)$ and $\mathcal{B}_{\tau_q}(z)$ denote their restrictions to Y and $\widetilde{Y}$, respectively. Define*

$$\begin{aligned} \mathcal{B}_{\mathrm{ind}}(z) &:= \mathcal{C}_{\tau_q} \mathcal{B}(z) \mathcal{C}_{\tau_q}^{-1}, \\ \Delta\mathcal{B}(z) &:= \mathcal{B}_{\tau_q}(z) - \mathcal{B}_{\mathrm{ind}}(z) \\ &= -\big[D_X E_{\tau_q}(X_*(z); z)|_Y\big] \mathcal{C}_{\tau_q}^{-1}. \end{aligned} \tag{4.16}$$

The consistently transported source and readout are

$$\mathsf{S}_{\mathrm{ind}}(z) := \mathcal{C}_{\tau_q} \mathsf{S}_X(z) : \mathbb{C}^m \longrightarrow \widetilde{Y}, \qquad \mathsf{R}_{\mathrm{ind}}(z) := \big(\mathsf{R}_X(z)|_Y\big) \mathcal{C}_{\tau_q}^{-1} : \widetilde{Y} \longrightarrow \mathbb{C}^m. \tag{4.17}$$

They preserve the unfiltered response represented on the comparison sector:

$$\mathsf{R}_{\mathrm{ind}} (I_{\widetilde{Y}} - \mathcal{B}_{\mathrm{ind}})^{-1} \mathsf{S}_{\mathrm{ind}} = \big(\mathsf{R}_X|_Y\big) [I_Y - (\mathcal{B}|_Y)]^{-1} \mathsf{S}_X \tag{4.18}$$

wherever the two inverses exist. Let

$$\widetilde{\mathsf{S}}_X(z) : \mathbb{C}^m \to \widetilde{Y}, \qquad \widetilde{\mathsf{R}}_X(z) : \widetilde{Y} \to \mathbb{C}^m \tag{4.19}$$

be the source and readout of the coarse response evaluated on $\bar{X}_*(z)$.

Let $K \Subset U$, *and let a positively oriented contour* γ *in the spectral* ζ *plane enclose exactly one algebraically simple eigenvalue of* $\mathcal{B}_{\mathrm{ind}}(z)$ *and no other spectrum, for every* $z \in K$. *Put*

$$\begin{aligned} \epsilon &:= \sup_{z\in K} \|\Delta\mathcal{B}(z)\| \le \sup_{z\in K} \|D_X E_{\tau_q}(X_*(z); z)|_Y\| \|\mathcal{C}_{\tau_q}^{-1}\|, \\ K_\gamma &:= \sup_{z\in K,\, \zeta\in\gamma} \|(\zeta - \mathcal{B}_{\mathrm{ind}}(z))^{-1}\|, \\ L_\gamma &:= \mathrm{length}(\gamma). \end{aligned} \tag{4.20}$$

If $K_\gamma\epsilon < 1$, *the contour remains in the resolvent set of the coarse family and its Riesz projection obeys*

$$\|\Pi_{\tau_q}(z) - \Pi_{\mathrm{ind}}(z)\| \le d_\Pi := \frac{L_\gamma}{2\pi} \frac{K_\gamma^2 \epsilon}{1 - K_\gamma \epsilon}. \tag{4.21}$$

Writing $B_* := \sup_{z\in K} \|\mathcal{B}_{\mathrm{ind}}(z)\|$ *and* $\Pi_* := \sup_{z\in K} \|\Pi_{\mathrm{ind}}(z)\|$, *the enclosed eigenvalues satisfy*

$$|\lambda_{\tau_q}(z) - \lambda_{\mathrm{ind}}(z)| \le d_\lambda := \epsilon(\Pi_* + d_\Pi) + 2B_* d_\Pi. \tag{4.22}$$

Suppose $\lambda_{\mathrm{ind}}(z_*) = 1$ *and* $\alpha := |\partial_z \lambda_{\mathrm{ind}}(z_*)| > 0$. *For sufficiently small* $d_\lambda > 0$, *assume the disk* $D_* := \{z : |z - z_*| \le 4d_\lambda/\alpha\}$ *lies in* K, *that* z_* *is the only zero of* $\lambda_{\mathrm{ind}} - 1$ *in* D_*, *and that*

$$|\lambda_{\mathrm{ind}}(z) - 1| \ge \frac{\alpha}{2}|z - z_*|, \qquad z \in \partial D_*. \tag{4.23}$$

Then $\lambda_{\tau_q} - 1$ *has exactly one zero* $\widetilde{z}_*$ *in* D_*, *and*

$$\delta_z := |\widetilde{z}_* - z_*| \le \frac{4d_\lambda}{\alpha}. \tag{4.24}$$

When $d_\lambda = 0$, *take* $\widetilde{z}_* = z_*$.

For $z \in D_*$ *set*

$$\begin{aligned} \mathcal{N}_{\mathrm{ind}}(z) &:= \mathsf{R}_{\mathrm{ind}}(z)\Pi_{\mathrm{ind}}(z)\mathsf{S}_{\mathrm{ind}}(z), \\ \widetilde{\mathcal{N}}(z) &:= \widetilde{\mathsf{R}}_X(z)\Pi_{\tau_q}(z)\widetilde{\mathsf{S}}_X(z). \end{aligned} \tag{4.25}$$

Introduce the same-point map discrepancies and uniform norms

$$\begin{aligned} d_R &:= \sup_{D_*} \|\widetilde{\mathsf{R}}_X - \mathsf{R}_{\mathrm{ind}}\|, \qquad d_S := \sup_{D_*} \|\widetilde{\mathsf{S}}_X - \mathsf{S}_{\mathrm{ind}}\|, \\ R_* &:= \sup_{D_*} \|\mathsf{R}_{\mathrm{ind}}\|, \quad \Pi_*^D := \sup_{D_*} \|\Pi_{\mathrm{ind}}\|, \quad \widetilde{\Pi}_* := \sup_{D_*} \|\Pi_{\tau_q}\|, \quad \widetilde{S}_* := \sup_{D_*} \|\widetilde{\mathsf{S}}_X\|, \end{aligned} \tag{4.26}$$

and define

$$d_{\mathrm{num}}^{(0)} := d_R \widetilde{\Pi}_* \widetilde{S}_* + R_* d_\Pi \widetilde{S}_* + R_* \Pi_*^D d_S. \tag{4.27}$$

Analyticity on D_* *gives finite constants*

$$L_{\mathrm{num}} := \sup_{z\in D_*} \|\partial_z \widetilde{\mathcal{N}}(z)\|, \quad d_{\lambda'}^{(0)} := \sup_{z\in D_*} |\partial_z \lambda_{\tau_q}(z) - \partial_z \lambda_{\mathrm{ind}}(z)|, \quad L_{\lambda'} := \sup_{z\in D_*} |\partial_z^2 \lambda_{\tau_q}(z)|. \tag{4.28}$$

In particular, L_{num} may be bounded by the product rule from the separate z-derivative bounds for $\widetilde{\mathsf{R}}_X$, Π_{τ_q}, and $\widetilde{\mathsf{S}}_X$. Put

$$D_{\mathrm{num}} := d^{(0)}_{\mathrm{num}} + L_{\mathrm{num}}\delta_z, \qquad D_{\lambda'} := d^{(0)}_{\lambda'} + L_{\lambda'}\delta_z. \tag{4.29}$$

If $D_{\lambda'} < \alpha/2$, the residues at their respective critical roots,

$$Z_{\mathrm{alg,ind}} := -\frac{\mathcal{N}_{\mathrm{ind}}(z_*)}{\partial_z \lambda_{\mathrm{ind}}(z_*)}, \qquad \widetilde{Z}_{\mathrm{alg}} := -\frac{\widetilde{\mathcal{N}}(\widetilde{z}_*)}{\partial_z \lambda_{\tau_q}(\widetilde{z}_*)}, \tag{4.30}$$

obey

$$\|\widetilde{Z}_{\mathrm{alg}} - Z_{\mathrm{alg,ind}}\| \le \frac{2D_{\mathrm{num}}}{\alpha} + \frac{2\|\mathcal{N}_{\mathrm{ind}}(z_*)\|}{\alpha^2} D_{\lambda'}. \tag{4.31}$$

Thus the response estimate requires reference-branch matching, a bounded comparison inverse, control of the nonnormal resolvent, and transversality. The residue estimate additionally transports the numerator and eigenvalue slope between the two actual critical roots.

The proof is given in Appendix C. Condition (4.14) is substantive. If it fails, an actual coarse branch $\widetilde{X}_*(z)$ and an independent bound

$$d_X := \sup_{z\in K} \|\widetilde{X}_*(z) - \mathcal{C}_{\tau_q} X_*(z)\| \tag{4.32}$$

must first be established. Provided the reevaluated families preserve the same comparison spaces, re-evaluate $\mathcal{B}_{\tau_q}$ and every coarse source or readout map at $\widetilde{X}_*$. If $D_X\mathfrak{F}_{\tau_q}$ is L_B-Lipschitz in X, the kernel bound then obeys

$$\epsilon \le \sup_{z\in K} \|D_X E_{\tau_q}(X_*(z); z)|_Y\| \|\mathcal{C}_{\tau_q}^{-1}\| + L_B d_X. \tag{4.33}$$

Base-point-dependent source and readout maps acquire their corresponding Lipschitz displacement terms. At a critical response, the residual $\|E_{\tau_q}(X_*; z)\|$ alone does not supply such a displacement bound, because $I - \mathcal{B}_{\tau_q}$ need not be invertible.

The word "flow" is used for several inequivalent operations in field theory. The Gaussian maps $\mathcal{H}_\tau$ form a heat semigroup ordered by resolution. A Wilsonian or functional RG trajectory orders effective descriptions by a cutoff scale k. Physical time parametrises actual evolution through distinguishable, calibrated clock readings. Hamiltonian evolution and open-system dynamical maps provide field-theoretic realisations of that evolution. An update count, Gaussian variance or cutoff scale does not by itself specify a physical clock. The remaining subsections specify the first two operations and their relation to the response comparisons above.

4.5 The Gaussian scale semigroup

Equation (2.11) gives

$$\mathcal{H}_0 = I, \qquad \mathcal{H}_{\tau+s} = \mathcal{H}_\tau \mathcal{H}_s, \qquad \partial_\tau \mathcal{H}_\tau = \Delta \mathcal{H}_\tau. \tag{4.34}$$

The parameter τ has dimensions of length squared and labels the amount of smoothing. Increasing τ defines a partial order of resolution: information at a fixed nonzero wave number is attenuated more strongly at larger τ. No physical clock is introduced by Eq. (4.34).

For every finite τ, the Fourier multiplier $e^{-\tau|q|^2}$ is pointwise nonzero. Consequently $\mathcal{H}_\tau$ is injective on $L^2(\mathbb{R}^d)$. It is not, however, continuously invertible on its range.

Operational irreversibility. For every $\tau > 0$, the inverse of $\mathcal{H}_\tau$ on $\operatorname{Ran}\mathcal{H}_\tau \subset L^2(\mathbb{R}^d)$ is unbounded. Hence arbitrarily small errors in coarse-grained high-frequency data can correspond to order-one errors before coarse-graining. Choose $f_n \in L^2$ with $\|f_n\|_2 = 1$ and Fourier support in the annulus $n \le |q| \le n+1$. Plancherel's theorem gives

$$\|\mathcal{H}_\tau f_n\|_2 \le e^{-\tau n^2}\|f_n\|_2 \xrightarrow[n\to\infty]{} 0. \tag{4.35}$$

If the inverse were bounded, then the inequality $1 = \|f_n\|_2 \le \|\mathcal{H}_\tau^{-1}\|\,\|\mathcal{H}_\tau f_n\|_2$ would contradict Eq. (4.35).

This supplies the precise sense in which Gaussian smoothing is operationally irreversible at finite precision. Algebraically the map remains injective; the loss of recoverability is the instability of its inverse. A partial trace is a separate state-level operation.

This is the familiar obstruction in deblurring. No finite Fourier mode is set exactly to zero, yet sufficiently fine structure is buried below any fixed observational tolerance, and inversion amplifies that uncertainty without bound.

4.6 Wilsonian integration and information loss

Wilsonian integration produces an effective functional for retained variables by integrating over the eliminated momentum modes [33]. At the level of quantum states, discarding a factor of a specified Hilbert space acts by partial trace; restricting observations can instead be described by a conditional expectation onto an observable algebra. For density matrices, a completely positive trace-preserving reduction contracts their quantum relative entropy [34]. In an interacting field theory, identifying such factors or algebras is part of the physical realisation of the reduction.

For a finite carrier with $A = A^*$, the random-phase channel

$$\mathcal{E}_v(\rho) = \mathbb{E}_{\theta\sim N(0,v)}[e^{-\mathrm{i}\theta A}\rho e^{\mathrm{i}\theta A}]$$

is completely positive and trace preserving. In an eigenbasis its multiplier is $e^{-v(a_i-a_j)^2/2}$. Random phases leave the populations unchanged and attenuate coherence between different eigenvalues; the squared eigenvalue difference determines the decay. Field-amplitude smoothing instead acts on each mode with $e^{-\tau a_i}$. Composition adds the variances.

For a Euclidean field representation, reflection positivity tests whether positive-time insertions define nonnegative state norms after reflection [35, 36]. This question involves the covariance across time slices. Let $A \ge 0$ be self-adjoint, $m > 0$, $H = (A + m^2)^{1/2}$, and let B be a bounded, time-independent operator on each time slice. For $C_B(t) = Be^{-|t|H}(2H)^{-1}B^*$ and $t_i \ge 0$,

$$\sum_{i,j}\langle f_i, C_B(t_i + t_j)f_j\rangle = \left\|\sum_i e^{-t_i H}(2H)^{-1/2}B^* f_i\right\|^2 \ge 0.$$

Each positive-time insertion thus contributes a vector to a common Hilbert space, and the reflected quadratic form is their Gram norm. The mass gap bounds the inverse square root; B need not commute with H. This proves reflection positivity of the declared free covariance. The time Gaussian $C(t) = e^{-t^2}$ fails the test: at $t_1 = 1/4, t_2 = 3/4$, its reflected matrix has determinant $e^{-5/2} - e^{-2} < 0$. Finite-dimensional evaluations of these covariances and of the random-phase channel use the software of Ref. [16].

The Gaussian averaging semigroup and an FRG regulator serve different roles. The former is the canonical averaging prescription within the composition class of Proposition 2.3 and is the one used in Theorem 2.5; the latter is a computational term chosen to interpolate between effective actions. Different admissible regulators can be used to estimate a channel, but regulator variation at finite truncation is an uncertainty diagnostic, not a family of alternative averaging prescriptions.

4.7 Composite-source functional flow

The dynamical kernel in Eq. (3.23) can be computed with several nonperturbative methods. For FRG calculations the operator block must be coupled to sources before the flow is differentiated. Let h_i couple to the renormalised composite insertions and introduce

$$Z_k[J,h] = \int \mathcal{D}\phi \, \exp\left[-S[\phi] - \Delta S_k[\phi] + J\cdot\phi + \sum_i h_i \cdot [\mathcal{O}_i]_R\right], \tag{4.36}$$

where $\Delta S_k = \frac{1}{2}\phi\cdot R_k\cdot\phi$ is written schematically, with the superfield generalisation understood. The modified Legendre transform is taken in J while retaining h as an external source [37, 38]. To exhibit the response, use bosonic variables, write $W_k = \log Z_k$, $\varphi = \delta_J W_k$, and take R_k independent of h. On a twice differentiable local Legendre branch,

$$\begin{aligned} \Gamma_k[\varphi,h] &= J\cdot\varphi - W_k[J,h] - \tfrac{1}{2}\varphi\cdot R_k\varphi, \\ \Gamma_{k,\varphi} + R_k\varphi &= J, \qquad W_{k,h} = -\Gamma_{k,h}. \end{aligned}$$

At fixed J, where $L_k := \Gamma_{k,\varphi\varphi} + R_k$ has a bounded inverse on the chosen response spaces, differentiation gives

$$\begin{aligned} \varphi_h &= -L_k^{-1}\Gamma_{k,\varphi h}, \\ W_{k,hh} &= -\Gamma_{k,hh} + \Gamma_{k,h\varphi}L_k^{-1}\Gamma_{k,\varphi h}. \end{aligned} \tag{4.37}$$

The first term is the change of the composite expectation at fixed mean field. The second is the response transmitted by adjustment of that field. In the notation $D + RL^{-1}S$, the same action gives $D = -\Gamma_{k,hh}$, $S = -\Gamma_{k,\varphi h}$ and $R = -\Gamma_{k,h\varphi}$. Thus source, readout and direct response are fixed together by the generating functional.

The choice of state variables determines where a collective singularity appears in this identity. For an even composite in a symmetric phase, for example, the mixed derivative with an odd fundamental mean field vanishes at the symmetric state, and $-\Gamma_{k,hh}$ carries the composite response. Including composite expectation values or propagators among the response variables exposes their adjustment in the inverse operator, as in composite-field and 2PI formulations [38, 39]. Application of the isolated-pole theorem then uses the analyticity of the direct and complementary terms in that particular representation.

At fixed composite sources the effective action obeys the Wetterich equation [37]

$$\partial_t \Gamma_k[\varphi, h] = \frac{1}{2}\,\mathrm{STr}\left[(\Gamma_k^{(2)}[\varphi, h] + R_k)^{-1}\partial_t R_k\right], \qquad t = \ln(k/k_0). \tag{4.38}$$

Functional derivatives with respect to h_i generate flows for composite insertions, including their mixing and contact terms. A formulation with explicit composite fields and scale-dependent bosonisation is given in Ref. [38]. Derivatives with respect to both h_i and the fundamental fields generate the vertices entering a projected response equation. After its self-consistent variables X have been specified, that equation defines a map $\mathfrak{F}_k(X; z)$, and the exact response operator is its Jacobian $D_X\mathfrak{F}_k$ as in Eq. (2.32). In the 2PI formalism, the self-energy and the Bethe–Salpeter kernel are obtained from successive functional derivatives of the same effective action [39]; the renormalisation of the resulting scalar two- and higher-point functions is treated in Ref. [40]. This gives an equivalent response representation only when the source pairing, propagators, renormalisation conditions, and continuation convention agree with those used to define $\mathfrak{F}_k$. A selected list of diagrams does not by itself establish that equivalence.

Equation (4.38) is exact but not closed on a finite collection of vertices. A practical computation chooses an operator basis, a field expansion, a momentum resolution, and a regulator; standard truncation schemes and their applications are reviewed in Ref. [41]. The resulting $\mathcal{B}_{k,N,\mathcal{R}}$ is a truncated response operator. Its omitted-vertex and channel-closure errors contribute to the derivative of the closure defect in Eq. (4.11). An eigenvalue reaching one in that operator is evidence within the stated implementation; stability under increasing N, improving momentum resolution, and varying R_k is required before extrapolating to the exact channel. The retained vertices and their closure specify which contributions enter each finite approximation.

The parameter $t = \ln(k/k_0)$ in Eq. (4.38) is an RG scale coordinate and orders features along a family of effective descriptions. A claim about Lorentzian formation additionally requires a real-time formalism, initial data, and a dynamical evolution law.

5 Generated examples and independent response tests

The examples follow a disturbance from its source through the unresolved variables to the observed response. An arithmetic merge first shows how those variables change a static susceptibility and retain a relaxation history. A reflecting-heat update then makes the collective feedback, critical mode and memory error explicit. In the field-theory examples, the mass spectrum determines how the response extends through space, while symmetry determines which source couplings are allowed.

5.1 An arithmetic fibre: merge, environment, and static response

This example separates preservation of a static source response from closure of the underlying update. The eliminated variables can be accounted for in the static response even though they must be retained, or represented by memory, for evolution.

The arithmetic merge construction of Ref. [9], on multiplicative and additive closure, motivates the ordered fixed-sum fibre used here. Relative to the finite-fibre expression resolution in that reference, the present fibre is its two-input additive value-level compression: it retains the ordered evaluated inputs while discarding their internal expression histories. We define and prove the finite data needed below. For $k \geq 2$ take the orthonormal basis $|m, k-m\rangle$, $1 \leq m \leq k-1$, and define N_1, N_2 by their displayed degrees. The total degree is $N_+ = kI$ on this fibre; the cross-term of its quadratic cost is

$$K_+ = N_+^2 - N_1^2 - N_2^2 = 2N_1N_2 = \mathrm{diag}\{2m(k-m) : 1 \leq m \leq k-1\}. \tag{5.1}$$

All these equalities follow on the basis. They specify an arithmetic cost, not a matter energy or a source susceptibility.

Write $C_k|m, k-m\rangle = c_m$. Invariance under every permutation of this basis forces equal coefficients, and $C_kC_k^* = 1$ forces $|c_m| = (k-1)^{-1/2}$. Requiring positive coefficients fixes the target phase. This proves locally the naturality of the normalised merge

$$C_k = \frac{(1, \ldots, 1)}{\sqrt{k-1}}. \tag{5.2}$$

A faithful orthogonal dilation $|m, k-m\rangle \mapsto |k\rangle \otimes f_m$ requires $k-1$ orthonormal environment vectors. That lower bound is attained by retaining the ordered input label, and all such minimal orthogonal dilations are unitarily equivalent. This is the faithful-environment statement of the arithmetic construction, not yet minimality for a chosen future observation protocol.

Since $N_+^2 = k^2 I$, its heat family descends exactly: $C_k e^{-tN_+^2} = e^{-tk^2} C_k$. The connected cost need not descend through the same merge. For $k = 4$ it gives

$$K_+ = \mathrm{diag}(6, 8, 6), \qquad C = \frac{(1,1,1)}{\sqrt{3}}, \qquad Ce^{-tK_+}(1,-1,0)^{\mathsf{T}} = \frac{e^{-6t} - e^{-8t}}{\sqrt{3}} \neq 0 \quad (t > 0). \tag{5.3}$$

The input vector lies in $\operatorname{Ker} C$, so Eq. (2.2) fails. Nor is compression a replacement semigroup: $f(t) = Ce^{-tK_+}C^* = (2e^{-6t} + e^{-8t})/3$ satisfies $f''(0) = 136/3 \neq 400/9 = f'(0)^2$, whereas a differentiable scalar semigroup with $f(0) = 1$ has $f''(0) = f'(0)^2$.

To define an actual static source response, use real amplitudes and the declared cost and source pairing

$$\Phi(X, J) = \tfrac{1}{2} X^{\mathsf{T}} K_+ X - JCX, \qquad K_+ X = C^* J, \qquad O(X) = CX. \tag{5.4}$$

Positive definiteness gives a unique minimiser. Its derivative with respect to the source gives

$$\mathscr{C}(4) = CK_+^{-1}C^* = \tfrac{1}{3}(\tfrac{1}{6} + \tfrac{1}{8} + \tfrac{1}{6}) = \frac{11}{72}, \qquad \mathcal{S}_+(4) = \mathscr{C}(4)^{-1} = \frac{72}{11}. \tag{5.5}$$

The Schur stiffness $\mathcal{S}_+(4)$ differs from $CK_+C^* = 20/3$. More generally,

$$C_k K_+^{-1} C_k^* = \frac{1}{k-1} \sum_{m=1}^{k-1} \frac{1}{2m(k-m)} = \frac{H_{k-1}}{k(k-1)}, \qquad H_j = \sum_{m=1}^{j} \frac{1}{m}.$$

The identity $1/[m(k-m)] = (1/m+1/(k-m))/k$ proves the sum. Minimising the quadratic cost at fixed $C_k X$ gives the reciprocal as the Schur stiffness, proving the required finite version of the exact arithmetic Schur theorem. Static elimination and the failure of memoryless heat descent are therefore compatible results. The effective stiffness measures the response after unresolved amplitudes have adjusted to the applied source. Their distinct relaxation rates still affect transient observations, so this static coefficient alone does not determine the coarse evolution.

For a concrete environment calculation at $k = 4$, put

$$v = (1,1,1)/\sqrt{3}, \qquad b = (1,-2,1)/\sqrt{6}, \qquad a = (1,0,-1)/\sqrt{2}.$$

In this basis

$$K_+ = \begin{pmatrix} 20/3 & -2\sqrt{2}/3 & 0 \\ -2\sqrt{2}/3 & 22/3 & 0 \\ 0 & 0 & 6 \end{pmatrix}. \tag{5.6}$$

The a coordinate is invisible to $C = v^*$ under the fixed heat family and affine source injections along v. The observational equivalence is exactly equality of $(X_1 + X_3, X_2)$: the two distinct exponentials e^{-6t}, e^{-8t} separate these two coordinates. They give a two-state realisation. A one-state linear realisation cannot produce both rates; equivalently its Hankel determinant would vanish, whereas $f(0)f''(0) - f'(0)^2 = 8/9 > 0$. Thus two states, or one retained coordinate and one residual, are minimal in the linear class for this restricted protocol. Allowing all fibre permutations changes the quotient: permuting a before heat action can make it visible, and the observation rows $Ce^{-tK_+}P$ span all three coordinates for fixed $t > 0$. The full three-state information must then be retained. The minimal observation space thus depends on the allowed protocol, whereas the faithful orthogonal environment preserves all input labels.

In the two visible coordinates (x, u), the heat equation has $\dot{x} = -20x/3 + (2\sqrt{2}/3)u$ and $\dot{u} = (2\sqrt{2}/3)x - 22u/3$. Hence

$$u(t) = e^{-22t/3}u(0) + \frac{2\sqrt{2}}{3}\int_0^t e^{-22(t-s)/3}x(s)\,ds.$$

The retained equation contains the initial residual and the memory kernel $(8/9)e^{-22t/3}$. The variable t denotes arithmetic heat evolution, not a physical clock. The cost pairing in Eq. (5.4) is explicitly supplied and does not identify an arithmetic source with a universal matter source.

5.2 A reflecting-window protocol with a visible critical mode

The mechanism combines spatial smoothing with an inhomogeneous modulation. Smoothing attenuates a nonconstant mode; modulation couples it to the constant mode, allowing a disturbance to return through the unresolved coordinate. We construct this update on a reflecting window and follow the resulting feedback through its static response, relaxation and memory.

The interval can be connected to the same Gaussian operation used in Section 2. The finite-window construction in Section 3 of Ref. [9] uses a calibrated local information

level and a global peak-equivalent capacity. We prove the required version here before choosing its boundary closure. The entropy calculation uses the continuous-density form of Rényi's entropy family (Ref. [42], Eq. (1.21)), with natural logarithms and a fixed Lebesgue reference measure; the endpoint rules are specified separately.

Calibration of the Gaussian correlation window. Fix a displacement unit and Lebesgue measure in that unit. For $b > 0$, the one-dimensional heat kernel of Eq. (2.10) gives the unit-peak profile

$$K_b(d) = \frac{g_{b^2/2}(d)}{g_{b^2/2}(0)} = e^{-d^2/(2b^2)}. \tag{5.7}$$

Let $\mathcal{I} : (0, 1] \to [0, \infty)$ be continuous, additive under products, and calibrated by $\mathcal{I}(e^{-1}) = 1$. Define the local endpoint by unit information loss and the global endpoint by complete peak-equivalent capacity:

$$\ell_{\mathrm{UV}} := \inf\{d > 0 : \mathcal{I}(K_b(d)) \geq 1\}, \qquad L_{\mathrm{IR}} := \frac{\int_{\mathbb{R}} K_b(d)\, dd}{K_b(0)}. \tag{5.8}$$

Then these endpoints are unique and

$$\ell_{\mathrm{UV}} = \sqrt{2}\, b, \quad L_{\mathrm{IR}} = \sqrt{2\pi}\, b, \quad \Delta_b = L_{\mathrm{IR}} - \ell_{\mathrm{UV}} = b(\sqrt{2\pi} - \sqrt{2}) > 0. \tag{5.9}$$

For $p_b = K_b / \int_{\mathbb{R}} K_b$, the same normalization gives $L_{\mathrm{IR}} = \exp H^{\mathrm{R}}_{\infty}(p_b)$, where $H^{\mathrm{R}}_{\infty}(p_b) = -\log \|p_b\|_{\infty}$ is the infinite-order Rényi entropy. The function $u \mapsto \mathcal{I}(e^{-u})$ is continuous and additive on $[0, \infty)$, hence linear; the unit calibration makes it u. Thus $\mathcal{I}(K_b(d)) = d^2/(2b^2)$, strictly increasing for $d > 0$, which gives the unique local endpoint. The Gaussian integral gives $\int_{\mathbb{R}} K_b = \sqrt{2\pi}\, b$, proving the global endpoint and window length. More explicitly, for $\alpha > 1$,

$$\int_{\mathbb{R}} p_b^{\alpha}\, dd = \frac{(\sqrt{2\pi}\, b)^{1-\alpha}}{\sqrt{\alpha}}, \qquad H^{\mathrm{R}}_{\alpha}(p_b) = \log(\sqrt{2\pi}\, b) + \frac{\log \alpha}{2(\alpha - 1)}.$$

Taking $\alpha \to \infty$ proves the asserted entropy characterization.

These endpoint rules use the full signed displacement line for capacity; there is no extra factor of one half. The resulting global carrier $[0, L_{\mathrm{IR}}]$ represents that mass at unit height, not the support of the Gaussian or an interval containing its entire probability. Resolving its part beyond unit local information gives the active window $[\ell_{\mathrm{UV}}, L_{\mathrm{IR}}]$. In the information coordinate its endpoints are 1 and π. The length is unique under the declared calibration and capacity rule, not under Gaussianity alone; a different information unit or endpoint rule would be different data. In particular, Shannon entropy would give the different Gaussian width $\sqrt{2\pi e}\, b$. Fixing the reference length unit above is essential when using a differential entropy as a width.

Reflecting closure and the operational scale. Translate the active window by $x = d - \ell_{\mathrm{UV}}$ to $[0, \Delta_b]$. Among separated homogeneous self-adjoint boundary conditions for the local Laplacian, preserving constants selects Neumann closure: an endpoint condition $af + cf' = 0$, $(a, c) \neq (0, 0)$, must have $a = 0$ because $f = 1$ is in the domain. Consequently

$f' = 0$ at each endpoint, giving zero diffusive flux there. Within this boundary class, preservation of a uniform profile therefore fixes the closure. The Neumann heat operator fixes the constant function and conserves mass by self-adjointness.

For the common-source realisation set $\Delta = \Delta_b$. The calculations below also hold for an independently specified $\Delta > 0$ with the same reflecting closure. Our convention is $H_\tau = e^{-\tau A}$; the normalisation $e^{-tA/2}$ in Ref. [9] has $\tau = t/2$. Its endpoint scale convention $t(d) = d^2/2$ therefore gives $\tau_{\mathrm{UV}} = b^2/2$ and $\tau_{\mathrm{IR}} = \pi b^2/2$. The protocol may choose either or any declared positive τ; these are averaging scales, not physical time.

Let $\mathcal{H} = L^2(0, \Delta)$. The nonnegative Neumann operator is $A = -\partial_x^2$ with domain $\{f \in H^2(0, \Delta) : f'(0) = f'(\Delta) = 0\}$. Its eigenfunctions and eigenvalues are

$$e_0 = \Delta^{-1/2}, \quad e_n = \sqrt{2/\Delta}\cos(n\pi x/\Delta), \quad E_n = (n\pi/\Delta)^2 \quad (n \geq 1), \qquad E_0 = 0. \tag{5.10}$$

Solving $-e'' = Ee$ with the two boundary conditions gives these modes; even extension to a 2Δ-periodic function and Fourier completeness give a complete orthonormal basis. Integration by parts gives $\langle f, Af\rangle = \|f'\|_2^2$. Equivalently the cosine diagonal realisation is self-adjoint with domain $\sum E_n^2 |f_n|^2 < \infty$. Thus $H_\tau = e^{-\tau A}$ is a bounded positive self-adjoint contraction, with $H_\tau e_n = e^{-\tau E_n} e_n$. Its kernel is

$$h_\tau(x, y) = \sum_{j \in \mathbb{Z}} [g_\tau(x - y + 2j\Delta) + g_\tau(x + y + 2j\Delta)],$$

the even periodisation of the line Gaussian. Fourier coefficients give the displayed Neumann eigenvalues and multiplier. This specifies the reflection operation; neither a finite interval nor reflection truncates its infinite spectrum. Line increments, reflection, and the following finite spectral projection are distinct steps.

Fix $\tau > 0$ and $\varepsilon \in \mathbb{R}$ with $0 < |\varepsilon| < 1$, let $q(x) = \cos(\pi x/\Delta)$, and let P_2 project onto $\mathrm{span}\{e_0, e_1\}$. Compose the declared operations

$$T = P_2 H_\tau M_{1+\varepsilon q} H_\tau P_2 = \begin{pmatrix} 1 & \gamma \\ \gamma & r^2 \end{pmatrix}, \qquad r = e^{-\tau E_1}, \qquad \gamma = \frac{\varepsilon r}{\sqrt{2}}. \tag{5.11}$$

The multiplication identities $\langle e_0, qe_0\rangle = \langle e_1, qe_1\rangle = 0$ and $\langle e_0, qe_1\rangle = 1/\sqrt{2}$ follow by elementary cosine integration. Heat action contributes the two diagonal factors $(1, r)$. The multiplier is bounded between $1 - |\varepsilon|$ and $1 + |\varepsilon|$, so $\langle v, Tv\rangle \geq (1 - |\varepsilon|)\|H_\tau v\|^2 > 0$ on this two-mode space. The finite projection completes the specified two-mode operation. Its amplitudes are Hilbert vectors, with positivity understood in the quadratic-form sense just proved. The factor r measures attenuation of the nonconstant profile in each heat step, while γ measures its coupling to the uniform profile through the spatial modulation.

Define the actual discrete update and observation by

$$X_{n+1} = \eta T X_n + e_0 h_n, \qquad O_n = e_0^* X_n, \qquad \eta \geq 0. \tag{5.12}$$

In the common-source specialization Δ is fixed by b through Eq. (5.9); otherwise it is a declared window parameter. The remaining τ, ε, η and the source pairing specify the protocol. The window length fixes E_1, the heat scale fixes r, the modulation fixes γ, and η sets the overall feedback strength per update. These relations connect the Gaussian calibration and boundary closure to the observed response.

Failure of one-state descent and minimal residual. Write $X_n = (x_n, u_n)$ and $a = \eta$, $\beta = \eta\gamma$, $d = \eta r^2$. For $\beta \neq 0$, the projection onto x fails fibre consistency, because $x_{n+1} = ax_n + \beta u_n + h_n$. Keeping u closes the actual system and

$$\begin{aligned} u_n &= d^n u_0 + \beta \sum_{j=0}^{n-1} d^{n-1-j} x_j, \\ x_{n+1} &= ax_n + h_n + \beta d^n u_0 + \beta^2 \sum_{j=0}^{n-1} d^{n-1-j} x_j. \end{aligned} \tag{5.13}$$

These identities follow by induction, including the empty sum at $n = 0$. If $0 \leq d < 1$, dropping lags $k \geq m$ gives the effective tail bound

$$|\text{discarded memory}| \leq \frac{\eta^2\gamma^2 d^m}{1-d} \sup_{j<n} |x_j|, \qquad |\text{initial residual}| = |\beta| d^n |u_0|. \tag{5.14}$$

The history bound is essential; critical forcing does not have a uniformly bounded history merely because $d < 1$.

For zero initial state, put $m_j = e_0^*(\eta T)^j e_0$. Direct multiplication gives $m_0 = 1$, $m_1 = \eta$, $m_2 = \eta^2(1+\gamma^2)$ and $m_0 m_2 - m_1^2 = \eta^2\gamma^2 > 0$ when $\eta > 0$. In an n-state linear time-invariant realisation a Hankel matrix factors as an observability matrix times a reachability matrix, so has rank at most n; this is the rank bound underlying linear minimal-realisation methods [2]. Two states are therefore necessary and sufficient for this transfer function. The one residual coordinate is minimal in that class. At $\eta = 0$ the determinant vanishes and no such two-state lower bound is claimed. Independently, observing x and its next unforced update recovers $u = (x_{n+1} - ax_n)/\beta$, so the observation-sufficient quotient for $\beta \neq 0$ separates both amplitude coordinates.

Actual stability, crossing, and response. The two eigenvalues, obtained from the characteristic polynomial, are

$$\lambda_\pm = \frac{1 + r^2 \pm \sqrt{(1-r^2)^2 + 4\gamma^2}}{2}, \qquad \eta_* = \lambda_+^{-1}. \tag{5.15}$$

Since $r^2 - \gamma^2 = r^2(1 - \varepsilon^2/2) > 0$ and the characteristic polynomial is negative at r^2 and at 1, one has $0 < \lambda_- < r^2 < 1 < \lambda_+$. Therefore $\|\eta T\| = \eta\lambda_+ < 1$ for $0 \leq \eta < \eta_*$. In the unmodulated limit $\varepsilon = 0$, the uniform mode has $\lambda_+ = 1$ and the other mode decays by r^2 per application of T. For nonzero modulation the two modes mix, raising λ_+ above one and lowering the critical gain η_*. The critical direction is therefore a collective profile of the full update. Reflection supplies the constant mode; modulation and feedback determine when its coupled response becomes critical. The actual update is a strict contraction there; constant forcing has a unique attracting fixed point. At η_* one eigenvalue reaches one, with control derivative $\partial_\eta(\eta\lambda_+) = \lambda_+ > 0$; the other is strictly below one and $d_* := \eta_* r^2 < 1$. Above η_* the largest mode grows. These are statements about the specified update, not about a stationary solver.

For exponential input $h_n = w^n h$, define the particular amplitude by $X_n = w^n X$. The transfer equation is $(wI - \eta T)X = e_0 h$. Put $w = 1 + z$; this is a discrete transfer variable, not squared spatial momentum. Differentiating the sourced update gives

$$\mathscr{C}_\eta(z) = e_0^*[(1+z)I - \eta T]^{-1} e_0 = \frac{1 + z - \eta r^2}{(1+z-\eta)(1+z-\eta r^2) - \eta^2\gamma^2}. \tag{5.16}$$

Schur elimination gives the inverse scalar denominator $1 + z - \eta - \beta^2/(1 + z - d)$, retaining the transformed hidden-state propagator. At η_*,

$$\mathscr{C}_{\eta_*}(z) = \frac{Z}{z} + \frac{1 - Z}{z + 1 - \eta_* \lambda_-}, \qquad Z = \frac{1 - \eta_* r^2}{1 - \eta_* \lambda_-} \in (0, 1). \tag{5.17}$$

The inequalities for λ_- prove the stated range of Z. Here $L(z) = (1 + z)I - \eta_* T$, its complementary inverse is bounded by $(1-\eta_*\lambda_-)^{-1}$ at zero, and $L'(0) = I$. Hence $G_0 = I$ on the one-dimensional critical space. The e_0 source sees that space, since an eigenvector with zero first component would require $\gamma = 0$. This verifies every hypothesis of Theorem 3.3 from the actual operations. The variable z is the transfer coordinate of this discrete update. In the QFT realisation below, invariant mass enters through the corresponding spectral representation. Here e_0 is the coarse uniform level and e_1 is the first spatial contrast. Heat attenuates the contrast, the inhomogeneous multiplier converts level into contrast and back, and the gain η closes the feedback loop. The critical threshold therefore belongs to a collective profile of the full update. At criticality the unforced fixed points form the critical line; constant nonzero forcing accumulates along its visible direction, so there is no unique forced stationary state at that endpoint.

The same decomposition displays critical slowing in the actual discrete update. Let $P_+ = (T - \lambda_- I)/(\lambda_+ - \lambda_-)$ be the orthogonal projector at λ_+ and $\mu_\pm = \eta\lambda_\pm$. Then $e_0^* P_+ e_0 = Z$, so zero initial state and constant input $h \neq 0$ give, for $0 < \eta < \eta_*$,

$$\frac{x_n}{h} = Z\frac{1 - \mu_+^n}{1 - \mu_+} + (1 - Z)\frac{1 - \mu_-^n}{1 - \mu_-}. \tag{5.18}$$

This follows from $X_n = \sum_{j=0}^{n-1} (\eta T)^j e_0 h$ and the two orthogonal spectral projectors. The slow contribution has relaxation length $(-\log \mu_+)^{-1} \sim (1 - \mu_+)^{-1}$ as $\eta \uparrow \eta_*$ in update steps, while its stationary susceptibility is $Z/(1-\mu_+)$. At criticality the first term becomes Zn: the source accumulates in the visible neutral direction. Thus the transfer-pole residue is also the weight of the slow response. A physical relaxation time additionally requires a calibration of these update steps.

The error from dropping the residual altogether can also be computed. With $C = (1, 0)$, $B = \eta T - zI$, $\bar{B} = a - z$, and the inherited scalar source and readout, the four defects are $E_B = (0, -\beta)$ and $E_S = E_R = E_D = 0$. On the common inverse domain, Theorem 4.1 gives

$$\bar{\mathscr{C}} - \mathscr{C} = -\frac{\beta^2}{(w - a)[(w - a)(w - d) - \beta^2]}, \qquad w = 1 + z. \tag{5.19}$$

Direct subtraction of $1/(w - a)$ and Eq. (5.16) verifies the same identity. Thus update closure failure alone produces a nonzero output error even when source, readout, and direct terms are individually compatible.

Schur reduction and finite-memory error. To expose terms hidden by a one-leg probe, allow two source columns $S = I_2$ and two readouts $R = I_2$, with zero full-system direct term. Let $w = 1 + z$, $H = (w - d)^{-1}$ and $L_{\text{eff}} = w - a - \beta^2 H$. Eliminating the second state gives

$$S_{\text{eff}} = (1, \beta H), \quad R_{\text{eff}} = (1, \beta H)^{\mathsf{T}}, \quad D_{\text{eff}} = \begin{pmatrix} 0 & 0 \\ 0 & H \end{pmatrix}, \quad \mathscr{C} = D_{\text{eff}} + R_{\text{eff}} L_{\text{eff}}^{-1} S_{\text{eff}}. \tag{5.20}$$

Solving the two linear equations proves the formula, including the off-diagonal observations and the nonzero direct term. In the time domain the second source also enters the hidden memory, as in Eq. (4.8).

For a finite-memory approximation use $H_m(w) = \sum_{k=0}^{m-1} d^k w^{-k-1}$ in all four terms, not just the effective denominator. On $|w| \geq w_0 > d$,

$$|H - H_m| \leq \frac{d^m}{w_0^m (w_0 - d)} =: \delta_m. \tag{5.21}$$

Thus the effective kernel, source, readout and direct-term errors are bounded respectively by $\beta^2 \delta_m$, $|\beta| \delta_m$, $|\beta| \delta_m$, and δ_m. Use $B_{\text{eff}} = I - L_{\text{eff}}$ to apply Theorem 4.1. On a compact invertible domain with $\sup |L_{\text{eff}}^{-1}| \leq M$, the condition $M \beta^2 \delta_m < 1$ gives $\sup |L_{\text{eff},m}^{-1}| \leq M/(1 - M\beta^2 \delta_m)$ by the Neumann series. Substitution gives an explicit convergent complete-response error bound. A small contour around the critical root, excluding the second pole and $w = d$, then gives the total-residue bound of Eq. (4.6). All tail bounds have supplied geometric majorants; they are not inferred from visually stable eigenvalues.

The pole displacement and residue error can be resolved more sharply at the critical parameter. The following consequence also distinguishes approximation of the memory kernel from approximation of its derivative.

Corollary 5.1 (Pole and residue under finite-memory truncation)**.** *Fix the two-mode model at $\eta = \eta_*$ and fix $d = \eta_* r^2 \in (0, 1)$ as $m \to \infty$. All $O(\cdot)$ constants below may depend on this fixed model, including d. Let $m \geq 1$ be an integer, and write $a = \eta_*$ and $\beta = \eta_* \gamma \neq 0$. In the transfer variable w, set*

$$L(w) = w - a - \frac{\beta^2}{w - d}, \qquad L_m(w) = w - a - \beta^2 H_m(w), \qquad Z = \frac{1}{L'(1)}. \tag{5.22}$$

Here $L(1) = 0$ and Z is the scalar residue in Eq. (5.17). For sufficiently large m, L_m has exactly one zero w_m in a fixed sufficiently small neighbourhood of 1. It is real and simple. The residue $Z_m = \operatorname{Res}_{w=w_m} L_m(w)^{-1}$ satisfies

$$w_m - 1 = -\frac{Z\beta^2}{1 - d}\, d^m + O(m d^{2m}), \tag{5.23}$$

$$Z_m - Z = \frac{Z^2 \beta^2}{1 - d}\, m d^m + O(d^m). \tag{5.24}$$

For the two-source response in Eq. (5.20), replace H by H_m in all four terms and denote its residue at w_m by Z_m. With $v = (1, \beta/(1 - d))^{\mathsf{T}}$ and $\mathsf{Z} = Z v v^{\mathsf{T}}$,

$$\|\mathsf{Z}_m - \mathsf{Z}\| = O(m d^m). \tag{5.25}$$

No uniform estimate as $d \uparrow 1$ is asserted. The control parameter remains η_*, and the residues are compared at their respective transfer poles.

Proof. The geometric sum gives the exact identity

$$L_m(w) - L(w) = \frac{\beta^2 d^m w^{-m}}{w - d}, \qquad L_m(1) = \frac{\beta^2 d^m}{1 - d}. \tag{5.26}$$

Choose a closed disk about 1 that excludes d, zero and the second zero of L, and on which $|w| > d$. The defect converges uniformly there. Rouché's theorem gives exactly one zero w_m, counted with multiplicity, for large m; reality of the coefficients makes it real. To obtain the sharper rate, put $t = d^m$. On $|w - 1| = Kt$ with any fixed $K > Z\beta^2/(1 - d)$, one has $w^{-m} = 1 + O(mt)$,

$$L(w) = Z^{-1}(w - 1) + O(t^2), \qquad L_m(w) - L(w) = \frac{\beta^2}{1 - d} t + O(mt^2).$$

Rouché's theorem on this smaller circle yields $w_m - 1 = O(t)$. Substitution in $L_m(w_m) = 0$ then proves Eq. (5.23).

Differentiating the defect at 1 gives

$$L_m'(1) - L'(1) = -\beta^2 d^m \left(\frac{m}{1 - d} + \frac{1}{(1 - d)^2} \right). \tag{5.27}$$

On $|w - 1| \le Kt$, the second derivative of the defect is $O(m^2 t)$, whereas L'' is bounded. Thus transport from 1 to w_m gives

$$L_m'(w_m) = Z^{-1} - \frac{\beta^2}{1 - d} mt + O(t + m^2 t^2).$$

Since $m^2 t \to 0$, taking the reciprocal proves Eq. (5.24). For the full matrix response, $H_m(w_m) = H(1) + O(t)$ by the geometric remainder and the root estimate. The direct term is analytic near w_m, and the residue equals $Z_m v_m v_m^{\mathsf{T}}$ with $v_m = (1, \beta H_m(w_m))^{\mathsf{T}} = v + O(t)$. Equation (5.25) follows. □

The nonzero leading coefficient in Eq. (5.24) shows that the factor m is present in the actual scalar residue error, not merely in an upper bound. This is an explicit instance of the derivative and displaced-root terms in Proposition 4.2: a memory defect of order d^m moves the pole by that order, while its derivative changes the residue by order md^m. Locating the critical response and predicting its strength therefore require different accuracies in the same finite-memory model. The extra derivative weights long lags more strongly, which accounts for the factor m from differentiating the tail.

Ref. [16] supplies supplementary finite-matrix implementations of this reduction. The response formula retains all four terms, with the row functional continued analytically in z. An elementary matter example is the sixteen-dimensional operator $D = \left(\begin{smallmatrix} 0 & rI_8 \\ rI_8 & 0 \end{smallmatrix}\right)$, $r > 0$, with one right coordinate designated neutral. Its charged compression has dimension fifteen. Wherever the complementary block is invertible, Proposition 3.1 makes direct inversion and elimination of the neutral coordinate identical. For sources and readouts with nonzero neutral components, this equality includes the neutral direct term. The coordinate labels alone assign neither a physical flavour nor a gravitational interpretation to this finite matrix.

Replicated matter and its quadratic source. For the companion compact-matter construction, consider a field-theoretic realisation with fixed family multiplicity. Its one-family carrier $\mathcal{E}_1$ is generated by the internal representation and covariant transport of Ref. [14]. The common family module in the companion application Ref. [15] is taken to be unitarily identified with $\mathcal{E}_g = \mathbb{C}^g \otimes \mathcal{E}_1$. At fixed common bosonic background with vanishing Higgs field, the regulated fermion integrals of the families are independent. With the same source variation of the domain, measure, spectral projectors and regulator in every copy,

$$K_g[h] = I_g \otimes K_1[h], \qquad W_g[h] = gW_1[h], \qquad \partial_a \partial_b W_g = g\big(\langle S_{1,a} S_{1,b}\rangle_c - \langle S_{1,ab}\rangle\big). \tag{5.28}$$

The generating functionals are evaluated at zero fermion sources; source-independent constants are immaterial. Their product makes the logarithms add, giving the second identity for the fermion contribution at that background. The third follows by twice differentiating the same Euclidean integral, including the measure identification and moving-domain contributions. The common physical TT compression therefore has the same factor g. This is the quadratic response used in the companion compact-endpoint calculation of Ref. [15], which takes $g = 3$. Higgs vertices and a displaced matter background are incorporated by differentiating their full action. This example fixes the direct term as well as the propagated source terms in the classification.

5.3 Scalar feedback and the fold value 4/27

For comparison with the transfer pole, consider the explicitly specified scalar feedback

$$\Pi_{\mathrm{d}} = \frac{\Pi_0}{(1 - V\Pi_{\mathrm{d}})^2}, \qquad F_x(y) = \frac{x}{(1-y)^2}, \quad y = V\Pi_{\mathrm{d}}, \quad x = V\Pi_0, \tag{5.29}$$

on $0 \le y < 1$, $x \ge 0$. Its fixed points satisfy $x = y(1-y)^2$. The derivative of this cubic is $(1-y)(1-3y)$, so it increases up to $y = 1/3$ and then decreases to zero. Thus

$$x_c = \frac{4}{27}, \quad y_c = \frac{1}{3}, \tag{5.30}$$

and $0 < x < x_c$ has exactly two admissible branches $y_- \in (0, 1/3)$ and $y_+ \in (1/3, 1)$. At $x = 0$ the only admissible fixed point is zero; at x_c the branches meet, and above it there is no fixed point in this domain. At a fixed point $F_x'(y) = 2y/(1-y)$, giving local attraction of y_- and repulsion of y_+. If $a < 1/3$ and $0 \le x \le a(1-a)^2$, then $F_x([0,a]) \subset [0,a]$ and $\sup_{[0,a]} F_x' \le 2a/(1-a) < 1$. The contraction theorem proves uniqueness and convergence within this invariant interval, not uniqueness among all fixed points. In particular the declared initialization $y_0 = 0$ selects y_- for $0 < x < x_c$.

Implicit differentiation of the same rule gives

$$\frac{dy}{dx} = \frac{1}{(1-y)(1-3y)}. \tag{5.31}$$

Writing $y = 1/3 + u$ gives $x = 4/27 - u^2 + u^3$. The susceptibility therefore diverges with inverse square-root magnitude as $x \uparrow x_c$ along either branch, with opposite signs. At

the endpoint $F'_x = 1$, so the strict contraction proof no longer applies. This is a fold in a control parameter, not automatically a simple pole in a propagation variable. Replacing the iteration by a preconditioned solver leaves the implicit derivative unchanged when sources are transformed as in Eq. (4.10), while solver stability can change. Even where iteration is the declared update here, a physical time or a physical branch-selection law requires its own model identification. Applying this feedback to an arithmetic or TT construction requires matching its state, source, normalisation and parameter maps. These maps determine whether the same fold mechanism is present.

Field-theory cross-checks

The following previously established channels test the additional QFT representation and individual parts of the general response theory. They examine spectral weight and symmetry constraints in physical representations, complementing the operation-based examples above.

5.4 A free field: preservation of an existing pole

For a canonically normalised free scalar field of mass m,

$$C_\phi(Q^2) = \frac{1}{Q^2 + m^2}. \tag{5.32}$$

Applying the external representation Eq. (2.12) gives

$$C_{\phi,\tau}(Q^2) = \frac{e^{-\tau Q^2}}{Q^2 + m^2}. \tag{5.33}$$

The multiplier is entire and nonzero at every finite Q^2. It therefore leaves the pole location and order unchanged. For $m = 0$ it also leaves the residue unchanged:

$$\lim_{Q^2 \to 0} Q^2 C_{\phi,\tau}(Q^2) = 1. \tag{5.34}$$

This verifies the pole-preservation statement surrounding Eq. (2.29); pole generation belongs to the dynamical step.

5.5 A free composite field: a bubble continuum

Let ϕ be a free real scalar of mass $m > 0$ and choose the normal-ordered composite

$$\mathcal{O}(x) = \frac{1}{\sqrt{2}} :\phi^2(x): . \tag{5.35}$$

Wick's contraction rule [43] gives the connected Euclidean two-point function

$$C_{\phi^2}(Q^2) = \int \frac{d^4p}{(2\pi)^4} \frac{1}{(p^2 + m^2)((p+Q)^2 + m^2)}. \tag{5.36}$$

After a momentum-independent subtraction, dimensional regularisation gives

$$C_{\phi^2,R}(Q^2) - C_{\phi^2,R}(0) = -\frac{1}{16\pi^2} \int_0^1 dx \, \ln\left[1 + \frac{x(1-x)Q^2}{m^2}\right]. \tag{5.37}$$

Analytic continuation has a branch point at $s = 4m^2$. With the spectral convention of Appendix A, the nonlocal part has

$$\rho_{\phi^2}(s) = \frac{1}{16\pi^2}\sqrt{1-\frac{4m^2}{s}}\,\theta(s-4m^2), \tag{5.38}$$

with the normalisation in Eq. (5.35). There is no isolated one-particle atom. This example separates a positive threshold density from an isolated pole. In the massless limit the continuum begins at $s = 0$. Its spatial response can be evaluated at separated points, where the subtraction at zero momentum is unnecessary. Write $r = |\mathbf{x}| > 0$. The four-dimensional massless propagator is $G(t,\mathbf{x}) = [4\pi^2(t^2+r^2)]^{-1}$, and Wick contraction with the normalisation in Eq. (5.35) gives $\langle \mathcal{O}(t,\mathbf{x})\mathcal{O}(0)\rangle_c = G(t,\mathbf{x})^2$. The response to a time-independent source is the zero-frequency kernel, obtained by integration over Euclidean time:

$$K_{\rm cont}(r) = \frac{1}{16\pi^4}\int_{-\infty}^{\infty}\frac{dt}{(t^2+r^2)^2} = \frac{1}{32\pi^3 r^3}, \qquad r > 0. \tag{5.39}$$

The substitution $t = r\tan\theta$ reduces the integral to $r^{-3}\int_{-\pi/2}^{\pi/2}\cos^2\theta\,d\theta$. Equivalently, the constant massless density in Eq. (5.38) gives the same result by integrating the static Yukawa kernels, $\int_0^\infty ds\, e^{-\sqrt{s}r}/(64\pi^3 r)$. Local subtraction terms are supported at $r = 0$ and leave this expression unchanged. Long range is therefore not synonymous with a particle pole. A continuum reaching zero mass superposes all small masses and produces the algebraic r^{-3} tail in this example. The zero-mass atom of Eq. (3.25) selects an isolated propagation channel and supplies the Coulombic $1/r$ term. The threshold tail is determined by the full spectral integral, whereas the isolated collective pole admits the local Fredholm decomposition.

5.6 An unbroken conserved current without a massless pole

Consider a free massive Dirac field and its exactly conserved global current $J_\mu = \bar\psi\gamma_\mu\psi$. The Ward identity makes the separated-point two-point function transverse,

$$\langle J_\mu(Q)J_\nu(-Q)\rangle = (Q_\mu Q_\nu - Q^2\delta_{\mu\nu})\Pi_V(Q^2) + \text{contact terms}. \tag{5.40}$$

It also fixes the charge form factor of a one-particle fermion at zero transfer. Nevertheless the vacuum spectral density of Π_V starts at the fermion–antifermion threshold,

$$\rho_V(s) \;\propto\; \left(1+\frac{2m^2}{s}\right)\sqrt{1-\frac{4m^2}{s}}\,\theta(s-4m^2), \tag{5.41}$$

and contains no massless atom. The example directly separates charge normalisation from the existence of a long-range vacuum channel.

5.7 A Goldstone channel: symmetry plus state information

Let an exact continuous global symmetry have conserved current J_A^μ, and let a local Lorentz-scalar field Φ satisfy

$$v_A \equiv \langle\Omega|\delta_A\Phi(0)|\Omega\rangle \neq 0. \tag{5.42}$$

The nonzero expectation value is state information: the vacuum does not share the symmetry of the action. Apply Eq. (3.22) to the mixed correlator. After subtracting polynomial contact contributions from that correlator, retain the order-parameter contact term on the right-hand side of the Ward identity. With the Fourier convention of Section 2, this gives

$$p_\mu G_A^\mu(p) = v_A, \qquad G_A^\mu(p) = \int d^d x\, e^{\mathrm{i}px} \langle \Omega | T J_A^\mu(x) \Phi(0) | \Omega \rangle. \tag{5.43}$$

Indeed, the Fourier transform of the divergence is $-\mathrm{i}p_\mu G_A^\mu$, and that of its Ward contact term is $-\mathrm{i}v_A$. For a Lorentz-invariant vacuum there is no independent vector, so Lorentz covariance gives $G_A^\mu(p) = p^\mu F_A(p^2)$. Consequently, modulo the contact terms already removed, the longitudinal form factor has the singular part

$$F_A(p^2) = \frac{v_A}{p^2 + \mathrm{i}0}, \tag{5.44}$$

which is the massless Goldstone pole [44]. The conclusion required both the exact identity and Eq. (5.42). With $v_A = 0$, the same Ward identity does not force the pole. For an operator carrying Lorentz indices, additional covariants may occur and the one-form-factor argument must be replaced by the corresponding tensor decomposition.

5.8 A separable response kernel

Let the response operator on $\mathcal{X}_\mathfrak{C}$ be rank one,

$$\mathcal{B}(z) = \lambda(z) |r\rangle\langle \ell|, \qquad \langle \ell | r \rangle = 1, \tag{5.45}$$

with λ analytic. The resolvent is obtained exactly,

$$[I - \mathcal{B}(z)]^{-1} = I + \frac{\lambda(z)}{1 - \lambda(z)} |r\rangle\langle \ell|. \tag{5.46}$$

Suppose $\lambda(0) = 1$ and $\partial_z \lambda(0) \neq 0$. For the response-space maps $\mathsf{S}_X : \mathbb{C}^m \to \mathcal{X}_\mathfrak{C}$ and $\mathsf{R}_X : \mathcal{X}_\mathfrak{C} \to \mathbb{C}^m$, the singular part is

$$C_{\mathrm{sing}}(z) = -\frac{\mathsf{R}_X |r\rangle\langle \ell| \mathsf{S}_X}{\partial_z \lambda(0) z} + O(1). \tag{5.47}$$

The pole disappears from the measured channel if either matrix factor vanishes, even though the kernel still has eigenvalue one. This solvable example contains every algebraic condition in Theorem 3.2. Only when the response is the unfiltered physical correlator and the hypotheses of Theorem 3.5 hold does the displayed residue equal the Hermitian positive-semidefinite atom Z_0. Thus a denominator zero alone is insufficient.

5.9 Spectral extraction and approximation control

In lattice, FRG, Dyson–Schwinger and Bethe–Salpeter calculations, the extraction of a spectral pole involves both the definition of the renormalised channel and control of its finite approximation. We discuss these requirements in turn.

Renormalised channel. The starting data are the action, vacuum or ensemble, spacetime signature and dimension. The operator basis, mixing matrix, EOM/BRST quotient and spin/internal projector specify the physical channel; the operator source–readout maps, response space, response projector and realisation maps specify S_X and R_X. In a gauge theory, the Slavnov–Taylor residuals and the treatment of ghost and gauge-fixing sectors are part of these data. Closure is tested by Eq. (3.2). If it fails, the calculation requires an enlarged block or Eq. (3.3).

Spectral observable. The spectral observable is the projected, connected Euclidean correlation matrix of centred insertions. Removed one-point products and subtracted contact polynomials have distinct origins and are recorded separately. For the Poincaré-invariant vacuum spectral branch of Appendix A, a candidate massless atom is tested through

$$Z_0 = \lim_{Q^2 \to 0} Q^2 \big[C(Q^2) - P(Q^2) \big] \tag{5.48}$$

or the corresponding positive residue matrix. The exchange of this limit with the spectral integral uses the continuum integrability condition of Proposition A.4. The spectral bridge requires this unfiltered projected matrix to be the same response used in the Fredholm factorisation, with identical vacuum, centring, source pairing, continuation and contact subtraction. An ensemble calculation must instead supply a spectral representation and diagnostic appropriate to that state. For a spectral-FRG calculation, the spectral representation and its continuation must be specified in the truncation. For example, Ref. [45] computes two- and four-point spectral functions in scalar ϕ^4 theory in three spacetime dimensions directly in a Lorentzian formulation.

Finite-volume scaling. In a massive theory, the large-volume behaviour of an isolated stable particle is treated in Ref. [46], and that of two-particle scattering levels in Ref. [47]. Applying a level-scaling test requires the finite-volume assumptions and spectral separation appropriate to the state being extrapolated. The massive-theory estimates do not supply a zero-mass extrapolation theorem. A pole claim requires a sequence of volumes and resolutions showing the scaling of the candidate energy, its matrix element, and its separation from multiparticle levels. When the infinite-volume continuum also reaches $s = 0$, the channel is *threshold-sensitive*: finite-volume level separation alone does not distinguish a zero-mass atom from continuous threshold weight. In the isolated-state setting, the result must survive the continuum and infinite-volume extrapolations.

Kernel test. In the two-particle realisation, the dimensionless operator $\mathcal{B} = \mathcal{K}_{2\mathrm{PI}} G_0$ is constructed in the same projected basis. Following its relevant eigenvalue branches in the complex external invariant z gives $\lambda(0)$ and $\partial_z \lambda(0)$. The associated left and right eigenvectors, together with the source and readout, give both matrix factors in Eq. (5.47). Onset along a control-parameter family also calls for a separate evaluation of $\partial_\eta \lambda(0; \eta_c)$, with higher-order analysis if it vanishes. At intermediate FRG scale this remains a flow diagnostic; a physical pole is a property of the endpoint response.

For a nonnormal discretised operator, sensitivity is assessed through the right and left eigenvector residuals and the condition number

$$\kappa_N = \frac{\|r_N\|\,\|\ell_N\|}{|\langle \ell_N, r_N\rangle|}. \tag{5.49}$$

The smallest singular value of $I-\mathcal{B}_N$, or an equivalent resolvent bound, should accompany the eigenvalue. A small value of $|1-\lambda_N(0)|$ without overlap, conditioning, and convergence information is not a pole determination.

Approximation analysis. Approximation errors are tested by controlled enlargement of the operator and momentum basis, regulator variation, quadrature refinement, and changes of finite-volume parameters. Stability under these changes tests sensitivity to the approximation. The observed spread is not a certified truncation bound; certification requires a remainder estimate relative to the limiting operator. Once the assumptions and normalisation of Proposition 2.3 are fixed, the Gaussian averaging prescription is held fixed during this analysis, while FRG regulator variation probes the truncation uncertainty.

For a continuum Fredholm claim, all approximants are realised on a common response space. On a compact set K in the z plane, a sufficient operator-norm convergence condition of the form

$$\sup_{z\in K} \|\widetilde{\mathcal{B}}_N(z) - \mathcal{B}(z)\| \longrightarrow 0, \tag{5.50}$$

together with convergence of $\partial_z\mathcal{B}_N$, $\mathsf{S}_{X,N}$, and $\mathsf{R}_{X,N}$ supplies the required approximation control. This condition is directly checkable when a derivation supplies an explicit remainder bound. Alternatively, one may bound the nested increments by a summable sequence and supply the computable tail estimate $\sum_{n\geq N}\varepsilon_n(K) \leq r_N(K) \to 0$ from Eqs. (C.28)–(C.29). This effective tail controls the continuum error once a consistency argument identifies the limit with the exact response Jacobian.

For a coarse-grained closure, the reference-state comparison starts with $E_{\tau_q}(X_*; z)$. If it is nonzero, an independent fixed-point displacement estimate is needed. Lipschitz constants for $D_X\mathfrak{F}_{\tau_q}$ and every base-point-dependent source or readout then convert that displacement into kernel and map-error terms. In the invertible-sector comparison, the remaining data are $\|D_X E_{\tau_q}\|$, $\|\mathcal{C}_{\tau_q}^{-1}\|$, the resolvent constants in Eqs. (4.20)–(4.31), and the transported-map discrepancies $\|\widetilde{\mathsf{S}}_X - \mathsf{S}_{\mathrm{ind}}\|$ and $\|\widetilde{\mathsf{R}}_X - \mathsf{R}_{\mathrm{ind}}\|$. A residue comparison concerns the reference root z_* and the perturbed root $\widetilde{z}_*$. Equation (4.24) and the analytic-variation bounds transport the Riesz numerator and $\partial_z\lambda$ between these points. This step is what permits the comparison of residues at their actual, generally different roots. If γ is a contour in the spectral ζ plane that encloses the eigenvalue one and remains in the resolvent set, the relevant Riesz projections are

$$\Pi_{\gamma,N}(z) = \frac{1}{2\pi\mathrm{i}} \oint_\gamma (\zeta - \widetilde{\mathcal{B}}_N(z))^{-1}\, d\zeta. \tag{5.51}$$

Their convergence is required alongside that of the eigenvalues. Isolated-eigenvalue approximation and its error bounds are treated in Chapter 5 of Ref. [7]; nonnormal resolvent sensitivity is the subject of Ref. [8]. When Eq. (5.50) is unavailable, checks for spectral

pollution and pseudospectral sensitivity remain necessary diagnostics, but do not by themselves certify convergence to the exact operator.

Numerical results and reproducibility. The action, normalisation conventions, regulator and basis determine the finite calculation; numerical tolerances and eigenvector residuals specify the accuracy of its evaluation. The software in Ref. [16] provides supplementary finite-matrix response calculations. The companion compact-spectrum application Ref. [15] uses common spectral data to evaluate effective couplings and mass scales; its spectral-tail bounds and endpoint sensitivities distinguish numerical approximation errors from changes in the scale prescription. For a scalar observable realised by a Wick polynomial, the one-particle and multiparticle contributions depend on its centred expansion and source pairing. Its use in Appendix A requires the positive vacuum representation and subtraction convention of that same observable. For a channel classification, the result comprises the four quantities in Eq. (3.27), subject to the compatibility assumptions in Section 3.7. A finite-truncation estimate must be distinguished from a limiting result supported by the convergence bounds above.

The five analytic examples separate pole preservation, continuum weight, current conservation without a pole, the Goldstone mechanism and source visibility at a kernel zero. Their closed-form responses allow individual parts of a numerical calculation to be compared with exact results. A different interacting channel still requires its own dynamical and convergence analysis.

6 Conclusions

Eliminating explicit variables preserves collective response when their effects on feedback, source coupling and observation are retained consistently. Hidden dynamical variables also leave their preparation and history in the reduced evolution. These are the physical contents of complete-response preservation and memory in the constructions studied here. Joint differentiable descent preserves $D + R(I - B)^{-1}S$; departures from it are measured by the four-defect identity of Theorem 4.1. Because the comparison map need not be invertible, this identity applies to reductions of the state space. The two resolvents in its update-defect term explain the enhanced sensitivity near criticality. Together with the displaced-root estimates, this identity carries irreversible reduction, reference-branch displacement and source–readout errors into a comparison of the same observable response and its residues.

The arithmetic and reflecting-heat examples show why agreement of static susceptibilities does not imply a memoryless coarse dynamics. A hidden state can be eliminated from a stationary response while remaining necessary for future observations. Keeping that state, retaining its memory kernel, and estimating a memory truncation are distinct reductions with different error terms. In the reflecting-heat example, finite-memory truncation makes this distinction quantitative: the pole displacement is $O(d^m)$, whereas the residue error has a nonzero leading term of order md^m at fixed $0 < d < 1$. The derivative of the memory remainder accounts for the different rates.

The two-mode system also relates its transfer pole to the response to constant forcing: the residue is the weight of the slow mode, whose relaxation length diverges at the critical update. In a spatial realisation with a constant contracted zero-mass residue, the Gaussian transform instead gives a smoothed $1/r$ kernel in three dimensions. The massless free composite gives a complementary example: its continuous spectral weight produces a static r^{-3} tail. The spatial decay thus records how the source couples to the low-mass spectrum. These interpretations use, respectively, the specified evolution and the static momentum representation of the same response.

Near an isolated critical point, the observed residue is controlled jointly by the slope of the zero, the source–readout overlaps and the resolvent bounds. The estimates here compare residues at their respective displaced roots. For the reciprocal Hilbert-space response of Theorem 3.3, self-adjointness at the endpoint, an isolated finite-dimensional critical subspace, a bounded complementary inverse and a positive crossing matrix give a positive-semidefinite finite-rank residue. When, in addition, all exact matching hypotheses of Theorem 3.5 hold, that coefficient is the zero-mass atom of the corresponding vacuum spectral measure. The free helicity covariance realises a positive null-shell residue. By contrast, a complete stress tensor obeys the TT exclusion of Proposition A.3 under its positive-metric, positive-energy, exact-covariance, conservation and unique-vacuum hypotheses.

The present estimates are local in the response parameter and compare specified reference branches on domains with isolated critical roots and the stated resolvent and transversality bounds. Within these domains, the four-map comparison quantifies how errors in a coarse description affect the observed pole and residue.

At the level of the framework, these constructions are tied together by a common physical criterion: a reduced description is adequate when it preserves the complete path from source to observation. The quotient specifies which distinctions can be forgotten, the response factorisation follows the disturbance through the retained state, Schur elimination records the feedback of omitted variables, and the Fredholm and spectral steps determine how a collective mode appears in the measured long-range response. In this organisation the same four response maps recur across generated operations, finite-memory models, and QFT channels.

A Response parameters and physical spectral measures

The spectral identification in Section 3 is conditional on matching the analytic response to the same unfiltered, connected vacuum covariance in a positive physical representation, with the projection, source pairing, continuation, subtraction and endpoint conventions fixed as in Theorem 3.5. We treat local physical insertions and covariances constructed directly on the physical helicity space. In each case we specify the positive measure, its continuation and its zero-mass atom. Under those exact matching hypotheses, these data identify a Laurent residue of the response with a vacuum spectral weight.

A.1 Response parameters and their representations

The internal Neumann eigenvalue E_n determines heat attenuation in Section 5.2; the transfer parameter there is $z = w - 1$, with w the discrete input frequency variable. The control η changes the actual update and is not z. The arithmetic heat parameter orders a cost semigroup, not a physical clock. Theorem 3.3 certifies a Laurent coefficient in the analytic parameter actually specified. For a finite self-adjoint A with orthonormal eigenvectors v_j, direct diagonalisation gives

$$S^*(wI - A)^{-1}S = \sum_j \frac{S^* v_j v_j^* S}{w - a_j}.$$

Each residue is positive, but this transfer spectral measure is not a Lorentzian invariant-mass measure. The latter uses $s = p^2$ and the Euclidean variable Q^2 under the assumptions below. Identifying these variables requires an actual representation, not a change of notation. Only a spatial Fourier variable gives the spatial Green function calculated later in this appendix.

The positive measure is obtained from connected Wightman two-point functions of centred physical operators. Time-ordered correlators, commutators, and their Euclidean continuations are related to that measure but do not carry identical positivity statements. The invariant-mass representation has its origins in Refs. [30, 31]; the vacuum framework used below is that of Ref. [48]. For tensor channels we state the physical-polarisation compression separately and check its positivity.

A.2 Physical realisations and their positive measures

Definition A.1 (Physical spectral realisation)**.** *A physical spectral realisation consists of a positive-energy unitary representation of translations, an invariant vacuum, centred source insertions and their connected covariance, with either of the following realisations.*

(i) Renormalised local physical operators act on the positive-metric vacuum Hilbert space. Their tensor Wightman measure is compressed onto the chosen physical polarisations.

(ii) The covariance is constructed on the reduced physical helicity space. The states carry the unitary Poincaré representation of those helicities, and sources are test sections of the corresponding polarisation bundle. Positivity, the spectrum condition and the continuation are established for this reduced covariance itself.

In both cases the invariant-mass disintegration, physical source pairing, Euclidean continuation and local subtraction terms are part of the realisation. The vacuum projection and integrability assumptions stated below apply to that same covariance.

A change of helicity frame is unitary on the source fibres. It preserves both the positive measure and the source-contracted covariance. For a directly constructed helicity covariance, the null-shell compression in Assumption A.2 is this fibre pairing. The following local-operator construction supplies the first realisation; Eq. (3.26) supplies the second.

For the invariant-mass spectral representation, work in a positive-metric physical representation with a Poincaré-invariant vacuum $|\Omega\rangle$ and the positive-energy spectrum condition. We assume that the joint spectral projection of the momentum operators at zero is $E_P(\{0\}) = |\Omega\rangle\langle\Omega|$: the vacuum is unique within the chosen representation. For the local realisation let O_i, $i = 1, \dots, n$, be a finite block of renormalised local physical operators, with $O_i^\dagger$ included when necessary, and set

$$v_i = \langle\Omega|O_i|\Omega\rangle, \qquad \Theta_i = O_i - v_i I. \tag{A.1}$$

The two-point matrix used throughout this appendix is connected:

$$\begin{aligned} W_{ij}(x) &= \langle\Omega|\Theta_i(x)\Theta_j^\dagger(0)|\Omega\rangle \\ &= \langle\Omega|O_i(x)O_j^\dagger(0)|\Omega\rangle - v_i\overline{v}_j. \end{aligned} \tag{A.2}$$

The omitted constant has Fourier transform $(2\pi)^d v_i\overline{v}_j\delta^{(d)}(p)$. This is a vacuum contribution at zero four-momentum, not a massless particle-shell contribution and not a contact polynomial. Since $E_P(\{0\})\Theta_j^\dagger|\Omega\rangle = 0$, the connected measure has no atom at $p = 0$. Its possible $\delta_0(ds)$ term instead arises from disintegration on the nonzero future null shell.

Thermal states and other ensembles may admit the response analysis of the main text, but their spectral representations generally depend separately on frequency and spatial momentum. The invariant-mass representation and the Q^2 atom diagnostic below are not asserted for those states.

Inserting the spectral resolution of the four-momentum operator yields a matrix-valued measure supported in the closed forward cone. For tensor operators write this Fourier-space Wightman measure as $d\Sigma_{ij,AB}(p)$, where A, B collect covariant tensor indices. Positivity is first a statement on the positive-metric physical Hilbert space: for every smooth compactly supported physical test section $f_i^A(p)$,

$$\sum_{i,j} \int_{\overline{V}_+} \overline{f_i^A(p)}\, d\Sigma_{ij,AB}(p)\, f_j^B(p) \geq 0. \tag{A.3}$$

Assumption A.2 (Positive physical-polarisation compression)**.** *For $s > 0$, a tensor or spin channel a is represented by a measurable orthonormal family $\{\varepsilon_{a,r}(p)\}_{r=1}^{N_a}$ in the positive-norm physical polarisation space. Its channel measure is obtained from the compression*

$$d\Sigma_{ij;rs}^{(a)}(p) := \overline{\varepsilon_{a,r}^A(p)}\, d\Sigma_{ij,AB}(p)\, \varepsilon_{a,s}^B(p), \tag{A.4}$$

followed by Lorentz-covariant disintegration with respect to $s = p^2$ and a positive contraction in the polarisation indices. Here a positive contraction means $X \mapsto \mathrm{tr}_{\mathrm{pol}}[H_a(p)X]$ for a specified measurable $0 \preceq H_a(p) \preceq I$ that transforms covariantly under a change of polarisation frame; $H_a = I$ gives the full irreducible block. If $s = 0$ is included, the endpoint compression is defined separately on the null shell by an orthonormal family spanning the specified physical helicities. The Euclidean tensor projector used in the channel is required, after analytic continuation, to induce this same helicity compression. Only the compressed forms are required to possess the zero-mass limit; no limit of the full massive-spin projector is assumed. For a scalar channel the compression is the identity. A measurable

change of physical polarisation frame is unitary on each polarisation fibre and does not affect positivity.

Under Assumption A.2, denote the invariant-mass part of the compressed spectral measure by $d\rho_{ij}^{(a)}(s)$, $s \geq 0$. A Hilbert-space compression and a positive contraction preserve Eq. (A.3). Hence, for every $c \in \mathbb{C}^n$ and every nonnegative Borel test function f,

$$\sum_{i,j} \bar{c}_i \int_0^\infty f(s)\, d\rho_{ij}^{(a)}(s) c_j \geq 0. \tag{A.5}$$

Thus $d\rho^{(a)}$ is positive semidefinite as a matrix-valued measure. This conclusion concerns the compressed physical channel, not an arbitrary scalar coefficient in a covariant tensor decomposition. The Fourier transform of a commutator contains the difference of positive- and negative-energy boundary values and is not itself a nonnegative scalar measure on all four-momenta.

A.3 The zero-mass stress-tensor channel

For a complete covariant stress tensor, positivity before any channel compression imposes an additional restriction at the null shell. This is stronger input than positivity of a projected response alone.

Proposition A.3 (Null-shell TT exclusion for a complete positive covariant stress tensor)**.** *In four-dimensional Minkowski spacetime, work in a positive-metric, positive-energy representation with a Poincaré-invariant vacuum $|\Omega\rangle$ satisfying $E_P(\{0\}) = |\Omega\rangle\langle\Omega|$, so that the vacuum is the unique zero-momentum state in the chosen representation. Let $T_{\mu\nu}$ be a Hermitian symmetric conserved tensor with an exact Lorentz tensor transformation law, not one defined only modulo gauge transformations, and assume its centred connected Wightman measure $d\Sigma_{\mu\nu,\rho\sigma}$ is locally finite and positive on all symmetric tensor test functions before TT compression. On the nonzero future null shell $H_0^+ = \{p : p^0 > 0,\ p^2 = 0\}$ it then has the form*

$$d\Sigma_{\mu\nu,\rho\sigma}(p)|_{H_0^+} = c\, p_\mu p_\nu p_\rho p_\sigma\, d\Omega_0(p), \qquad c \geq 0, \tag{A.6}$$

where $d\Omega_0$ is a fixed invariant measure on that shell. Its contraction with every conserved source is zero. Since the connected measure has no atom at $p = 0$, every conserved TT source matrix has zero invariant-mass atom at $s = 0$.

Proof. The nonzero null shell is a single Lorentz orbit. Its restricted measure has a covariant matrix density $H(p)$ relative to $d\Omega_0$: to see this, average compensated Lorentz translates against a smooth compactly supported Haar weight of integral one. Covariance leaves the measure unchanged. Since the orbit map is a submersion, this convolution smooths a distributional section along the orbit and hence gives the asserted density. Positivity of the measure gives $H(p) \succeq 0$. This argument is confined to one orbit and does not require a density in the mass variable s.

At a reference null momentum ℓ, factor $H(\ell) = VV^\dagger$, with V of full column rank; the case $H = 0$ is immediate. If $D(g)$ is the symmetric-tensor representation of an element

of the stabiliser of ℓ, covariance gives $D(g)H(\ell)D(g)^\dagger = H(\ell)$. Thus $D(g)V = VU(g)$, and a left inverse of V shows $U(g)U(g)^\dagger = I$. Null rotations are unipotent in D, so their restrictions U are both unitary and unipotent, hence the identity. Every column of V is therefore fixed by all null rotations.

For the component calculation, raise the tensor indices and choose a null tetrad (ℓ, n, e_1, e_2) with $\ell \cdot n = 1$ and $e_i \cdot e_j = -\delta_{ij}$. The null-rotation generators satisfy $N_k \ell = 0$, $N_k n = e_k$, and $N_k e_i = \delta_{ki}\ell$. Write a symmetric tensor, with juxtaposition denoting tensor product, as

$$u = a\ell\ell + bnn + c_1(\ell n + n\ell) + \sum_i d_i(\ell e_i + e_i\ell) + \sum_i f_i(ne_i + e_i n) + \sum_{i,j} g_{ij} e_i e_j, \qquad g_{ij} = g_{ji}.$$

Applying N_k to both indices and setting every coefficient to zero gives $b = 0$, $f_i = 0$, $d_i = 0$, and $g_{ij} = -c_1\delta_{ij}$. Consequently $u = a\ell\ell + c_1\eta$, where $\eta = \ell n + n\ell - \sum_i e_i e_i$. Conservation of the measure places every column of V in the transverse subspace; $\ell_\mu u^{\mu\nu} = c_1 \ell^\nu = 0$ then forces $c_1 = 0$. The range of $H(\ell)$ is therefore spanned by $\ell\ell$. Positivity and Lorentz transport give Eq. (A.6) with $c \geq 0$. For a conserved source J, $p_\mu J^{\mu\nu} = 0$ annihilates this measure. The absence of a connected atom at $p = 0$, proved above, completes the invariant-mass endpoint statement. □

This exclusion concerns the complete positive, exactly covariant tensor in the stated vacuum. It does not determine its spin-two spectrum at $s > 0$, nor does it follow by substituting $s = 0$ into a massive-spin projector. A tensor response of a generated system or a discrete transfer residue need not realise this Wightman tensor measure; the exclusion applies to such an object only if that representation and all the proposition's hypotheses are established. The general response theorems do not require this tensor representation.

A.4 Time-ordered and Euclidean correlators

Choose a spacelike subtraction point $Q_*^2 > 0$ and an integer $n_{\mathrm{sub}} \geq 0$ large enough for ultraviolet convergence. The projected Euclidean form factors then have the matrix-valued subtracted dispersion representation

$$C_{ij}^{(a)}(Q^2) = P_{ij}^{(a)}(Q^2) + (Q_*^2 - Q^2)^{n_{\mathrm{sub}}} \int_0^\infty \frac{d\rho_{ij}^{(a)}(s)}{(s+Q^2)(s+Q_*^2)^{n_{\mathrm{sub}}}}. \tag{A.7}$$

Here $P_{ij}^{(a)}$ is the finite subtraction polynomial, including the allowed local contact terms. For $n_{\mathrm{sub}} = 0$ the dispersive term in Eq. (A.7) reduces to the ordinary Stieltjes transform; a scheme-dependent local polynomial may still be present. Changing n_{sub} or Q_*^2 redistributes analytic terms between the integral and the polynomial but does not alter the spectral measure or an isolated pole residue. The Euclidean axioms of Ref. [35], with the reconstruction result corrected and extended in Ref. [36], provide a Hilbert-space interpretation when the full reconstruction hypotheses hold. Reflection positivity is one of these hypotheses, rather than a sufficient condition by itself.

Equation (A.7) keeps distinct the external Euclidean variable Q^2 and the Lorentzian spectral variable s. A Gaussian factor in external momentum multiplies the Euclidean correlator; it is not, by that operation alone, a redefinition of $d\rho(s)$ or a new Wightman theory.

A.5 Zero-mass atoms and the Euclidean diagnostic

Decompose the spectral measure in a channel as

$$d\rho(s) = Z_0\,\delta_0(ds) + d\rho_{\mathrm{c}}(s), \qquad \rho_{\mathrm{c}}(\{0\}) = 0, \tag{A.8}$$

where Z_0 is a positive-semidefinite matrix. The atom contributes Z_0/Q^2 to the singular part of Eq. (A.7); the remaining terms from its subtraction factor are analytic at the origin.

Proposition A.4 (Euclidean extraction of the atom)**.** *For every vector c, let $d\nu_c(s) = c^\dagger d\rho_{\mathrm{c}}(s)c/(s+Q_*^2)^{n_{\mathrm{sub}}}$. Suppose ν_c is locally finite at $s=0$, has no atom there, and obeys $\int_1^\infty s^{-1}d\nu_c(s) < \infty$. Then, entrywise and as a quadratic form,*

$$Z_0 = \lim_{Q^2\downarrow 0} Q^2\big[C(Q^2) - P(Q^2)\big]. \tag{A.9}$$

Proof. The atom term in Eq. (A.7), multiplied by Q^2, tends to Z_0. For a quadratic form c, the remaining term is

$$Q^2(Q_*^2 - Q^2)^{n_{\mathrm{sub}}} \int_{(0,\infty)} \frac{d\nu_c(s)}{Q^2+s}. \tag{A.10}$$

Apart from the bounded prefactor $(Q_*^2 - Q^2)^{n_{\mathrm{sub}}}$, the integrand is $Q^2/(Q^2+s)$. On $(0,1]$ it is bounded by one and converges pointwise to zero, so local finiteness, absence of an atom at zero, and dominated convergence apply. On $[1,\infty)$ it is bounded by Q^2/s, and the assumed weighted ultraviolet bound makes the tail vanish. Polarisation of the quadratic-form result gives the entrywise limit. The polynomial term is removed before the limit, and the result follows. □

Proposition A.4 applies to the unfiltered, connected vacuum correlator of the centred insertions. Theorem 3.5 connects it to a response residue after matching the vacuum, centring, operator projection, source pairing, continuation, subtraction, and endpoint correlator.

If the continuum falls outside the hypotheses of Proposition A.4, the endpoint requires a separate spectral analysis. Even when the limit exists, a branch cut can begin at $s=0$ and coexist with an atom. The decomposition (A.8) supplies the physical distinction that visual smoothness of Euclidean data cannot resolve.

The connection with long-range response is direct. For $d > 2$, away from coincident points,

$$\int \frac{d^dQ}{(2\pi)^d}\,\frac{e^{\mathrm{i}Q\cdot x}}{Q^2} = \frac{\Gamma(d/2-1)}{4\pi^{d/2}}\,\frac{1}{|x|^{d-2}}. \tag{A.11}$$

Thus a nonzero Z_0 produces a power-law tail, whereas the subtraction polynomial contributes only distributions supported at coincident points. In $d=2$ the corresponding

Green function is logarithmic. A continuum beginning at $s = 0$ can also generate long-distance power laws, but it is not an isolated atom and must be analysed with its threshold density.

A.6 Action of the Gaussian multiplier

Smearing both insertions with $\mathcal{H}_{\tau_Q/2}$ gives the external diagnostic map

$$C_{\tau_Q}(Q^2) = \mathcal{G}^{\mathrm{ext}}_{\tau_Q} C(Q^2) = e^{-\tau_Q Q^2} C(Q^2). \tag{A.12}$$

If $C(Q^2) = Z_0/Q^2 + C_{\mathrm{rem}}(Q^2)$ near the origin, with $Q^2 C_{\mathrm{rem}}(Q^2) \to 0$, then

$$C_{\tau_Q}(Q^2) = \frac{Z_0}{Q^2} + C_{\mathrm{rem}}(Q^2) - \tau_Q Z_0 + o(1). \tag{A.13}$$

The atom residue is unchanged. Conversely, multiplication of a regular form factor by an entire function cannot generate a pole. This proves the pole-preservation statement in Section 2.

The statement concerns the fixed-correlator multiplier. A positive KL measure is attached to the original Wightman correlator; the filtered object need not itself be a Stieltjes function. Moreover, $e^{-\tau_Q Q^2} P(Q^2)$ is no longer a contact polynomial; in position space it is a short-range smearing of a contact distribution. For this reason the physical contact subtraction and spectral measure are defined from the unfiltered correlator before Eq. (A.12) is applied. Pole preservation may be checked on the filtered object, whereas physical positivity must be checked on the original Wightman matrix.

The scale dependence of a full interacting correlator also enters through the effective vertices and response operator generated by integrating fluctuations.

A.7 Non-Hermitian and gauge-theory operators

For a non-Hermitian operator O, positivity is formulated with its centred insertion $\Theta = O - \langle O \rangle I$ and $\langle \Theta(x)\Theta^\dagger(0) \rangle$ or, for a mixing block, with the enlarged connected matrix containing Θ and $\Theta^\dagger$.

In a covariantly gauge-fixed formulation the state space before the physical quotient has indefinite metric, and elementary gauge-potential or ghost correlators need not possess positive spectral measures. The positivity used in Eq. (A.5) applies to physical BRST cohomology classes and to the physical-polarisation compression of Assumption A.2. Slavnov–Taylor identities control the gauge dependence and mixing of their representatives. Gauge invariance and the weight of a spectral atom are then assessed in the physical cohomology class and polarisation channel.

B Compatible operations, observations, and operator realisations

For the channel reduction in Section 3, invariance of a retained subspace and descent through a projection impose different block identities. We make this distinction explicit, then treat its renormalised-operator realisation and prove Proposition 3.6.

B.1 Projection, restriction, and quotient compatibility

For a linear update B and projector P, put $Q = I - P$. Invariance of the retained subspace requires $QBP = 0$, whereas descent through the coarse map $X \mapsto PX$ requires $PBQ = 0$. The first prevents escape from an initially retained state; the second prevents hidden states from changing the next observation. Indeed, $PX = PY$ means $X - Y \in \operatorname{Ran} Q$, so $PBX = PBY$ for all such pairs exactly when $PBQ = 0$. A readout descends through P exactly when $RQ = 0$. Neither a dimension cutoff nor smoothing proves either identity.

In the finite Neumann protocol the spectrum and bounded heat action are proved directly in Section 5.2; multiplication and projection are explicit bounded operations. Their finite composition defines the system studied, rather than asserting nonlinear closure for arbitrary omitted modes. For a gauge or other equivalence relation, one must likewise show that the update preserves equivalent states and that observations agree on them before passing to the quotient. The physical operator construction below implements this statement only in its stated BRST and separated-point setting.

Internal coordinates and external source coordinates have different covariance laws even without QFT. For a bounded state isomorphism T and a nonsingular source–output basis matrix M, set

$$B' = TBT^{-1}, \quad S' = TSM^*, \quad R' = MRT^{-1}, \quad D' = MDM^*.$$

Substitution in $D + R(I - B)^{-1}S$ proves $\mathscr{C}' = M\mathscr{C}M^*$. Full visible pole locations and orders and the rank of a residue are unchanged; positive residues remain positive. Pure internal similarity leaves the output unchanged. This argument supplies general response covariance; the proof below additionally identifies Wightman and renormalised-operator data in a QFT realisation.

B.2 Mixing and composite sources

For operators with common exact quantum numbers, renormalisation is a matrix problem,

$$O_R = ZO_B, \qquad \mu\frac{dO_R}{d\mu} = -\gamma O_R, \qquad \gamma = -\left(\mu\frac{dZ}{d\mu}\right)Z^{-1}, \tag{B.1}$$

with the sign convention used in Eq. (3.33). If the source term is $h_R^T O_R = h_B^T O_B$, then $h_B = Z^T h_R$. Two or more composite insertions generally require additional local counterterms polynomial in the sources. Those counterterms are the source-space origin of the contact polynomial $P_{ij}(Q^2)$ in Eq. (A.7).

Lorentz representation, internal charges, ghost number, and canonical dimension can be used to construct finite computational spaces $\mathcal{V}_N$. Membership in such a space does not imply invariance under mixing. In an effective field theory, increasing the dimension bound enlarges the space; conclusions at finite N are truncation dependent unless a power-counting or operator-norm estimate controls the omitted sector.

B.3 BRST, EOM, and descendant sectors

In an anomaly-free gauge theory, first form the on-shell space $\overline{\mathcal{V}} = \mathcal{V}/\mathcal{I}_{\mathrm{EOM}}$. The BRST differential s_{B} preserves the EOM ideal and therefore induces a nilpotent differential on $\overline{\mathcal{V}}$

under the standard regularity assumptions. A convenient off-shell basis used to determine counterterms contains three types of representatives:

$$O_G \in \operatorname{Ker} \mathsf{s}_{\mathrm{B}}, \qquad O_A = \mathsf{s}_{\mathrm{B}}\Psi, \qquad O_E = \mathcal{E}_F^R. \tag{B.2}$$

Here $\mathcal{E}_F^R$ is the renormalised insertion associated with a local change of integration variables. For a regulated bosonic integral with flat measure its defining identity is

$$\mathcal{E}_F = F^a \frac{\delta S}{\delta \phi^a} - \frac{\delta F^a}{\delta \phi^a}, \qquad \langle \mathcal{E}_F\, \mathcal{O} \rangle = \left\langle F^a \frac{\delta \mathcal{O}}{\delta \phi^a} \right\rangle. \tag{B.3}$$

Repeated field labels include spacetime integration. The identity follows by integrating the variation of $F^a \mathcal{O} e^{-S}$. For a general measure the second term is its field-space divergence, with the graded signs for fermionic variables. Renormalisation includes the Jacobian and operator-mixing counterterms so that this Schwinger–Dyson identity holds for renormalised insertions. With local variation support, its right-hand side has contact support in the other insertion coordinates. These are the EOM representatives used in $\mathcal{I}_{\mathrm{EOM}}$; derivatives of source-dependent insertions and the associated local source counterterms are retained in the direct response. The stress-tensor application uses the same renormalised representative in Ref. [13]. Under the standard assumptions for local composite-operator renormalisation, the renormalisation matrix can be chosen triangular among three sectors: physical cohomology classes, BRST-exact operators, and equation-of-motion operators [21]. Section 8.6 of Ref. [22] gives the local BRST cohomological formulation. BRST-exact insertions vanish between physical cohomology classes, while EOM insertions reduce to contact terms in separated-point correlators. After the mixing problem is closed, these statements justify taking the ghost-number-zero cohomology in Eq. (3.21); they do not permit the corresponding counterterms to be omitted while Z is being determined.

This cohomological quotient is an algebraic construction. Functional-analytic claims made with bounded channel projectors or source–readout maps use the explicit Banach completion and norm included in the channel data of Definition 3.4. The required maps must extend continuously to that completion; no canonical operator norm is presumed.

Infrared singularities can obstruct decoupling at exceptional kinematics. The on-shell gluonic matrix elements at zero momentum transfer in Ref. [23] provide a concrete example. The classification therefore forms the quotient at nonexceptional kinematics, retains all contact terms, and takes a zero-momentum limit only after the projected renormalised correlator has been constructed.

Total derivatives occupy a different position. When the boundary term vanishes, integration by parts gives

$$\int d^d x\, h\, \partial_\mu V^\mu = -\int d^d x\, (\partial_\mu h) V^\mu. \tag{B.4}$$

Thus a total derivative is redundant for a constant-source integrated insertion, or at zero insertion momentum when that limit is well defined. A nonconstant source generally leaves the source-gradient term on the right-hand side and does not make the insertion redundant. For a local insertion carrying momentum q, however,

$$[\partial_\mu V^\mu](q) = -\mathrm{i} q_\mu V^\mu(q), \tag{B.5}$$

and the descendant can contribute away from $q = 0$ or mix with other local operators. It is retained whenever the external kinematics resolve it.

B.4 Symmetry blocks and anomalies

Let $R(g)$ be the action of an exact global or spacetime symmetry on the renormalised operator space. A projector $\mathcal{P}$ defines a closed channel only if its range is stable under the mixing matrix and the unbroken symmetry action. When the regulator temporarily violates a symmetry, modified Ward or Slavnov–Taylor identities must be solved so that the physical $k \to 0$ block obeys the unmodified identity.

An anomalous current is not placed in the same block as an exactly conserved current. Its divergence contains a renormalised local insertion, and that insertion participates in the mixing problem. Similarly, explicit mass breaking and spontaneous breaking by the state are separate data: the former modifies the operator identity, while the latter leaves the identity intact and changes its matrix elements.

For the stress tensor, improvement transformations of the form

$$T_{\mu\nu} \longmapsto T_{\mu\nu} + (\partial_\mu\partial_\nu - \eta_{\mu\nu}\Box)X \tag{B.6}$$

with X a renormalised Lorentz scalar preserve conservation. A physical TT polarisation satisfies $p_\mu\epsilon^{\mu\nu} = 0$ and $\eta_{\mu\nu}\epsilon^{\mu\nu} = 0$, and hence

$$\epsilon^{\mu\nu}(-p_\mu p_\nu + \eta_{\mu\nu}p^2) = 0.$$

The improvement therefore vanishes under TT compression, including at the massless endpoint when the physical helicity compression of Appendix A exists. Contact terms are treated using the same subtraction convention. Spin-zero contributions and the overlaps of an individual representative with physical states need not be invariant.

Let $D_X = (\partial_\mu\partial_\nu - \eta_{\mu\nu}\Box)X$ be retained among the representatives of a closed block. When $(T, D_X, \dots) \mapsto (T+D_X, D_X, \dots)$ is a finite nonsingular constant basis change on that block, the covariance statement of Proposition 3.6 preserves the pole locations and orders of the full correlation matrix and the ranks of its isolated residues. No such invariance is asserted for a single entry after the descendant sector has been discarded.

B.5 Proof of external and internal covariance

Proof of Proposition 3.6. Let $O' = MO$ with constant, finite, and nonsingular M. In the same reference vacuum the centred insertions obey $\Theta' = M\Theta$. The connected Wightman matrices therefore satisfy

$$W'(x) = MW(x)M^\dagger. \tag{B.7}$$

Uniqueness of the distributional spectral decomposition then gives $d\rho' = M\,d\rho\,M^\dagger$, and analytic continuation gives $C' = MCM^\dagger$ including the transformed contact polynomial.

Suppose near $z = z_0$ the correlator matrix has a Laurent expansion

$$C(z) = \sum_{r=-p}^{\infty} C_r(z - z_0)^r, \qquad C_{-p} \neq 0. \tag{B.8}$$

Then $C'(z) = \sum_r MC_rM^\dagger(z-z_0)^r$. Because M and $M^\dagger$ are invertible, $MC_{-p}M^\dagger \neq 0$. The pole location and order are unchanged. For a simple spectral pole, $Z' = MZM^\dagger$ has the same rank as Z, and

$$c^\dagger Z' c = (M^\dagger c)^\dagger Z (M^\dagger c) \geq 0 \tag{B.9}$$

whenever Z is positive semidefinite.

A linear symmetry constraint $DC = 0$ is carried to $(DM^{-1})C' = 0$ after multiplication on the right by $M^\dagger$. Thus the dimensions of its forbidden, constrained, and allowed subspaces are unchanged.

Now let $T : \mathcal{X}_{\mathfrak{C}} \to \mathcal{X}'_{\mathfrak{C}}$ be a bounded response-state isomorphism. Direct substitution gives

$$\mathsf{R}_{X,T}(I - \mathcal{B}_T)^{-1}\mathsf{S}_{X,T} = \mathsf{R}_X T^{-1} T (I-\mathcal{B})^{-1} T^{-1} T \mathsf{S}_X = \mathsf{R}_X (I-\mathcal{B})^{-1}\mathsf{S}_X. \tag{B.10}$$

Thus T changes internal coordinates by similarity but does not transform the physical output matrix. The response projector transforms as $P_{X,T} = TP_XT^{-1}$, so $Q_{X,T}\mathcal{B}_T P_{X,T} = T(Q_X\mathcal{B}P_X)T^{-1}$; response closure is invariant. Combining T with the external basis change M instead gives

$$\begin{aligned}&\mathsf{R}'_X(I-\mathcal{B}')^{-1}\mathsf{S}'_X \\ &\quad = M\mathsf{R}_X T^{-1}T(I-\mathcal{B})^{-1}T^{-1}T\mathsf{S}_X M^\dagger = M\mathsf{R}_X(I-\mathcal{B})^{-1}\mathsf{S}_X M^\dagger,\end{aligned} \tag{B.11}$$

and the same congruence holds for C_{reg}.

Similarity preserves the response spectrum and algebraic multiplicities. At a simple critical eigenvalue, $|r'\rangle = T|r\rangle$ and $\langle \ell'| = \langle \ell| T^{-1}$, whence

$$\mathsf{R}'_X|r'\rangle = M\mathsf{R}_X|r\rangle, \qquad \langle \ell'|\mathsf{S}'_X = \langle \ell|\mathsf{S}_X M^\dagger. \tag{B.12}$$

Invertibility of M preserves vanishing or nonvanishing of both overlaps, and Eq. (3.9) gives $Z'_{\mathrm{alg}} = MZ_{\mathrm{alg}}M^\dagger$. These observations prove Proposition 3.6. □

B.6 Scale-dependent bases and finite schemes

For $O' = M(\mu)O$, differentiation of Eq. (3.28) and use of Eq. (B.1) give

$$\mu\frac{dO'}{d\mu} = -\left[M\gamma M^{-1} - \left(\mu\frac{dM}{d\mu}\right)M^{-1}\right]O', \tag{B.13}$$

which proves Eq. (3.33). Physical poles of the exact correlator remain unchanged by a finite nonsingular scheme transformation, but the coordinates of the residue, Wilson coefficients, and anomalous-dimension matrix change. At finite truncation, two implementations need not be related by an exact similarity transformation because the transformed operator basis can have components outside $P_N\mathcal{V}_{\mathrm{phys}}$ and its realised response can leave the retained range of a response projector $P_{X,N}$. The off-block response operators

$$\Delta_N^{\mathrm{out}} = (I_X - P_{X,N})\mathcal{B}P_{X,N}, \qquad \Delta_N^{\mathrm{in}} = P_{X,N}\mathcal{B}(I_X - P_{X,N}) \tag{B.14}$$

measure the direct coupling omitted by the compression. If they do not vanish, either the Schur complement Eq. (3.3) or a convergence argument such as Proposition C.1 is required. Regulator and basis variation are approximation diagnostics, not proofs of exact invariance.

C Descent, Gaussian, kernel, and error proofs

The proofs in this appendix supply three analytic steps used in the main text: differentiability through a Banach quotient, the Gaussian averaging estimates, and control of isolated response residues. In the last step, the resolvent identity first controls the spectral projection; root displacement and variation of the source and readout then enter the residue estimate of Proposition 4.2.

C.1 Regular descent through a Banach quotient

Proof of regular Banach descent. Put $K = \operatorname{Ker} C$ and let $q : \mathcal{X} \to \mathcal{X}/K$ be the quotient map, with

$$\|qu\|_{\mathcal{X}/K} = \inf_{k\in K} \|u + k\|_{\mathcal{X}}.$$

Since K is closed, $\mathcal{X}/K$ is Banach. The map

$$\widehat{C} : \mathcal{X}/K \longrightarrow \mathcal{Y}, \qquad \widehat{C}(qx) = Cx,$$

is a bounded bijection and hence a bounded isomorphism. The quotient map q is open: for every $\rho > 0$, $q(B_{\mathcal{X}}(0,\rho)) = B_{\mathcal{X}/K}(0,\rho)$. Thus $q(U)$ is open, and so is $C(U) = \widehat{C}(q(U))$. Fibre consistency defines a unique map $\widetilde{F} : q(U) \times V \to \mathcal{Y}$ by

$$\widetilde{F}(qx, h) = CF(x, h), \qquad \bar{F}(y, h) = \widetilde{F}(\widehat{C}^{-1}y, h).$$

Fix $(x, h) \in U \times V$ and choose $\rho > 0$ with $B_{\mathcal{X}}(x, \rho) \subset U$. If $k \in K$, then $CF(x+tk, h) = CF(x, h)$ for all sufficiently small t; differentiation at $t = 0$ gives

$$CD_xF(x, h)k = 0.$$

It follows that

$$\widetilde{A}_{x,h}(qu) := CD_xF(x, h)u$$

is a well-defined bounded operator on $\mathcal{X}/K$. Indeed, taking the infimum over representatives gives

$$\|\widetilde{A}_{x,h}\| \le \|C\| \, \|D_xF(x, h)\|.$$

Let $\eta \in \mathcal{X}/K$ satisfy $\|\eta\| < \rho/2$. By the definition of the quotient norm there is a representative u with $qu = \eta$ and $\|u\| \le 2\|\eta\| < \rho$. For j small enough that $h + j \in V$, fibre consistency and the joint Fréchet expansion at (x, h) give

$$\begin{aligned} &\widetilde{F}(qx + \eta, h + j) - \widetilde{F}(qx, h) - \widetilde{A}_{x,h}\eta - CD_hF(x, h)j \\ &\qquad = C\big[F(x + u, h + j) - F(x, h) - D_xF(x, h)u - D_hF(x, h)j\big]. \end{aligned} \tag{C.1}$$

The right-hand side is $o(\|u\| + \|j\|) = o(\|\eta\| + \|j\|)$. This estimate is uniform over the chosen near-minimal representatives because $\|u\| \le 2\|\eta\|$; its left-hand side depends only on η. Hence $\widetilde{F}$ is Fréchet differentiable at (qx, h).

It remains to verify continuity of the derivative. If $(\zeta, h') \to (qx, h)$ in $q(U) \times V$, choose u_ζ with $q(x + u_\zeta) = \zeta$ and $\|u_\zeta\| \le 2\|\zeta - qx\|$. For ζ sufficiently close to qx, $x + u_\zeta \in U$.

Both $CD_xF(x+u_\zeta, h')$ and $CD_xF(x,h)$ annihilate K, and therefore their induced quotient operators satisfy

$$\|\widetilde{A}_{x+u_\zeta,h'} - \widetilde{A}_{x,h}\| \le \|C\|\,\|D_xF(x+u_\zeta,h') - D_xF(x,h)\| \longrightarrow 0.$$

The same argument applied to CD_hF proves continuity of the source derivative. Thus $\widetilde{F}$, and consequently $\bar{F}$, is C^1. Composition with $\widehat{C}^{-1}$ gives precisely Eq. (2.3). The induced derivatives do not depend on a representative, either by the construction above or by uniqueness of the Fréchet derivative. Replacing CF by a fibre-constant C^1 observation O proves the final assertion in the same way. □

This proof also identifies the bounded isomorphism $\widehat{C} : \mathcal{X}/\operatorname{Ker} C \to \mathcal{Y}$ supplied by the quotient norm. Its inverse is an inverse on equivalence classes, not an inverse of the information-losing map C. The error identity of Theorem 4.1 applies more generally to any bounded linear comparison map.

C.2 Differentiating exact block elimination

For fixed complementary projectors, abbreviate the blocks by B_{ij} and put $H = (I - B_{22})^{-1}$. A dot denotes a total derivative along the declared parameter and reference branch. Then $\dot{H} = H\dot{B}_{22}H$, and the four Schur derivatives are

$$\begin{aligned}
\dot{B}_{\text{eff}} &= \dot{B}_{11} + \dot{B}_{12}HB_{21} + B_{12}\dot{H}B_{21} + B_{12}H\dot{B}_{21},\\
\dot{S}_{\text{eff}} &= \dot{S}_1 + \dot{B}_{12}HS_2 + B_{12}\dot{H}S_2 + B_{12}H\dot{S}_2,\\
\dot{R}_{\text{eff}} &= \dot{R}_1 + \dot{R}_2HB_{21} + R_2\dot{H}B_{21} + R_2H\dot{B}_{21},\\
\dot{D}_{\text{eff}} &= \dot{D} + \dot{R}_2HS_2 + R_2\dot{H}S_2 + R_2H\dot{S}_2.
\end{aligned} \tag{C.2}$$

They follow by differentiating $(I - B_{22})H = I$ and the four product formulae. If the projectors depend on the parameter, first differentiate the projected blocks themselves, including every projector derivative. Together with $\dot{B} = \partial_a B + D_X B\,\dot{X}_*$ and the corresponding identities for S, R, D, these formulae retain the actual reference-branch dependence. The product rule applies to all four maps, including their source and direct-response contributions.

C.3 Uniqueness of the Gaussian semigroup

Proof of Proposition 2.3. Write

$$\varphi_\tau(q) = \int_{\mathbb{R}^d} e^{\mathrm{i}q\cdot x}\, d\mu_\tau(x). \tag{C.3}$$

Weak continuity and convolution give $\varphi_{\tau+s}(q) = \varphi_\tau(q)\varphi_s(q)$ and $\varphi_\tau(q) \to 1$ as $\tau \downarrow 0$. Orthogonal invariance includes $x \mapsto -x$, so φ_τ is real. Moreover, $\varphi_\tau(q) = \varphi_{\tau/(2n)}(q)^{2n} \ge 0$. It is in fact strictly positive. If $\varphi_\tau(q) = 0$ for some $\tau > 0$, then

$$0 = \varphi_\tau(q) = \varphi_{\tau/n}(q)^n$$

would imply $\varphi_{\tau/n}(q) = 0$ for every positive integer n, in contradiction with $\varphi_{\tau/n}(q) \to \varphi_0(q) = 1$. The continuous real logarithm is therefore defined, and the scalar semigroup has the form

$$\varphi_\tau(q) = e^{-\tau\psi(q)} \tag{C.4}$$

with $\psi(q) \geq 0$ continuous.

Square-root self-similarity also gives $\varphi_\tau(q) = \varphi_1(\sqrt{\tau}q)$. Comparison with Eq. (C.4) yields $\psi(rq) = r^2\psi(q)$ for $r > 0$. Orthogonal invariance then gives $\psi(q) = D|q|^2$. Finite nonzero covariance makes D finite and strictly positive. Hence

$$\widehat{\mu}_\tau(q) = e^{-D\tau|q|^2}, \tag{C.5}$$

which proves Proposition 2.3. □

The exact square-root scaling fixes the homogeneity degree at two; removing the second-moment assumption alone does not change that conclusion. The stated covariance condition fixes a finite, nonzero scale. Without orthogonal invariance, the Gaussian covariance need not be isotropic. Non-Gaussian stable semigroups require a different scaling exponent, $\tau^{1/\alpha}$ with $0 < \alpha < 2$, and relaxation of the finite-second-moment assumption. Dropping exact self-similarity permits symmetric jump components, including ones of finite variance; the retained orthogonal invariance excludes a nonzero drift.

C.4 Tensor lift and suppression away from zero transfer

Proof of Theorems 2.4 and 2.5. The Fourier transform converts convolution in the ith coordinate into multiplication by $e^{-a_i\tau|p_i|^2}$. These smooth multipliers and all their derivatives are polynomially bounded, so they act continuously on tempered distributions. Since the coordinate convolutions commute, their tensor product gives

$$\prod_{i=1}^{n} e^{-a_i\tau|p_i|^2} = \exp\left[-\tau\sum_{i=1}^{n} a_i|p_i|^2\right]. \tag{C.6}$$

For the routed statement, $\widehat{F}_{\mathrm{red}}$ is part of the data after the overall delta distribution has been factored out. Write $\mathsf{W} = \mathrm{diag}(a_1 I_d, \ldots, a_n I_d)$ and let $T(P, \boldsymbol{\xi}) = AP + L\boldsymbol{\xi}$ be the adapted linear coordinate isomorphism of Theorem 2.4. With $g_\tau(p) = e^{-\tau p^{\mathsf{T}}\mathsf{W}p}$, the distributional identity in Eq. (2.17) means that for every $\Phi \in \mathcal{S}(\mathbb{R}^d \times \mathbb{R}^r)$,

$$\langle \widehat{F}_{\mathrm{full}}, \Phi\rangle = (2\pi)^d \langle \widehat{F}_{\mathrm{red}}, \Phi(0,\cdot)\rangle, \qquad \int_{\mathbb{R}^d} \delta^{(d)}(P) f(P)\, dP = f(0). \tag{C.7}$$

Multiplication by the smooth Gaussian is therefore computed without restricting an arbitrary distribution to $P = 0$:

$$\begin{aligned}\langle g_\tau(T)\widehat{F}_{\mathrm{full}}, \Phi\rangle &= (2\pi)^d \left\langle \widehat{F}_{\mathrm{red}}, g_\tau(T(0,\cdot))\Phi(0,\cdot)\right\rangle \\ &= (2\pi)^d \left\langle \widehat{F}_{\mathrm{red}}, e^{-\tau\boldsymbol{\xi}^{\mathsf{T}}L^{\mathsf{T}}\mathsf{W}L\boldsymbol{\xi}}\Phi(0,\boldsymbol{\xi})\right\rangle.\end{aligned} \tag{C.8}$$

Since $T(0,\boldsymbol{\xi}) = L\boldsymbol{\xi}$ parametrises all of $\operatorname{Ker}\Pi$, this is exactly the reduced action in Eq. (2.19), with $M_L = L^{\mathsf{T}}\operatorname{diag}(a_1 I_d, \dots, a_n I_d)L$. This matrix is positive semidefinite. On a subspace V on which $M_L \succeq cI$, one has $\boldsymbol{\xi}^{\mathsf{T}} M_L \boldsymbol{\xi} \geq c\|\boldsymbol{\xi}\|^2$. This proves Theorem 2.4, including its bound and the absence of a conclusion on $\operatorname{Ker} M_L$.

For Theorem 2.5(i), $q^{\mathsf{T}} M q \geq \delta^2$ on Ω_δ, and hence

$$\|e^{-\tau q^{\mathsf{T}} M q} A\|_{L^\infty(\Omega_\delta)} \leq e^{-\tau\delta^2}\|A\|_{L^\infty(\Omega_\delta)}. \tag{C.9}$$

For part (ii), put $\chi_\tau(q) = e^{-\tau q^{\mathsf{T}} M q}\chi(q)$. Each derivative of χ_τ is a finite sum of a polynomial in q and τ, multiplied by $e^{-\tau q^{\mathsf{T}} M q}$ and a derivative of χ. The support is fixed inside Ω_δ, so every Schwartz seminorm is bounded by a polynomial in τ times $e^{-\tau\delta^2}$ and tends to zero. Continuity of the tempered distribution A gives

$$\langle A_\tau, \chi\rangle = \langle A, \chi_\tau\rangle \longrightarrow 0. \tag{C.10}$$

Finally, Eq. (2.26) gives $\|A_\tau^{\mathrm{full}}\| \leq Ce^{-\tau(\delta^2 - a)}$, proving part (iii). □

C.5 Response factorisation and the closure defect

Proof of Propositions 2.6 and 4.2. Step 1: response factorisation. Apply the Banach-space implicit-function theorem to $G(X, J) = X - \mathfrak{F}_{\tau_q}(X; z, \eta) - \mathsf{S}_{X,\tau_Q}(z)J$. The derivative $D_X G = I - \mathcal{B}$ is invertible by hypothesis, so a unique differentiable solution branch exists locally. Differentiating Eq. (2.30) at $J = 0$, with $\delta X = D_J X\,\delta J$ and $\mathcal{B} = D_X\mathfrak{F}_{\tau_q}(X_*; z, \eta)$, one obtains

$$(I - \mathcal{B})\delta X = \mathsf{S}_{X,\tau_Q}\delta J. \tag{C.11}$$

Where $I - \mathcal{B}$ is invertible, composing its solution with the readout and differentiating the direct term in Eq. (2.31) gives Eq. (2.33). Under the hypotheses of Theorem 3.2, the same formula has the meromorphic continuation used below. This proves Proposition 2.6.

Step 2: transport of the closure defect. Assume $\mathcal{C}_{\tau_q}$ is bounded and linear and that the two maps in Eq. (4.11) are Fréchet differentiable. The chain rule gives

$$D_X E_{\tau_q}(X; z) = \mathcal{C}_{\tau_q} D_X\mathfrak{F}(X; z) - D_X\mathfrak{F}_{\tau_q}(\mathcal{C}_{\tau_q}X; z)\mathcal{C}_{\tau_q}. \tag{C.12}$$

Thus an exact commuting coarse-graining has the response intertwining Eq. (4.13); in an approximate closure, $D_X E_{\tau_q}$ measures its failure at the same base point. Compactness, analyticity, and threshold isolation are additional properties of the realised response family.

For Proposition 4.2, Eq. (4.14) and Eq. (4.12) first give $\mathfrak{F}_{\tau_q}(\bar{X}_*; z) = \bar{X}_*$. Thus the derivative at $\bar{X}_*$ is the response Jacobian at an actual coarse solution, rather than at a transported trial point. On the comparison spaces, multiply Eq. (C.12) on the right by $\mathcal{C}_{\tau_q}^{-1}$ to obtain

$$\big[D_X E_{\tau_q}(X_*(z); z)|_Y\big]\mathcal{C}_{\tau_q}^{-1} = \mathcal{B}_{\mathrm{ind}}(z) - \mathcal{B}_{\tau_q}(z) = -\Delta\mathcal{B}(z), \tag{C.13}$$

This proves the first norm estimate.

Step 3: spectral projection and root displacement. For $R(\zeta, z) = (\zeta - \mathcal{B}_{\mathrm{ind}}(z))^{-1}$, the condition $K_\gamma \epsilon < 1$ permits the Neumann expansion of the perturbed resolvent. The resolvent identity gives, uniformly on $K \times \gamma$,

$$\|R_{\tau_q}(\zeta, z) - R(\zeta, z)\| \le \frac{K_\gamma^2 \epsilon}{1 - K_\gamma \epsilon}. \tag{C.14}$$

Integrating $(2\pi\mathrm{i})^{-1}[R_{\tau_q}(\zeta, z) - R(\zeta, z)]\, d\zeta$ around γ proves Eq. (4.21). The same resolvent estimate holds for $\mathcal{B}_{\mathrm{ind}} + t\Delta\mathcal{B}$, $0 \le t \le 1$. The associated Riesz projections form a norm-continuous path of finite-rank idempotents and therefore have constant rank. Since the rank at $t = 0$ is one, Π_{τ_q} also has rank one. The finite-rank trace identities

$$\lambda_{\mathrm{ind}} = \mathrm{tr}(\mathcal{B}_{\mathrm{ind}} \Pi_{\mathrm{ind}}), \qquad \lambda_{\tau_q} = \mathrm{tr}(\mathcal{B}_{\tau_q} \Pi_{\tau_q}) \tag{C.15}$$

then imply Eq. (4.22): the term $\Delta\mathcal{B}\Pi_{\tau_q}$ has rank at most one, while $\mathcal{B}_{\mathrm{ind}}(\Pi_{\tau_q} - \Pi_{\mathrm{ind}})$ has rank at most two.

On ∂D_*, Eq. (4.23) gives $|\lambda_{\mathrm{ind}}(z) - 1| \ge 2d_\lambda$, whereas Eq. (4.22) bounds the perturbation by d_λ. Rouché's theorem therefore gives exactly one zero $\widetilde{z}_*$ of $\lambda_{\tau_q} - 1$ in D_* and proves Eq. (4.24).

Step 4: residue at the displaced root. At a common value of z, insert and subtract the two intermediate products $\mathsf{R}_{\mathrm{ind}} \Pi_{\tau_q} \widetilde{\mathsf{S}}_X$ and $\mathsf{R}_{\mathrm{ind}} \Pi_{\mathrm{ind}} \widetilde{\mathsf{S}}_X$. The triangle inequality, Eq. (4.21), and the uniform constants in Eqs. (4.26)–(4.27) give

$$\sup_{z \in D_*} \|\widetilde{\mathcal{N}}(z) - \mathcal{N}_{\mathrm{ind}}(z)\| \le d_{\mathrm{num}}^{(0)}. \tag{C.16}$$

The fundamental theorem of calculus along the line segment from z_* to $\widetilde{z}_*$ gives

$$\|\widetilde{\mathcal{N}}(\widetilde{z}_*) - \widetilde{\mathcal{N}}(z_*)\| \le L_{\mathrm{num}} \delta_z. \tag{C.17}$$

Combining the last two estimates yields

$$\|\widetilde{\mathcal{N}}(\widetilde{z}_*) - \mathcal{N}_{\mathrm{ind}}(z_*)\| \le D_{\mathrm{num}}. \tag{C.18}$$

The same argument applied to $\partial_z \lambda_{\tau_q}$ gives

$$|\partial_z \lambda_{\tau_q}(\widetilde{z}_*) - \partial_z \lambda_{\mathrm{ind}}(z_*)| \le d_{\lambda'}^{(0)} + L_{\lambda'} \delta_z = D_{\lambda'}. \tag{C.19}$$

Since $D_{\lambda'} < \alpha/2$, $|\partial_z \lambda_{\tau_q}(\widetilde{z}_*)| \ge \alpha/2$. The elementary difference-of-quotients estimate now gives

$$\left\| -\frac{\widetilde{\mathcal{N}}(\widetilde{z}_*)}{\partial_z \lambda_{\tau_q}(\widetilde{z}_*)} + \frac{\mathcal{N}_{\mathrm{ind}}(z_*)}{\partial_z \lambda_{\mathrm{ind}}(z_*)} \right\| \le \frac{2 D_{\mathrm{num}}}{\alpha} + \frac{2 \|\mathcal{N}_{\mathrm{ind}}(z_*)\| D_{\lambda'}}{\alpha^2}, \tag{C.20}$$

which is Eq. (4.31). This proves Proposition 4.2. □

C.6 Fredholm resolvent near a simple spectral zero

Proof of Theorem 3.2. Under the hypotheses of Theorem 3.2, the analytic Fredholm theorem implies that $[I - \mathcal{B}(z)]^{-1}$ is a meromorphic operator-valued function on U; see Chapter VII of Ref. [6]. If $1 \notin \operatorname{spec} \mathcal{B}(0)$, bounded invertibility is open in operator norm. The inverse is analytic near $z = 0$, and insertion into Eq. (2.33) proves Theorem 3.2(i).

Suppose now that $\lambda(0) = 1$ is algebraically simple. Analytic perturbation theory supplies, after reducing U if necessary, analytic continuations $\lambda(z)$, $|r(z)\rangle$, and $\langle \ell(z)|$ satisfying

$$\mathcal{B}(z)|r(z)\rangle = \lambda(z)|r(z)\rangle, \quad \langle \ell(z)|\mathcal{B}(z) = \lambda(z)\langle \ell(z)|, \quad \langle \ell(z)|r(z)\rangle = 1. \tag{C.21}$$

The associated Riesz projection is

$$\Pi_\lambda(z) = |r(z)\rangle\langle \ell(z)|. \tag{C.22}$$

On the complementary invariant subspace, $I - \mathcal{B}(z)$ remains invertible near the origin. Hence

$$[I - \mathcal{B}(z)]^{-1} = \frac{\Pi_\lambda(z)}{1 - \lambda(z)} + \mathcal{R}_\lambda(z), \tag{C.23}$$

where $\mathcal{R}_\lambda$ is analytic. The simple-zero condition gives

$$1 - \lambda(z) = -\partial_z \lambda(0) z + O(z^2). \tag{C.24}$$

Substitution in Eq. (2.33) yields

$$C_{\mathrm{sing}}(z) = -\frac{\mathsf{R}_X(0)|r(0)\rangle\langle \ell(0)|\mathsf{S}_X(0)}{\partial_z \lambda(0) z}. \tag{C.25}$$

The outer product is nonzero exactly when both displayed factors are nonzero. This proves part (ii) and Eq. (3.9).

Equation (C.25) is an algebraic matrix residue in the chosen response convention and proves part (iii). Its identification with a Wightman atom follows under the matching hypotheses of Theorem 3.5. For an approximate nonnormal kernel, those hypotheses must be tested independently of the critical eigenvalue. □

C.7 Uniform truncation and Riesz projections

Compactness of a finite matrix gives no convergence statement for the operator it approximates. The next proposition is a standard consequence of norm-resolvent control and analytic perturbation [6, 7]. Its final assertion specifies the additional simple-zero hypothesis needed to pass from spectral projections to response residues at moving roots.

Proposition C.1 (Convergence of an isolated spectral subspace)**.** *Let $K \Subset U$ be compact, and let $\widetilde{\mathcal{B}}_N(z)$ and $\mathcal{B}(z)$ act on one Banach space and be operator-norm continuous on a neighbourhood of K. Suppose*

$$\sup_{z \in K} \|\widetilde{\mathcal{B}}_N(z) - \mathcal{B}(z)\| \longrightarrow 0. \tag{C.26}$$

Let γ be a closed contour in the spectral ζ plane which lies in the resolvent set of $\mathcal{B}(z)$ for every $z \in K$ and is uniformly separated from its spectrum. Then, for all sufficiently large N, γ also lies in the resolvent set of $\widetilde{\mathcal{B}}_N(z)$ and

$$\sup_{z\in K} \|\Pi_{\gamma,N}(z) - \Pi_\gamma(z)\| \longrightarrow 0. \tag{C.27}$$

For convergence of critical roots and residues, assume additionally that the kernel families are analytic on a common open neighbourhood of a closed disk $D \subset \operatorname{int} K$. Let γ enclose exactly one algebraically simple eigenvalue $\lambda(z)$ and no other spectrum for every $z \in D$, and suppose $z_* \in \operatorname{int} D$ is the only zero of $\lambda - 1$ in D, with

$$\lambda(z_*) = 1, \qquad \lambda'(z_*) \neq 0.$$

Assume the kernel derivatives and the analytic source and readout maps converge uniformly on D in their specified operator norms. Then, for all sufficiently large N, the corresponding branch $\lambda_N - 1$ has exactly one zero $z_{*,N}$ in D, this zero is simple, and $z_{*,N} \to z_*$. The algebraic residues at their respective roots obey

$$-\frac{\mathsf{R}_{X,N}(z_{*,N})\Pi_{\gamma,N}(z_{*,N})\mathsf{S}_{X,N}(z_{*,N})}{\lambda_N'(z_{*,N})} \longrightarrow -\frac{\mathsf{R}_X(z_*)\Pi_\gamma(z_*)\mathsf{S}_X(z_*)}{\lambda'(z_*)}.$$

This is a residue of the selected critical response; analytic direct terms do not contribute. Algebraic simplicity in the spectral variable alone does not give the required first-order zero in z.

Proof. Indeed, the exact resolvents are uniformly bounded on the compact set $K \times \gamma$. Equation (C.26) and a Neumann series give existence and uniform convergence of the approximate resolvents. Contour integration then proves Eq. (C.27). The explicit norm-continuity hypothesis gives the asserted compact-set bound even for nonnormal kernels. For the additional assertion, nearby projections have the same rank one. Analytic perturbation on these spectral subspaces gives uniform convergence of λ_N to λ and, using the kernel-derivative convergence, of λ_N' to λ' [6, 7]. Since $\lambda - 1$ is nonzero on ∂D, Rouché's theorem gives exactly one approximate zero in D, counted with multiplicity, hence a simple zero. Applying the same argument to arbitrarily small disks about z_* gives $z_{*,N} \to z_*$. Uniform derivative convergence and continuity imply $\lambda_N'(z_{*,N}) \to \lambda'(z_*) \neq 0$; the denominators are therefore bounded away from zero for large N. Uniform convergence of the projections, sources and readouts gives convergence of the numerators at these moving roots, proving the residue limit. This is the qualitative version of the cross-root comparison in Proposition 4.2, specifically Eqs. (4.24) and (4.31). □

The norm hypothesis in Eq. (C.26) can be certified either by an explicit remainder estimate $\sup_{z\in K} \|\mathcal{B}(z) - \widetilde{\mathcal{B}}_N(z)\| \le r_N(K)$ with $r_N(K) \to 0$, or by the following a posteriori construction when all approximants act on the same Banach space.

A computable summable-increment criterion. Let $U \subset \mathbb{C}$ be open and let $\widetilde{\mathcal{B}}_N : U \to \mathcal{K}(\mathcal{X}_{\mathfrak{C}})$ be analytic compact-operator families. Suppose that for every $K \Subset U$ there are certified bounds $\varepsilon_N(K) \geq 0$ and explicitly computable numbers $r_N(K) \geq 0$ with certified decay to zero such that

$$\begin{aligned} \sup_{z\in K} \|\widetilde{\mathcal{B}}_{N+1}(z) - \widetilde{\mathcal{B}}_N(z)\| &\leq \varepsilon_N(K), \\ \sum_{n=N}^{\infty} \varepsilon_n(K) \leq r_N(K), &\qquad r_N(K) \longrightarrow 0. \end{aligned} \tag{C.28}$$

Then the sequence converges locally uniformly to an analytic compact family $\mathcal{B}_\infty$, with the computable tail estimate

$$\sup_{z\in K} \|\mathcal{B}_\infty(z) - \widetilde{\mathcal{B}}_N(z)\| \leq \sum_{n=N}^{\infty} \varepsilon_n(K) \leq r_N(K). \tag{C.29}$$

If a consistency argument for the truncation identifies $\mathcal{B}_\infty = D_X\mathfrak{F}(X_*; z, \eta)$, this tail bounds the truncation error in Eq. (C.26). The same construction applies to the source and readout maps when their increments have corresponding effective tail bounds. Local uniform convergence of the analytic kernel families also yields convergence of their z derivatives on smaller compact subsets of U. This is the standard Banach-valued Weierstrass argument: Eq. (C.28) makes the sequence Cauchy in $C(K, \mathcal{L}(\mathcal{X}_{\mathfrak{C}}))$, summation gives Eq. (C.29), and norm closure preserves compactness. Local uniform convergence preserves analyticity, while the Cauchy formula on a slightly larger compact set gives derivative convergence. The explicit majorants $r_N(K)$ make the tail effective; summability by itself gives convergence but not a computable error bound.

For example, certified computable constants $A(K) \geq 0$ and $0 < \theta(K) < 1$ with $\varepsilon_n(K) \leq A(K)\theta(K)^n$ give $r_N(K) = A(K)\theta(K)^N/[1 - \theta(K)]$. This geometric majorant supplies both the tail bound and its rate of decay.

If a threshold enters U, the meromorphic decomposition Eq. (C.23) may fail. When operator-norm convergence is unavailable, stable-looking eigenvalues of finite nonnormal matrices also require a spectral-pollution analysis. These cases call for an additional spectral argument beyond the isolated-pole theorem.

Declarations

Conflict of interest. The author declares no competing interests.

Data and code availability. This work uses no empirical dataset. The analytic examples are specified by the equations and conventions in the manuscript and can be reproduced directly from them. Related finite-dimensional calculations use *Coarse-Graining Genesis Framework V5* [16], available at https://doi.org/10.5281/zenodo.22067005.

Funding. No funding was received for this work.

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